Amaro

de Matos, João Amaro - Theoretical Foundations of Corporate

- Ross (1977): managers' signalling through fee schedules: Ch. 3, §3.2.1 (p. 79–82)
- Myers–Majluf: signals conditioned by investment opportunities: Ch. 3, §3.2.3 (p. 85–89)

Hence, without an indirect information transfer, venture capital markets may fail to exist.

Signaling becomes a crucial ingredient in a world with asymmetric information. In this context, firms are assumed to be of different types but will always wish to signal that they are of the best type. If firms of all types are led to use the same signaling strategy in all states, this signaling is said to be a *pooling* equilibrium. Otherwise, the equilibrium is said to be *separating*. Notice that a separating equilibrium does not necessarily distinguish among all the types. If that were the case, the equilibrium would be termed *fully separating*.

In this section, several models will be presented, according to which firms try to signal their values to the market indirectly through their debt levels, for instance. This is an alternative way to explain why firms look for an optimal debt level.

3.2.1 Managers' Signaling because of Fee Schedules

In the model of Ross (1977), the signal for the value of the firm is the total amount of debt issued with face value B. As discussed before, the signal must imply some cost for the management, supposedly offset by the benefits of signaling in equilibrium. Ross proposes a fee schedule including a punishment for the case of bankruptcy.

Two additional assumptions are required: the first to allow for explicit calculations of the cost of bankruptcy, and the second to clarify the kind of informational asymmetry incorporated.

Assumption 3.2.2. *The returns X of the firm are represented by a random variable uniformly distributed on $[0, k]$, where k characterizes the type of the firm and runs over the interval $[c, d]$.*

Assumption 3.2.1 here takes the specific form:

Assumption 3.2.3. *Managers–insiders know their own firm's k type.*

The model is defined within two points in time. Taking $\gamma_\circ$ and γ_1 as two real numbers, and $V_\circ$ as the time-0 value of the firm, the incentive schedule has two components: one for time 0 and another for time 1. The first one is proportional to the value of the firm at time 0, taken to present value at time 1. For a constant interest rate r, the component reads

$$M_\circ = (1 + r)\gamma_\circ V_\circ. \tag{3.7}$$

The second component is taken proportional to the *expected* return minus a penalty L in case of bankruptcy—that is, in case $X < B$,

$$M_1 = \gamma_1 E(X - L\mathbf{1}_{\{X<B\}}),$$

where $\mathbf{1}_{\{X<B\}}$ denotes a random variable assuming the value 1 if $X < B$ and a value of 0 otherwise. The higher the debt, the higher the risk of bankruptcy, and therefore the more efficient the management should be to avoid the penalty.

If the type of the firm is known with certainty (say, $k = t$), the density of probability of the returns is constant and equal to $1/t$ in the interval $[0, t]$, and the value of the firm at time 1 is written

$$(1 + r)V_\circ = \int_0^t x \frac{dx}{t} = \frac{t}{2},$$

which implies that

$$V_\circ = \frac{t}{2(1 + r)}. \tag{3.8}$$

If the essence of this approach is that the debt level B should be chosen so as to signal the type t through some function, say, $t = a(B)$, then the incentive schedule for the manager

$$M = M_\circ + M_1$$

must be seen by an outsider as

$$M = \frac{1}{2}\gamma_\circ a(B) + \gamma_1\left[\frac{1}{2}t - L\frac{B}{t}\right]. \tag{3.9}$$

To write M in the form (3.9), the expression (3.8) was substituted in equation (3.7) with $a(B)$ in the place of t and M_1 was rewritten using the fact that

$$E(X) = \int_0^t x \frac{dx}{t} = \frac{t}{2} \text{ and } E(\mathbf{1}_{\{X<B\}}) = \int_0^B \frac{dx}{t} = \frac{B}{t},$$

with B taken to be less[12] than t.

The signal of the true t is therefore a function $a(B)$ at time 0. The manager chooses B so as to maximize its schedule M. The debt level B must therefore satisfy the first-order condition[13]

$$\frac{1}{2}\gamma_\circ a'(B) = \gamma_1 \frac{L}{t}. \tag{3.10}$$

However, the management will also want the signal to be efficient, which is to say that the market perceives in equilibrium the correct type t through $a(B)$. Hence, the condition

$$a(B) = t \tag{3.11}$$

must be *imposed*. Substituting (3.11) into (3.10) gives in differential form

$$a(B)da(B) = 2\frac{\gamma_1}{\gamma_\circ}LdB.$$

Integrating both sides, it follows that

$$\frac{1}{2}a(B)^2 = 2\frac{\gamma_1}{\gamma_\circ}LB + b, \tag{3.12}$$

which can be solved for the signaling function $a(B)$ up to the constant of integration b:

$$a(B) = 2\sqrt{\frac{\gamma_1 L}{\gamma_\circ}}\left[B + b\frac{\gamma_\circ}{2\gamma_1 L}\right]^{1/2}. \tag{3.13}$$

To find the constant b, a *boundary condition* is necessary. In the context of this model, such a condition follows from the fact that it does not pay to the lower type $k = c$ to signal, that is, $c = a(0)$, from what follows from either equation (3.12) or equation (3.13)

$$b = \frac{c^2}{2}.$$

Substitution of this value of b into (3.13) together with the equilibrium condition (3.11) gives

$$t = \sqrt{\frac{4\gamma_1 L}{\gamma_\circ}B + c^2}$$

or, equivalently,

$$B = \frac{\gamma_\circ}{4\gamma_1 L}\left[t^2 - c^2\right]. \tag{3.14}$$

For this signaling to be an equilibrium, it should satisfy two conditions:

1. It gives managers no incentive to signal a *false type* as long as the probability of bankruptcy is different from 1, and
2. It permits full discrimination of the types.

In equilibrium all types of agents other than c will pay a cost, since they will implicitly accept a positive probability of incurring a penalty.

The optimum level of debt is used to identify the class to which the firm belongs. Note that given the type of the firm, its value is directly given by equation (3.8), independently of its capital structure which is compatible with the MM framework.

What this model states is that, under asymmetric information, by changing B the management changes the *perceived* value of the firm to the uninformed investors. The fact that these economic agents are aware of this forces the optimal relation between t and B to be unique and given by (3.14).[14]

However, as noted by Kim (1989), this model has a weakness. It does not impose any penalty on bondholders or stockholders if the optimal debt level is exceeded. Hence, these agents may have an incentive to give a "side payment" to managers in order to adopt a higher debt ratio than the equilibrium one, which makes the signaling mechanism in Ross's model noncredible.

One more important criticism is discussed in the paper: Managers cannot deal in their own shares. Otherwise, they would have incentive to signal erroneously, enhancing the value of the firm, and then to cash in.

3.2.2 When Managers Are Also Investors

Leland and Pyle (1977) developed a model in the same signaling context as Ross, in which risk averse managers are also investors in the firm. Consider entrepreneurs starting a project with a random return $X = \overline{X} + \epsilon$, with $E(\epsilon) = 0$, needing a capital investment K. Here, these entrepreneurs, who are going to manage the project, are the only individuals who know the expected return $\overline{X} = E(X)$. At time 0 they sell a fraction $(1 - \alpha)$ of the project and retain a fraction α. The uninformed investors use this proportion as a signal of the value of the firm. Underlying this idea is, clearly, Assumption 3.2.1. As the management better knows the value of the firm, the fact that it holds a reasonable part of the equity is a good signal, whereas no participation will naturally be considered as a bad signal. As the managers are risk averse, they would like to diversify but cannot, since they have to hold on to their shares. That is their cost of signaling, and that is why the signal is credible. Managers of good projects are then easily identified with this procedure because their claim on the firm is worth more. That is why they can afford to signal in that way and hold onto a greater portion of the shares.

In this model, Assumption 3.2.1 that managers have superior information takes the simple form

Assumption 3.2.4. *Only the management knows the value of the expected return X.*

Uninformed investors will have to *infer* in equilibrium the expected return through a function of α, say, $X(\alpha)$. Consequently, the *perceived* value of the firm, say, a function $V(\alpha)$, will also depend on α. As in Ross's model,

From Proposition 3.2.1, it follows that $X(\alpha)$ increases with debt; therefore equation (3.15) implies that the perceived value $V(\alpha)$ increases with the debt level B, just as in Ross's model.

Notice once again that the MM results are not really questioned here. As in Ross's model, V and B may be parameterized through the signaling variable α, in the context of the transfer of information. It is *not* a causal relationship that links V and B and therefore the MM propositions are preserved in all their purity, as they should be.

3.2.3 Signals Conditioned by Investment Opportunities

Another model that describes how informational advantages of the managers may affect the decision to issue stock or not is developed by Myers and Majluf (1984).

Supposing that there is an investment opportunity, management can try to raise internal funds to finance the investment or to raise external funds by issuing either debt or equity.

To issue equity is problematic. Taking into account the informational advantage of the management, having too many securities in the market leads the market players to suspect overpricing, and the prices of the securities fall, lowering the value of the company. Hence, there is a cost that the management will be willing to incur only if the benefits of the new investment are large enough to offset it.

As a consequence, a firm that needs to issue equity to invest will pass up some good investment projects with positive net present value. In this sense, it would be preferable to be able to be financed by internal funds, since there would be no opportunity cost. Thus, internal funds have an intrinsic value. If the management could costlessly communicate to the outsider investors that equity is being issued in order to raise funds to finance excellent investment from which the outsiders could well benefit, and not to expropriate them, the value of the internal funds would disappear. Also, the prices would not fall and the cost of signaling would not exist. However, such direct transfer of information is not credible, as aforementioned, and the only way of avoiding the costs of raising external funds is to have enough internal funds. Also, it follows from the model of Myers and Majluf that, on average, risky debt implies lower costs than equity. Hence, a hierarchy in the financing mechanism is established by this model in the sense that internal funds are preferred to external ones, and, among these, debt is preferred to equity issue.

Note that this hierarchy strongly contradicts the free cash flow hypothesis of Jensen (1986) according to which shareholders would appreciate low levels of cash and a high level of debt. Among other advantages, this strategy lowers the cost of monitoring the management. In contrast,

Myers and Majluf claim that it is good to have internal funds, so that profitable investments will not be passed up. Therefore, these two hypotheses here described are direct competitors.[16]

To see how all these conclusions are derived, the model must be formalized. Let I be the required investment and S denote the *slack* of the firm (i.e., large holdings of cash or marketable securities) or the ability to issue default-risk-free debt.

If $S \geq I$, it will not be necessary to issue any equity at all in order to finance the investment. The only interesting case happens when $S < I$, where the required amount of equity to be issued is $E = I - S$.

As opposed to the other models previously discussed, this is a three-date model with three distinct times—say, time 0, time 1, and time 2. Let us introduce the following notation. At time 0, the possible value of the assets of the firm at time 1 is described by a random variable $\widetilde{A}$. Similarly, let $\widetilde{B}$ denote the possible net present value of the investment opportunity at time 1.

At time 0, the information is symmetric. All the agents, managers and market, have the same information—namely, the distributions of $\widetilde{A}$ and $\widetilde{B}$ and the value of S.

At time 1, the asymmetry of information between managers and outsider investors is specified. Assumption 3.2.1 takes the form

Assumption 3.2.5. *At time* 1*, the management knows the realizations of* $\widetilde{A}$ *and* $\widetilde{B}$*, denoted respectively by* a *and* b*, which are unknown to the outsider investors. Negative values of* a *and* b *are ruled out.*

At this stage, the only thing that the outsiders know is that the amount of equity to be issued is $E = \max[0, I - S]$. Also, at time 1, let P be the market value of the shares of old shareholders if the stock is not issued and P' be the price of the same shares if it is issued. At time 2, finally, all the information is disclosed.

Let V° be the value of the firm for the old shareholders. If the management, knowing the values of a and b, decides not to issue equity, then the company is still owned by the same old shareholders with assets of value a and slack S. Thus

$$V^\circ = a + S.$$

If, on the other hand, knowing the value of a and b, the management decides to issue equity E, then the ownership of the company is shared between the old and new shareholders. The total value of the company will include the issued equity E and the net present value of the investment opportunity apart from the assets of value a and slack S. But the old shareholders will have only a proportion $P'/(P' + E)$ of all of this.

Thus, in this case

$$V^{\circ} = \frac{P'}{P' + E}(E + S + a + b).$$

To explain the decision of the managers, the following assumption is required:

Assumption 3.2.6. *The management acts in the best interest of the old shareholders. Moreover, old shareholders are assumed to be passive—that is, they do not adjust their portfolios in response to the issue–invest decision of the firm.*

If that is so, the decision to issue new equity is taken only if the old shareholders are better off in such case—that is, only if

$$\frac{P'}{P' + E}(E + S + a + b) \geq a + S.$$

This inequality may be rewritten as

$$b \geq -E + \frac{E}{P'}(S + a) \tag{3.20}$$

and should naturally be restricted by Assumption 3.2.5 to the region $a \geq 0$, $b \geq 0$.

Hence, inequality (3.20) divides the domain of the joint probability distribution of $\widetilde{A}$ and $\widetilde{B}$ into two disjoint regions identified in Figure 3.1: a region M' in which all pairs (a, b) satisfy the inequality and the firm issues new equity and invests in the new project, and a region M in which no pair (a, b) satisfies the inequality and there is no investment. In the first region, the amount of equity issued is $E = I - S$, whereas in the second, it is zero.

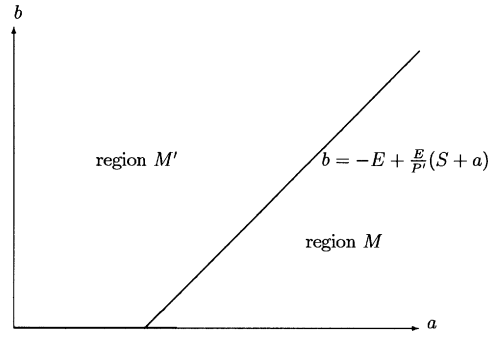


Figure 3.1. *The Regions of Investment and No-Investment Decisions According to Myers and Majluf's Model*

It is simple to see that the price P for the shares when $(a, b) \in M$ is greater than the price P' when $(a, b) \in M'$. The equilibrium prices should be given by[17]

$$P = S + E(\widetilde{A}|M) \tag{3.21}$$

and

$$P' = S + E(\widetilde{A} + \widetilde{B}|M').$$

For any $(a, b) \in M$, the inequality (3.20) is not satisfied and hence

$$b < -E + \frac{E}{P'}(S + a) \text{ or } a > P'(1 + b/E) - S.$$

Since $b/E \geq 0$ from Assumption 3.2.5, it follows that for any $(a, b) \in M$, $a > P' - S$. Thus $E(\widetilde{A}|M) > P' - S$ and (3.21) gives directly $P > P'$.

As advanced by Myers and Majluf themselves, in the case where the managers are also investors, the work of Leland and Pyle discussed already provides a natural extension of this model. In that case, the interest of the management in buying the new issues would not be a signal that the shares are overvalued, but rather that there is a good investment opportunity.

Up to now, only equity issue has been considered in this model. However, other forms of external funds could be raised, such as debt. To consider the relative benefits of issuing either debt or equity, it is necessary to describe the gains or losses to the old shareholders at disclosure of information. The amount of equity issued has the value $E = I - S$ at time 1 and is assumed to have a value $E + \Delta E$ at time 2. Here ΔE denotes the capital gain when information is revealed to the market.

Similarly, if debt is issued, the same argument holds with B replacing E and with ΔB in the place of ΔE. Capital gains may have both negative and positive values, reflecting the fact that both debt and equity considered here are risky securities.[18]

However, under the restrictive assumptions such as those in the paper of Galai and Masulis (1976), the values of ΔE and ΔB will have the same signal but $|\Delta E| > |\Delta B|$. Put differently, capital gains or losses are always less in absolute value when issuing debt than when issuing equity. This is because equity is riskier than debt.

The variation at time 2 of the additional payoff to the old shareholders from the fact of issuing and investing is seen from (3.20) to be $b - \Delta E$ in the case of equity and $b - \Delta B$ in the case of debt.

Hence, it is always in the interest of the old stockholders to issue debt rather than new stock. The opposite would happen only when $\Delta E < \Delta B$ and this, by the preceding arguments, implies that both ΔE and ΔB are negative. In other words, equity issue would be preferred to debt issue by

stockholders only if there is a capital loss. But that is a horrible signal, so there would be no investors to buy the new stock.

Hence, the less risky the security, the better off are the old shareholders. From this point of view, internal funds are preferred to external funds and, among the latter type, debt dominates equity.

This hierarchy expresses what is usually known as the "pecking order" hypothesis about the preferences over different types of funds. This result can be seen as constructed from the fact that debt's and equity's different information sensitivities imply different adverse selection costs. The point is that the pecking order holds because debt is less sensitive to management's private information and hence is less costly to issue.[19]

Masulis (1988) reports a work[20] that provides counterexamples to the assumption that the pecking order hypothesis can be generalized from Myers and Majluf's model.

An alternative justification for issuing new stock is to assume some additional informational asymmetry on the parameters that value the options of issuing debt or equity. For instance, a large asymmetry of information regarding the variance rate may tempt the managers to issue equity, in the case that investors overestimate it, other things being equal. The management might do that to signal to the investors that the firm is safer than prospective debtholders might think, which would make the issue of debt very expensive.

3.2.4 Stock Repurchase as a Signal

As an almost inverse problem to the one just described, Vermaelen (1981,1984) studies the case of stock repurchase. In this case, the management would use its informational advantage to buy back shares of its company from the market. Intuitively, and following the same line of reasoning developed in the previous models, this would suggest to the market that securities are *underpriced*. Thus, after a stock repurchase announcement, one would expect the share prices to rise as a reaction.

It is interesting to notice that firms spend their cash and/or debt capacity to buy back stock. The sources for future investment may therefore be reduced and firms might be restricted to funds at the lowest levels of the pecking order (i.e., their better-valued stock). Hence, stock repurchases strongly contradict the pecking order theory.

If the management wants to signal that the stock is underpriced, there are two simultaneous signals to give. One is the price offered to buy back the stock, P_T, and the other is the number of shares to be repurchased, N_p.

Clearly, this process increases the debt-equity ratio. The issues to be discussed in this section are how stock repurchases affect the prices of

Vermaelen (1981,1984) provides evidence that on average the premium π is significantly higher than the value of information, which is inconsistent with the hypothesis of the existence of costless fully revealing mechanisms. Also, it is reported that the insider-holdings variable is statistically significant, which contradicts the hypothesis that full ex-post settling up is possible and confirms the initial assumption that stock repurchases reveal information.

Finally, it is important to stress that when firms announce a capital structure change, they usually raise money to invest and/or reveal inside information. Some discussion of the dynamic pecking order theory (proposed by Loughran and Ritter (1997)) should also be mentioned. Their basic idea is that firms tend to issue shares when they are overvalued and buy them back when they are undervalued. This means, although the firm has some long-run target leverage ratio, financial managers may once in a while deviate from this target for opportunistic reasons. For example, you may want to issue equity, even if you are below your target debt-equity ratio, because you believe your shares are undervalued. This type of opportunistic timing behavior is consistent with long-term event studies and makes it almost impossible to test or optimal capital structure theories.

3.2.6 Exercises

1. Consider the signaling model of Ross (1977). Explain what happens to the optimal decision of debt (3.14) if $\gamma_0 = 0$. And what happens if $\gamma_1 = 0$?
2. In the context of Leland and Pyle (1977), suppose that the expected utility of the manager is given by $E[U(W_1)] = G[E(W_1) - \frac{b}{2}\sigma^2_{W_1}]$, where G is an increasing function, $\sigma^2_{W_1}$ is the variance of the wealth At time 1, and the risk adjustment parameter λ is given by $\lambda = \frac{E[M]-(1+r)V_M}{\sigma^2_M}\operatorname{cov}(\varepsilon, M)$. In this case, under what conditions does the level of debt increase with the proportion α of the equity owned by the managers?
3. Prove Proposition 3.2.2.
4. Instead of Assumption 3.2.6, assume that the management acts in the interest of all shareholders. How would that affect the investment decision of the management? If management acts in the interest of the new shareholders, what is the optimal equity issuance rule?
5. Myers and Majluf (1984) assume that there are no issuing costs and prove that $P' < P$. What happens to the investment decision of the management and to the conclusion that $P' < P$ in the presence of issuing costs?
6. Prove that in Section 3.2.4, P_T is always higher than $\overline{P}$.

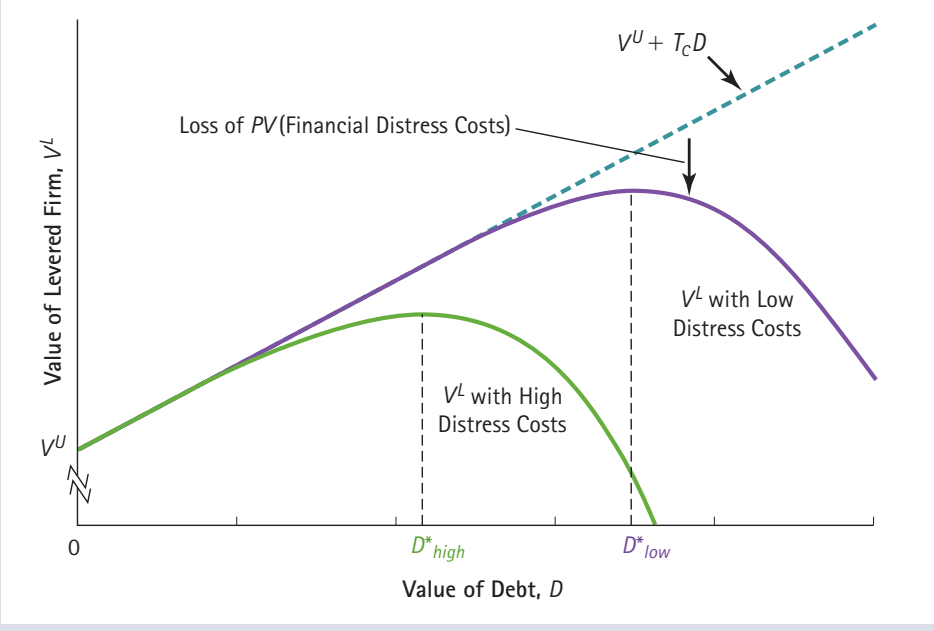
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- Agency costs and information: lemons, pecking order: Ch. 16, §16.6 (p. 482–487)
- Signalling with payout policy: Ch. 17, §17.5 (p. 518–522)

FIGURE 16.8
Optimal Leverage with Taxes and Financial Distress Costs

As the level of debt increases, the tax benefits of debt increase until the interest expense exceeds the firm's EBIT. However, the probability of default, and hence the present value of financial distress costs, also increases. The optimal level of debt, D^*, occurs when these effects balance out and the value of the levered firm is maximized. D^* will be lower for firms with higher costs of financial distress.



optimal debt choice for a firm with high costs of financial distress is indicated by D^*_{high}. Not surprisingly, with higher costs of financial distress, it is optimal for the firm to choose lower leverage.

The tradeoff theory helps to resolve two important facts about leverage:

1. The presence of financial distress costs can explain why firms choose debt levels that are too low to fully exploit the interest tax shield.
2. Differences in the magnitude of financial distress costs and the volatility of cash flows can explain the differences in the use of leverage across industries.

Concept Check

9. According to the tradeoff theory, how should a financial manager determine the right capital structure for a firm?
10. Why would managers in one industry choose different capital structures than those in another industry?

16.6 Additional Consequences of Leverage: Agency Costs and Information

Taxes and financial distress costs are not the only capital market imperfections that arise in practice. In this section, we continue to relax the assumption of perfect capital markets to see other ways in which capital structure can affect a firm's value. We begin by discussing how leverage alters managers' incentives and changes their investment decisions. We then address the complexities of stakeholders in the firm having different information.

Agency Costs

agency costs The costs that arise when there are conflicts of interest between stakeholders.

Agency costs are costs that arise when there are conflicts of interest between stakeholders. In Chapter 1, we briefly mentioned the agency problem that arises when managers put their own interests ahead of shareholders' interest. Here, we discuss how debt can mitigate this problem. However, it may also distort equity holders' preferences for the types of projects pursued by the firm.

Managerial Entrenchment. Although managers often do own shares of the firm, in most large corporations they own only a very small fraction of the outstanding shares. Shareholders, through the board of directors, have the power to fire managers. In practice, they rarely do so unless the firm's performance is exceptionally poor.

management entrenchment A situation arising as a result of the separation of ownership and control, in which managers may make decisions that benefit themselves at investors' expense.

This separation of ownership and control creates the possibility of **management entrenchment**; facing little threat of being fired and replaced, managers are free to run the firm in their own best interests. As a result, managers may make decisions that benefit themselves at investors' expense. Managers may reduce their effort, spend excessively on perks such as corporate jets, or undertake wasteful projects that increase the size of the firm (and their paychecks) at the expense of shareholder value, often called "empire building." If these decisions have a negative NPV for the firm, they are a form of agency cost.

These agency costs are most likely to arise when equity ownership is highly diluted (so that no individual shareholder has an incentive to monitor management closely) and when the firm has a great deal of cash available for managers to spend on wasteful projects. Debt can therefore help in two ways. First, by borrowing rather than raising funds by issuing shares, ownership of the firm may remain more concentrated, improving the monitoring of management. Second, by forcing the firm to pay out cash to meet interest and principal payments, debt reduces the funds available at management's discretion. For managers to engage in wasteful investment, they must have the cash to invest. Only when cash is tight will managers be motivated to run the firm as efficiently as possible. Thus, leverage can provide incentives for managers to run the firm efficiently and effectively. These benefits provide an additional incentive to use debt rather than equity financing.

Airlines Use Financial Distress to Their Advantage

The need to generate cash flows sufficient to make interest payments may also tie managers' hands and commit them to pursue sound business strategies with greater vigor than they would without the threat of financial distress. For example, when American Airlines was in labor negotiations with its unions in April 2003, the firm was able to win wage concessions by explaining that higher costs would push it into bankruptcy. (A similar situation enabled Delta Airlines to persuade its pilots to accept a 33% wage cut in November 2004.) Without the threat of financial distress, American's managers might not have reached agreement with the union as quickly or achieved the same wage concessions.

A firm with greater leverage may also become a fiercer competitor and act more aggressively in protecting its markets because it cannot risk the possibility of bankruptcy. This commitment to aggressive behavior can scare off potential rivals. (However, this argument can also work in reverse, as a firm weakened by too much leverage might become so financially fragile that it crumbles in the face of competition, allowing other firms to erode its markets.)

Equity-Debt Holder Conflicts. When a firm has leverage, a conflict of interest exists if investment decisions have different consequences for the value of equity and the value of debt. Such a conflict is most likely to occur when the risk of financial distress is high. In some circumstances, managers may take actions that benefit shareholders but harm the firm's creditors and lower the total value of the firm.

To illustrate, put yourself in the place of the shareholders of a company in distress that is likely to default on its debt. You could continue as normal, in which case you will very likely lose the value of your shares and control of the firm to the bondholders.

Alternatively, you could:

1. Roll the dice and take on a risky project that could save the firm, even though its expected outcome is so poor that you normally would not take it on.
2. Conserve funds rather than invest in new, promising projects.
3. Cash out by distributing as much of the firm's capital as possible to the shareholders before the bondholders take over.

Why would you roll the dice? Even if the project fails, you are no worse off because you were headed to default anyway. If it succeeds, then you avoid default and retain ownership of the firm. This incentive leads to **excessive risk-taking**, a situation that occurs when a company is near distress and shareholders have an incentive to invest in risky negative-NPV projects that will destroy value for debt holders and the firm overall. Anticipating this behavior, security holders will pay less for the firm initially. This cost is likely to be highest for firms that can easily increase the risk of their investments.

excessive risk-taking A situation that occurs when a company is near distress and shareholders have an incentive to invest in risky negative-NPV projects that will destroy value for debt holders and the firm overall.

On the other hand, when default is very likely, a firm may pass up a good project before default because some or most of the benefit will go to bondholders (by reducing the risk or extent of default). For example, if a firm can invest \$100 in a project that will increase the value of its debt by \$30 and the value of its equity by \$90, then the project has a positive NPV of \$30 + \$90 − \$100 = \$20. Equity holders will not want to fund it as their gain (\$90) is less than the project's cost of \$100. In this case, there is an **under-investment problem**: Shareholders choose not to invest in a positive-NPV project because the firm is in financial distress and the value of undertaking the investment opportunity will accrue to bondholders rather than themselves. This failure to invest is costly for debt holders and for the overall value of the firm, because it is giving up the NPV of the missed opportunities. The cost is highest for firms that are likely to have profitable future growth opportunities requiring large investments.

under-investment problem A situation in which shareholders choose not to invest in a positive-NPV project because the firm is in financial distress and the value of undertaking the investment opportunity will accrue to bondholders rather than to shareholders.

The ultimate form of under-investment is cashing out. Knowing that they are likely to lose the firm to the bondholders, shareholders have an incentive to withdraw as much capital from the firm as possible before it enters bankruptcy and transfers to the debt holders. An extreme example of this would be to sell all the assets of the firm and distribute the proceeds to shareholders as a special dividend. The firm would then enter bankruptcy as a worthless shell. As we discussed in Chapter 15, bondholders can anticipate this problem and often require restrictions on the size of dividends and source of funds for those dividends.

Our discussion suggests that we can create a more complete tradeoff model of capital structure by incorporating the possible agency costs and benefits of debt in addition to tax benefits and financial distress costs. This tradeoff model representing optimal leverage is illustrated in Figure 16.9. When debt is too low, adding leverage increases the value of the firm by providing tax benefits and motivating managers to run the firm more efficiently (and avoid wasteful investment). But if debt is too high, the firm will incur financial distress costs and suffer from excessive risk taking and under-investment. The optimal level of debt D^* is the point that maximizes firm value by balancing these positive and negative consequences of leverage.

Financial Distress and Rolling the Dice, *Literally*

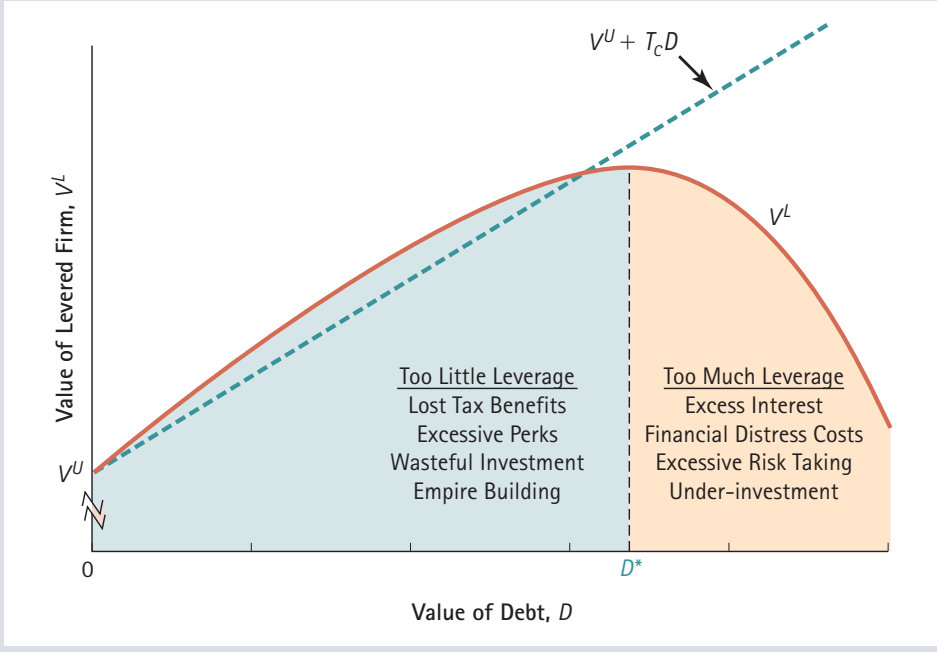
Shortly after launching FedEx, founder Frederick Smith was running out of cash and on the verge of shuttering the company's operations. The problem was massive fixed costs (airplanes, sorting facilities, delivery personnel, etc.) and low initial volume. "By 1973, Smith was so desperate for cash that he flew to Las Vegas to play the blackjack tables. He wired the \$27,000 he won back to FedEx."* The company used the winnings to pay its fuel bills, and the rest is history.

*"Frederick W. Smith: No Overnight Success," *Business Week,* September 20, 2004.

FIGURE 16.9

Optimal Leverage with Taxes, Financial Distress, and Agency Costs

As the level of debt increases, the value of the firm increases from the interest tax shield ($T_C D$) as well as improvements in managerial incentives. If leverage is too high, however, the present value of financial distress costs, as well as the agency costs from debt holder–equity holder conflicts, dominates and reduces firm value. The optimal level of debt, D^*, balances these benefits and costs of leverage.



Debt and Information

asymmetric information A situation in which parties have different information. It can arise when, for example, managers have superior information to that of outside investors regarding the firm's future cash flows.

signaling theory of debt The use of leverage as a way to signal good information to investors.

market timing When managers sell new shares because they believe the stock is overvalued, and rely on debt and retained earnings (and possibly repurchasing shares) if they believe the stock is undervalued.

Our final market imperfection is related to the role of information. Throughout this chapter, we have assumed that managers, stockholders, and creditors have the same information. We have also assumed that securities are fairly priced: The firm's shares and debt are priced according to their true underlying value. These assumptions may not always be accurate in practice. Due to **asymmetric information**, managers' information about the firm and its future cash flows is likely to be superior to that of outside investors. This asymmetric information may motivate managers to alter a firm's capital structure.

Leverage as a Credible Signal. To convince the stock market that it has great projects, a firm may commit to large future debt payments. If the projects are great, then the firm will have no trouble making the debt payments. But if the firm is making false claims and then does not grow, it will have trouble paying its creditors and will experience financial distress. This distress will be costly for the firm and also for its managers, who will likely lose their jobs. Thus, the managers can use leverage as a way to convince investors that they do have information that the firm will grow, even if they cannot provide verifiable details about the sources of growth. Investors will interpret the additional leverage as a credible signal of the managers' confidence. The use of leverage as a way to signal good information to investors is known as the **signaling theory of debt**.

Market Timing. When managers have better information than outside investors regarding the value of the firm, they may attempt to engage in **market timing** by selling new shares when they believe the stock is overvalued, and relying on debt and retained

earnings (and possibly repurchasing shares) if they believe the stock is undervalued. Managers who successfully time the market in this way benefit long-term shareholders by trading the firm's shares whenever they are mispriced. The firm's capital structure would change as the managers issued new equity (decreasing the leverage) or new debt (increasing the leverage) while attempting to time the market. As a result, the firm's capital structure may deviate above or below the optimal level described by the tradeoff theory, depending on management's view of the share price relative to its true value.

Adverse Selection and the Pecking Order Hypothesis. Suppose managers do try to issue equity when it is overpriced relative to its true value. Knowing this, how would investors react? Recall from Chapter 14 the adverse selection or lemons effect in seasoned equity offerings. Fearful that they are being sold a "lemon," investors will discount the price they are willing to pay for the stock.

The adverse selection problem has implications for capital structure. Managers do not want to sell equity if they have to discount it to find buyers. Therefore, they may seek alternative forms of financing. Debt issues may also suffer from adverse selection. Because the value of low-risk debt is not very sensitive to managers' private information about the firm (but is instead determined mainly by interest rates), the discount needed to attract buyers will be smaller for debt than for equity. Of course, a firm can avoid underpricing altogether by financing investment using its cash (retained earnings) when possible. The **pecking order hypothesis** therefore states that to avoid this "lemons cost":

pecking order hypothesis The idea that managers will have a preference to fund investment using retained earnings, followed by debt, and will only choose to issue equity as a last resort.

> *Managers will have a preference to fund investment using retained earnings, followed by debt, and will only choose to issue equity as a last resort.*

This hypothesis has implications for firms' capital structure. When firms are profitable and generate sufficient cash to fund their investments, they will not issue debt or equity, but just rely on retained earnings. Thus, highly profitable firms will have little debt in their capital structure. Only firms that need to raise external capital will have significant debt financing. According to the pecking order hypothesis, firms should almost never issue equity. In reality, it is likely that multiple forces will shape the firm's capital structure, and firms will issue equity when the agency costs or financial distress costs of debt are too great.

EXAMPLE 16.5 The Pecking Order of Financing Alternatives

Problem

Axon Industries needs to raise $9.5 million for a new investment project. If the firm issues one-year debt, it may have to pay an interest rate of 8%, although Axon's managers believe that 6% would be a fair rate given the level of risk. However, if the firm issues equity, the managers believe the equity may be underpriced by 5%. What is the cost to current shareholders of financing the project out of retained earnings, debt, and equity?

Solution

Plan

We can evaluate the financing alternatives by comparing what the firm would have to pay to get the financing versus what its managers believe it should pay if the market had the same information they do.

Execute

If the firm spends $9.5 million out of retained earnings, rather than paying that money out to shareholders as a dividend, the cost of financing the project is $9.5 million.

Using one-year debt costs the firm $\$9.5 \times (1.08) = \$10.26$ million in one year, which has a present value based on management's view of the firm's risk of $\$10.26 \div (1.06) = \$9.68$ million.

If equity is underpriced by 5%, then to raise $9.5 million the firm will need to issue shares that are actually worth $10 million. (For example, if the firm's shares are each worth $50, but it sells them for

0.95 × $50 = $47.50 per share, it will need to sell $9.5 million ÷ $47.50/share = 200,000 shares. These shares have a true value of 200,000 shares × $50/share = $10 million.) Thus, the cost of financing the project with equity will be $10 million.

Evaluate

Comparing the three options, retained earnings are the cheapest source of funds, followed by debt, and finally by equity. The ranking reflects the effect of differences in information between managers and investors that result in a lemons problem when they issue new securities, and particularly when they issue new equity.

Concept Check

11. How can too much debt lead to excessive risk-taking?
12. What is the "pecking order" hypothesis?

16.7 Capital Structure: Putting It All Together

In this chapter, we have examined a number of factors that might influence a firm's choice of capital structure. What is the bottom line for a financial manager devising the optimal capital structure for a firm? The optimal capital structure depends on market imperfections such as taxes, financial distress costs, agency costs, and asymmetric information, as follows:

1. *Make use of the interest tax shield if your firm has consistent taxable income.* The interest tax shield allows firms to repay investors and avoid the corporate tax.
2. *Balance the tax benefits of debt against the costs of financial distress when determining how much of the firm's income to shield from taxes with leverage.* Whereas the risk of default is not itself a problem, financial distress may lead to other consequences that reduce the value of the firm.
3. *Consider short-term debt for external financing when agency costs are significant.* Too much debt can motivate managers and equity holders to take excessive risks or under-invest in a firm. When free cash flows are high, too little leverage may encourage wasteful spending.
4. *Increase leverage to signal managers' confidence in the firm's ability to meet its debt obligations.* Investors understand that bankruptcy is costly for managers.
5. *Be mindful that investors are aware that you have an incentive to issue securities that you know are overpriced.* Thus, when an issue is announced, investors will lower their valuation of that security. This effect is most pronounced for equity issues because the value of equity is most sensitive to the manager's private information.
6. *Rely first on retained earnings, then debt, and finally equity.* This pecking order of financing alternatives will be most important when managers are likely to have a great deal of private information regarding the value of the firm.
7. *Do not change the firm's capital structure unless it departs significantly from the optimal level.* Actively changing a firm's capital structure (for example, by selling or repurchasing shares or bonds) entails transactions costs. Most changes to a firm's debt-equity ratio are likely to occur passively, as the market value of the firm's equity fluctuates with changes in the firm's stock price.

*29. Info Systems Technology (IST) manufactures microprocessor chips for use in appliances and other applications. IST has no debt and 100 million shares outstanding. The correct price for these shares is either $14.50 or $12.50 per share. Investors view both possibilities as equally likely, so the shares currently trade for $13.50.

IST must raise $500 million to build a new production facility. Because the firm would suffer a large loss of both customers and engineering talent in the event of financial distress, managers believe that if IST borrows the $500 million, the present value of financial distress costs will exceed any tax benefits by $20 million. At the same time, because investors believe that managers know the correct share price, IST faces a lemons problem if it attempts to raise the $500 million by issuing equity.

a. Suppose that if IST issues equity, the share price will remain $13.50. To maximize the long-term share price of the firm once its true value is known, would managers choose to issue equity or borrow the $500 million if:
   i. They know the correct value of the shares is $12.50?
   ii. They know the correct value of the shares is $14.50?
b. Given your answer to part (a), what should investors conclude if IST issues equity? What will happen to the share price?
c. Given your answer to part (a), what should investors conclude if IST issues debt? What will happen to the share price in that case?
d. How would your answers change if there were no distress costs, but only the tax benefits of leverage?

TABLE 17.5
Selected Firms with Large Cash Balances

Company	Cash ($ billion)	Percentage of Market Capitalization
Toyota	41.7	37.5%
Cisco	39.1	29.5%
China Mobile	39.0	19.1%
Microsoft	37.2	16.4%
Google	30.1	19.4%
Apple	23.2	9.8%
Oracle	18.5	15.1%
Intel	18.3	15.1%
Johnson & Johnson	18.0	12.0%
Pfizer	17.3	14.4%

Source: Yahoo! Finance, July 2010.

incentives may differ from those of shareholders. Managers may prefer to retain and maintain control over the firm's cash rather than pay it out. The retained cash can be used to fund investments that are costly for shareholders but have benefits for managers (for instance, pet projects and excessive salaries), or it can simply be held as a means to reduce leverage and the risk of financial distress that could threaten managers' job security. According to the managerial entrenchment theory of payout policy, managers pay out cash only when pressured to do so by the firm's investors.

Concept Check

7. Is there an advantage for a firm to retain its cash instead of paying it out to shareholders in perfect capital markets?
8. How do corporate taxes affect the decision of a firm to retain excess cash?

17.5 Signaling with Payout Policy

One market imperfection that we have not yet considered is asymmetric information. When managers have better information than investors regarding the future prospects of the firm, their payout decisions may signal this information. In this section, we look at managers' motivations when setting a firm's payout policy, and we evaluate what these decisions may communicate to investors.

Dividend Smoothing

Firms can change dividends at any time, but in practice they vary the sizes of their dividends relatively infrequently. For example, General Motors (GM) has changed the amount of its regular dividend only 8 times over a 20-year period. Yet during that same period, GM's earnings varied widely, as shown in Figure 17.7.

The pattern seen with GM is typical of most firms that pay dividends. Firms adjust dividends relatively infrequently, and dividends are much less volatile than earnings. This practice of maintaining relatively constant dividends is called **dividend smoothing**. Firms also increase dividends much more frequently than they cut them. For example, from 1971 to 2001, only 5.4% of dividend changes were decreases.[8] In a classic survey of corporate executives, John Lintner suggested that these observations resulted from (1) management's belief that investors prefer stable dividends with sustained growth, and (2)

dividend smoothing The practice of maintaining relatively constant dividends.

[8] F. Allen and R. Michaely, "Payout Policy," in G. Constantinides, M. Harris, and R. Stulz, eds., *Handbook of the Economics of Finance* (2003).

management's desire to maintain a long-term target level of dividends as a fraction of earnings.[9] Thus, firms raise their dividends only when they perceive a long-term sustainable increase in the expected level of future earnings, and cut them only as a last resort. While this is perhaps a good description of how firms *do* set their dividends, as we have shown in this chapter there is no clear reason why firms *should* smooth their dividends. One explanation is that it contributes to signaling with dividends, as discussed in the next section.

How can firms keep dividends smooth as earnings vary? As we have already discussed, firms can maintain almost any level of dividend in the short run by adjusting the number of shares they repurchase or issue and the amount of cash they retain. However, due to the tax and transaction costs of funding a dividend with new equity issues, managers do not wish to commit to a dividend that the firm cannot afford to pay out of regular earnings. For this reason, firms generally set dividends at a level they expect to be able to maintain based on the firm's earnings prospects.

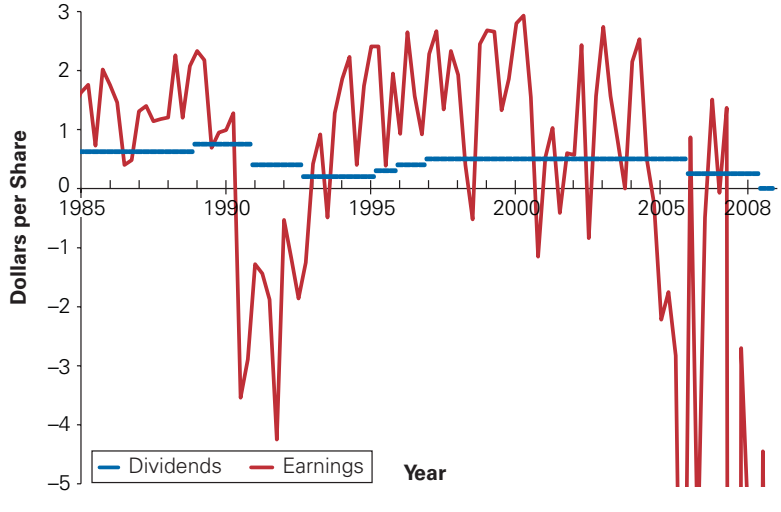
Dividend Signaling

If firms smooth dividends, the firm's dividend choice will contain information regarding management's expectations of future earnings.

1. When a firm increases its dividend, it sends a positive signal to investors that management expects to be able to afford the higher dividend for the foreseeable future.
2. When managers cut the dividend, it may signal that they have given up hope that earnings will rebound in the near term and so need to reduce the dividend to save cash.

FIGURE 17.7
GM's Earnings and Dividends per Share, 1985–2008

Compared to GM's earnings, its dividend payments were relatively stable until it finally suspended them in 2008 prior to filing for bankruptcy the next year. (Data adjusted for splits, earnings exclude extraordinary items.)



Source: Compustat and Capital IQ.

[9] J. Lintner, "Distribution of Incomes of Corporations Among Dividends, Retained Earnings and Taxes," *American Economic Review* 46 (1956): 97–113.

dividend signaling hypothesis The idea that dividend changes reflect managers' views about a firm's future earnings prospects.

The idea that dividend changes reflect managers' views about a firm's future earnings prospects is called the **dividend signaling hypothesis**.

While an increase of a firm's dividend may signal management's optimism regarding its future cash flows, it might also signal a lack of investment opportunities. For example, Microsoft's move to initiate dividends in 2003 was largely seen as a result of its declining growth prospects as opposed to a signal about its increased future profitability.[10] Conversely, a firm might cut its dividend to exploit new positive-NPV investment opportunities. In this case, the dividend decrease might lead to a positive—rather than negative—stock price reaction (see the box on Royal & SunAlliance's dividend cut). In general, we must interpret dividends as a signal in the context of the type of new information managers are likely to have.

Royal & SunAlliance's Dividend Cut

In some quarters, Julian Hance must have seemed like a heretic. On November 8, 2001, the finance director of Royal & SunAlliance, a U.K.-based insurance group with £12.6 billion ($20.2 billion) in annual revenue, did the unthinkable—he announced that he would cut the firm's dividend.

Many observers gasped at the decision. Surely, they argued, cutting the dividend was a sign of weakness. Didn't companies cut their dividend only when profits were falling?

Quite the contrary, countered Hance. With insurance premiums rising around the world, particularly following the World Trade Center tragedy, Royal & SunAlliance believed that its industry offered excellent growth opportunities.

"The outlook for business in 2002 and beyond makes a compelling case for reinvesting capital in the business rather than returning it to shareholders," explained Hance.

The stock market agreed with him, sending Royal & SunAlliance's shares up 5% following its dividend news. "Cutting the dividend is a positive move," observed Matthew Wright, an insurance analyst at Credit Lyonnais. "It shows the company expects future profitability to be good."

Source: Justin Wood, http://CFOEurope.com, December 2001.

Signaling and Share Repurchases

Share repurchases, similar to dividends, may also signal managers' information to the market. However, several important differences distinguish share repurchases and dividends:

1. Managers are much less committed to share repurchases than to dividend payments. As we noted earlier, when firms announce authorization for an open market share repurchase, they generally announce the maximum amount they plan to spend on repurchases. The actual amount spent, however, may be far less. Also, it may take several years to complete the share repurchase.
2. Firms do not smooth their repurchase activity from year to year, as they do with dividends. Thus, announcing a share repurchase today does not necessarily represent a long-term commitment to repurchase shares. In this regard, share repurchases may be less of a signal than dividends are about future earnings of a firm.
3. The cost of a share repurchase depends on the market price of the stock. If managers believe the stock is currently over-valued, a share repurchase will be costly to the firm. That is, buying the stock at its current (over-valued) price is a negative-NPV investment. By contrast, repurchasing shares when managers perceive the stock to be under-valued is a positive-NPV investment. Managers will clearly be more likely to repurchase shares if they believe the stock to be under-valued.

[10] See "An End to Growth?" *The Economist* (July 22, 2004): 61.

INTERVIEW WITH JOHN CONNORS



John Connors was Senior Vice President and Chief Financial Officer of Microsoft. He retired in 2005 and is now a partner at Ignition Partners, a Seattle venture capital firm.

QUESTION: ***Microsoft declared a dividend for the first time in 2003. What goes into the decision of a company to initiate a dividend?***

ANSWER: Microsoft was in a unique position. The company had never paid a dividend and was facing shareholder pressure to do something with its $60 billion cash buildup. The company considered five key questions in developing its distribution strategy:

1. Can the company sustain payment of a cash dividend in perpetuity and increase the dividend over time? Microsoft was confident it could meet that commitment and raise the dividend in the future.
2. Is a cash dividend a better return to stockholders than a stock buyback program? These are capital structure decisions: Do we want to reduce our shares outstanding? Is our stock attractively priced for a buyback, or do we want to distribute the cash as a dividend? Microsoft had plenty of capacity to issue a dividend *and* continue a buyback program.
3. What is the tax effect of a cash dividend versus a buyback to the corporation and to shareholders? From a tax perspective to shareholders, it was largely a neutral decision in Microsoft's case.
4. What is the psychological impact on investors, and how does it fit the story of the stock for investors? This is a more qualitative factor. A regular ongoing dividend put Microsoft on a path to becoming an attractive investment for income investors.
5. What are the public relations implications of a dividend program? Investors don't look to Microsoft to hold cash but to be a leader in software development and provide equity growth. So they viewed the dividend program favorably.

QUESTION: ***How does a company decide whether to increase its dividend, have a special dividend, or repurchase its stock to return capital to investors?***

ANSWER: The decision to increase the dividend is a function of cash flow projections. Are you confident that you have adequate cash flow to sustain this and future increases? Once you increase the dividend, investors expect future increases as well. Some companies establish explicit criteria for dividend increases. In my experience as a CFO, the analytic framework involves a set of relative comparables. What are the dividend payouts and dividend yields of the market in general and of your peer group, and where are we relative to them? We talk to significant investors and consider what is best for increasing shareholder value long-term.

A special dividend is a very efficient form of cash distribution that generally involves a nonrecurring situation, such as the sale of a business division or a cash award from a legal situation. Also, companies without a comprehensive distribution strategy use special dividends to reduce large cash accumulations. For Microsoft, the 2004 special dividend and announcement of the stock dividend and stock buyback program resolved the issue of what to do with all the cash and clarified our direction going forward.

QUESTION: ***What other factors go into dividend decisions?***

ANSWER: Powerful finance and accounting tools help us to make better and broader business decisions. But these decisions involve as much psychology and market thinking as math. You have to consider non-quantifiable factors such as the psychology of investors. Not long ago, everyone wanted growth stocks; no one wanted dividend-paying stocks. Now dividend stocks are in vogue. You must also take into account your industry and what the competition is doing. In many tech companies, employee ownership in the form of options programs represents a fairly significant percentage of fully diluted shares. Dividend distributions reduce the value of options.*

At the end of the day, you want to be sure that your cash distribution strategy helps your overall story with investors.

*We discuss options in Chapter 21. The key point here is that employee stock options increase in value when the stock price goes up, but as we learned in this chapter, a dividend reduces the stock price on the ex-dividend day.

Thus, share repurchases may signal that managers believe the firm to be under-valued (or at least not severely over-valued). Share repurchases are a credible signal that the shares are underpriced, because if they are overpriced a share repurchase is costly for cur-

rent shareholders. If investors believe that managers have better information regarding the firm's prospects and act on behalf of current shareholders, then investors will react favorably to share repurchase announcements.

In a 2004 survey, 87% of CFOs agreed that firms should repurchase shares when their stock price is a good value relative to its true value.[11] Investors also appear to interpret share repurchases as a positive signal. The average market price reaction to the announcement of an open market share repurchase program is about 3% (with the size of the reaction increasing in the portion of shares outstanding sought).[12]

Concept Check

9. What possible signals does a firm give when it cuts its dividend?
10. Would managers be more likely to repurchase shares if they believe the stock is under- or over-valued?

17.6 Stock Dividends, Splits, and Spin-Offs

In this chapter, we have focused on a firm's decision to pay cash to its shareholders. But a firm can pay another type of dividend that does not involve cash: a stock dividend. In this case, each shareholder who owns the stock before it goes ex-dividend receives additional shares of stock of the firm itself (a stock split) or of a subsidiary (a spin-off). Here, we briefly review these two types of transactions.

Stock Dividends and Splits

stock dividend (stock split) When a company issues a dividend in shares of stock rather than cash to its shareholders.

In a **stock dividend** or **stock split**, the company issues additional shares rather than cash to its shareholders. If a company declares a 10% stock dividend, each shareholder will receive one new share of stock for every ten shares already owned. Stock dividends of 50% or higher are generally referred to as stock splits. For example, with a 50% stock dividend, each shareholder will receive one new share for every two shares owned. Because a holder of two shares will end up holding three new shares, this transaction is also called a 3:2 ("3-for-2") stock split. Similarly, a 100% stock dividend is equivalent to a 2:1 stock split.

With a stock dividend, a firm does not pay out any cash to shareholders. As a result, the total market value of the firm's assets and liabilities, and therefore of its equity, is unchanged. The only thing that is different is the number of shares outstanding. The stock price will therefore fall because the same total equity value is now divided over a larger number of shares.

Unlike cash dividends, stock dividends are not taxed. Thus, from both the firm's and shareholders' perspectives, there is no real consequence to a stock dividend. The number of shares is proportionally increased and the price per share is proportionally reduced so that there is no change in value.

Stock Splits and Share Price. Why, then, do companies pay stock dividends or split their stock? The typical motivation for a stock split is to keep the share price in a range thought to be attractive to small investors. Stocks generally trade in lots of 100 shares, and in any case do not trade in units of less than one share. As a result, if the share price rises significantly, it might be difficult for small investors to afford one share, let alone 100. Making the stock more attractive to small investors can increase the demand for and the

[11] A. Brav, J. Graham, C. Harvey, and R. Michaely, "Payout Policy in the 21st Century," *Journal of Financial Economics* 77 (3) (2005): 483–527.

[12] See D. Ikenberry, J. Lakonishok, and T. Vermaelen, "Market Underreaction to Open Market Share Repurchases," *Journal of Financial Economics* 39 (2) (1995): 181–208.

7. Natsam Corporation has $250 million of excess cash. The firm has no debt and 500 million shares outstanding with a current market price of $15 per share. Natsam's board has decided to pay out this cash as a one-time dividend.
   a. What is the ex-dividend price of a share in a perfect capital market?
   b. If the board instead decided to use the cash to do a one-time share repurchase, in a perfect capital market, what is the price of the shares once the repurchase is complete?
   c. In a perfect capital market, which policy in part (a) or (b) makes investors in the firm better off?

8. Suppose the board of Natsam Corporation decided to do the share repurchase in Problem 7(b), but you as an investor would have preferred to receive a dividend payment. How can you leave yourself in the same position as if the board had elected to make the dividend payment instead?

The Tax Disadvantage of Dividends

9. The HNH Corporation will pay a constant dividend of $2 per share, per year, in perpetuity. Assume all investors pay a 20% tax on dividends and that there is no capital gains tax. The cost of capital for investing in HNH stock is 12%.
   a. What is the price of a share of HNH stock?
   b. Assume that management makes a surprise announcement that HNH will no longer pay dividends but will use the cash to repurchase stock instead. What is the price of a share of HNH stock now?

10. You purchased CSH stock for $40 and it is now selling for $50. The company has announced that it plans a $10 special dividend.
    a. Assuming 2010 tax rates, if you sell the stock or wait and receive the dividend, will you have different after-tax income?
    b. If the capital gains tax rate is 20% and the dividend tax rate is 40%, what is the difference between the two options in part (a)?

Payout Versus Retention of Cash

11. Assume perfect capital markets. Kay Industries currently has $100 million invested in short-term Treasury securities paying 7%, and it pays out the interest payments on these securities as a dividend. The board is considering selling the Treasury securities and paying out the proceeds as a one-time dividend payment.
    a. If the board went ahead with this plan, what would happen to the value of Kay stock upon the announcement of a change in policy?
    b. What would happen to the value of Kay stock on the ex-dividend date of the one-time dividend?
    c. Given these price reactions, will this decision benefit investors?

12. Redo Problem 11, but assume that Kay must pay a corporate tax rate of 35%, and that investors pay no taxes.

13. Redo Problem 11, but assume that investors pay a 15% tax on dividends but no capital gains taxes, and that Kay does not pay corporate taxes.

Signaling with Payout Policy

Use the following information to answer Problems 14 through 18:

AMC Corporation currently has an enterprise value of $400 million and $100 million in excess cash. The firm has 10 million shares outstanding and no debt. Suppose AMC uses

its excess cash to repurchase shares. After the share repurchase, news will come out that will change AMC's enterprise value to either $600 million or $200 million.

14. What is AMC's share price prior to the share repurchase?

15. What would AMC's share price be after the repurchase if its enterprise value goes up? What would AMC's share price be after the repurchase if its enterprise value declines?

***16.** Suppose AMC waits until after the news comes out to do the share repurchase. What would AMC's share price be after the repurchase if its enterprise value goes up? What would AMC's share price be after the repurchase if its enterprise value declines?

17. Suppose AMC management expects good news to come out. Based on your answers to Problems 15 and 16, if management wants to maximize AMC's ultimate share price, will they undertake the repurchase before or after the news comes out? When would management undertake the repurchase if they expect bad news to come out?

***18.** Given your answer to Problem 17, what effect would you expect an announcement of a share repurchase to have on the stock price? Why?

Stock Dividends, Splits, and Spin-Offs

19. FCF Co. has 20,000 shares outstanding and a total market value of $1 million, $300 thousand of which is debt and the other $700 thousand is equity. It is planning a 10% stock dividend.
   a. What is the stock price before the dividend and what will it be after the dividend?
   b. If an investor owns 1000 shares before the dividend, what will the total value of her investment in FCF be before and after the dividend?

20. Suppose the stock of Host Hotels & Resorts is currently trading for $20 per share.
   a. If Host issues a 20% stock dividend, what would its new share price be?
   b. If Host does a 3:2 stock split, what would its new share price be?

21. If Berkshire Hathaway's A shares are trading at $120,000, what split ratio would it need to bring its stock price down to $50?

22. After the market close on May 11, 2001, Adaptec, Inc., distributed a dividend of shares of the stock of its software division, Roxio, Inc. Each Adaptec shareholder received 0.1646 share of Roxio stock per share of Adaptec stock owned. At the time, Adaptec stock was trading at a price of $10.55 per share (cum-dividend), and Roxio's share price was $14.23 per share. In a perfect market, what would Adaptec's ex-dividend share price be after this transaction?



In your role as a consultant at a wealth management firm, you have been assigned a very powerful client who holds one million shares of Amazon.com purchased on February 28, 2006. In researching Amazon, you discovered that they are holding a large amount of cash, which was surprising because the firm has only relatively recently begun operating at a profit. The client is considering approaching Amazon's board of directors with a plan for half of the cash the firm has accumulated, but can't decide whether a share repurchase or a special dividend would be better. You have been asked to determine which initiative would generate the greater amount of money after taxes, assuming that with a share repurchase your client would keep the same proportion of ownership. Because both dividends and capital gains are taxed at the same rate (15%), your client has assumed that

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Financial Markets And Corporate Strategy{Mark Grinblatt, She

- Information content of the debt–equity choice: Ch. 19, §19.5 (p. 671–678)

scrutiny is likely to lead to a negative adjustment in the firm's stock price. Hence, the incentive to attract attention is greatest for those firms that are the most undervalued, which suggests that one might expect to see positive stock price reactions to any announcements that attract considerable attention.

The positive stock price reactions observed at the time stock dividends and stock splits are announced support the idea that stock prices respond to announcements of managerial decisions that do no more than attract attention to the firm.[7] Unlike cash dividends, stock dividends and splits affect neither the firm's cash flows nor its investment alternatives. Yet, observed stock returns at the time of stock dividend and stock split announcements are of approximately the same magnitude as the returns at the time increases in cash dividends are announced.

19.5 The Information Content of the Debt-Equity Choice

This section examines the type of information conveyed to investors by a firm's debt-equity choice. The debt-equity choice conveys information to investors for two reasons. First, because of financial distress costs, managers will avoid increasing a firm's leverage ratio if they have information indicating that the firm could have future financial difficulties. Hence, a debt issue can be viewed as a signal that managers are confident about the firm's ability to repay the debt. The second reason has to do with the reluctance of managers to issue what they believe are underpriced shares. Hence, an equity issue might be viewed as a signal that the firm's shares are not underpriced and therefore may be overpriced.

A Signaling Model Based on the Tax Gain/Financial Distress Cost Trade-Off

To understand why the debt-equity choice conveys information, assume, as a first approximation, that firms select their capital structures by trading off the tax benefits of debt financing (see Chapter 14) against the various costs of financial distress (see Chapters 16 and 17). In this setting, firms desire higher debt levels when expected cash flows are higher because they can better utilize the tax benefits of debt. In addition, for any given debt level, the probability of incurring the costs of financial distress is lower if expected cash flows are higher.

Because expected future cash flows determine the firm's optimal capital structure, the capital structure choice of better informed managers is likely to convey information to shareholders. While the information content of the capital structure decision would not affect the decisions of managers concerned only with intrinsic value, it would affect the decisions of managers who also are concerned about the current share prices of their firms. Indeed, managers whose objectives are heavily weighted toward the maximization of current share price are likely to avoid reducing leverage even when doing

[7]Grinblatt, Masulis, and Titman (1984), who proposed this "attention model," provided evidence that stock returns around the time of stock dividend and stock split announcements are of approximately the same magnitude as stock returns around the time of dividend increases. For further discussion and evidence relating to this hypothesis, see Brennan and Hughes (1991). In addition, Allen, Bernardo, and Welch (2000) argue that increased cash dividends can lead to increased scrutiny because stocks with higher dividend yields tend to attract more institutional investors (see our discussion of dividend clienteles in Chapter 15). Firms with greater institutional holdings are likely to be more closely monitored both because of the reduction of the free-rider problem discussed in Chapter 18 and because firms with greater institutional holdings generally receive greater analyst coverage.

so improves the intrinsic value of their shares but conveys information that reduces their current value. They may similarly choose to increase leverage beyond the point that maximizes intrinsic value.[8]

Result 19.8 An increase in a firm's debt ratio is considered a favorable signal because it indicates that managers believe the firm will be generating taxable earnings in the future and that they are not overly concerned about incurring financial distress costs. Managers understand that their firm's stock price is likely to respond favorably to higher leverage ratios and may thus have an incentive to select higher leverage ratios than they would otherwise prefer.

CUC International Borrows to Pay a Special Dividend[9]

In March 1989, CUC International's board of directors ratified a leveraged recapitalization plan, which involved paying out a special dividend of $5 per share, financed in part by a loan from GE Capital. The total size of the dividend payment ($100 million) represented over half of the market value of CUC's equity prior to the announcement. Walter Forbes, the company's chairman and CEO, admitted that part of the motivation for the recapitalization was the favorable signal of an increased debt ratio. Forbes said:

> "We judged that borrowing a moderate amount of debt to finance the special dividend would add an appropriate amount of leverage to our capital structure as well as providing, through the repayment of the debt, a clear signal of CUC's ability to generate cash."

It is clear that one of CUC International's motivations for increasing its debt ratio was to send a signal to investors. However, an investor may question whether such a signal is credible given that the motivation for the debt increase was to boost the firm's stock price. The following result describes conditions under which a financial signal conveys favorable information credibly:

Result 19.9 For a financial decision to credibly convey favorable information to investors, firms with poor prospects must find it costly to mimic the decisions made by firms with favorable prospects.

The Credibility of the Debt-Equity Signal. As Example 19.6 illustrates, issuing debt satisfies the requirement for a credible signal, as specified in Result 19.9, because additional debt is likely to have a much greater effect on the probability of bankruptcy for firms with unfavorable future prospects than for firms with favorable prospects.

Example 19.6: The Information Content of Leverage Changes

Analysts following Prairie Technologies are uncertain about whether Prairie has successfully reduced its production costs. If it has been successful, Prairie's future earnings are expected to range from $50 million to $60 million per year. However, if the company has not been successful, its earnings will be in the range of $25 million to $30 million. Prairie announces a debt for equity swap that increases its interest payments to $40 million per year. What information is conveyed by this decision?

Answer: Analysts can infer that the cost reductions have been successful. Otherwise, such an increase in leverage would eventually expose the firm to substantial bankruptcy risk and the associated financial distress costs. Hence, the market responds to the announcement by bidding up Prairie's stock price. Since the firm is certain to have the cash flow to meet these interest payments, the signal did not reduce Prairie's long-term value.

[8] Ross (1977) developed a theory of capital structure along these lines.

[9] This case study is based on Paul Healy and Krishna Palepu, "Using Capital Structure to Communicate with Investors: The Case of CUC International," *Journal of Applied Corporate Finance* (Winter 1996), pp. 30–44.

Example 19.6 illustrates a situation in which a firm with favorable prospects is able to use debt financing to signal its value without risking bankruptcy. In more realistic cases, a firm that wishes to signal its value will have to take on much more debt than it otherwise would have found optimal.

Consider, for example, a firm whose managers have strong incentives to increase current stock prices and would be willing to take on a high debt level to achieve this goal. As a result, outside investors will not find moderately high leverage ratios to be credible signals of high values. Hence, if the firm does have favorable prospects, it will have to use much more debt financing than it would otherwise use in order to convince investors of its higher value. We illustrate this concept in Example 19.7.

Example 19.7: CEO Incentives and the Credibility of Financial Signals

Textron's CEO knows that the firm's assets are worth either $500 million, $400 million, or $300 million, depending on the demand for their product. Because each possibility is equally likely, the average of these three numbers, $400 million, is the firm's intrinsic value. However, investors are not as optimistic about the firm's future earnings as the CEO and believe that the firm will have respective values of $450 million, $350 million, or $250 million in the three product demand scenarios given above, implying that the firm's current value is $350 million (assuming no financial distress costs). The $50 million discrepancy between the firm's intrinsic value and its current value presents a problem because Textron's CEO plans on selling a large block of stock in the near future. As a consequence, before selling the stock, the CEO would like to signal to investors that Textron's value is $50 million higher than investors currently believe it is.

The CEO has announced his beliefs about Textron's prospects and investors know that the only alternative to their own beliefs is the more optimistic beliefs of the CEO. However, the mere announcement of more optimistic beliefs is not a very credible signal to investors. They know the CEO would be delighted to engineer a temporary increase in Textron's stock price before unloading his block of shares.

Assume that the CEO has announced his intention to sell half of his shares and thus weights intrinsic value and current value equally. Also assume that financial distress costs reduce the value of the firm by $60 million in whichever product demand scenario such distress occurs. The CEO has concluded that he may be able to credibly signal the more optimistic prospects by issuing sufficient debt and using the proceeds to retire equity. From the CEO's perspective, the issuance of debt with a promised payment in excess of $400 million is precluded. There is no need to risk financial distress that reduces intrinsic value by more than one gains in current value (net of financial distress costs). Moreover, debt financing with a promised payment of less than $250 million would not be a very credible signal in that—using either investor beliefs or the more optimistic CEO beliefs—financial distress never occurs.

Analyze what happens to investor beliefs if the CEO issues debt (and retires an equivalent amount of equity) with a promised payment (1) between $250 million and $350 million or (2) between $350 million and $400 million.

Answer: (1) Debt issuance between $250 million and $350 million is not a credible signal to investors that the higher firm values will be realized. Since Textron's CEO weights the current and intrinsic values of Textron equally, he would be willing to take on this amount of debt if doing so would signal the higher value, even if this signal were false. To see this, note that the CEO gains $50 million in current market value and loses only $20 million $\left(\frac{1}{3} \times \$60 \text{ million}\right)$ in intrinsic value from being financially distressed in the lowest product demand scenario. Investors, aware of the incentive to be tricked by a CEO who sees cash flows as pessimistically as they do, will not believe that a debt signal of this magnitude is credible.

(2) A debt obligation between $350 million and $400 million would put the firm in financial distress $\frac{2}{3}$ of the time if investor beliefs are correct but only $\frac{1}{3}$ of the time if the more

optimistic announced beliefs of the CEO were true. No CEO with pessimistic beliefs would take on this much debt since, even if investors believe the CEO, the gain in current value (net of financial distress costs) is $30 million ($50 million less $\frac{1}{3}$ of $60 million), while the loss in intrinsic value to the deceptive manager is $40 million ($\frac{2}{3}$ of $60 million). By contrast, the loss in intrinsic value to a manager who truly holds optimistic beliefs is $20 million ($\frac{1}{3}$ of $60 million). Thus, the signal of debt is credible in this case because managers with optimistic beliefs find it profitable to issue debt in amounts between $350 million and $400 million at the same time that managers with pessimistic beliefs find it unprofitable to signal by mimicking the same action.

The amount of debt financing a firm must use to credibly signal a high value depends on its manager's incentive to increase the firm's current stock price. To understand this, consider two CEOs, Jane and Janet. Jane, who plans to retire soon and sell her holdings of her firm's stock, has a strong incentive to temporarily increase her firm's stock price. Janet, on the other hand, plans to stay on as CEO for 10 years at her firm. She also is interested in boosting her firm's current share price, but she is much more concerned about the firm's long-term success and, in addition, is worried about losing her job if the firm has trouble meeting future interest payments.

The interpretation of the signal offered by a leverage increase depends on whether the firm one is looking at is run by a CEO like Jane or a CEO like Janet. When Janet increases her firm's leverage, investors will infer that she is confident that the firm will be able to generate the cash flows to pay back the debt. They understand that she has little incentive to give a false signal, and she has a lot to lose if the firm subsequently fails to make the required interest payments. Investors are likely to react much differently to a leverage increase initiated by Jane. They understand that Jane has a strong incentive to appear optimistic, even when she isn't, and that the cost to her of overleveraging her firm is not substantial. Hence, an equivalent leverage increase will result in a lower stock price response to the leverage signal for Jane's firm than for Janet's.

Adverse Selection Theory

Consider a health insurance company offering two different policies. One policy is very expensive, but it pays 100 percent of all of your medical bills. The second policy is much less expensive, but it pays only 80 percent of your medical bills. How do you expect individuals to choose between the two policies?

Most economists predict that individuals will not *randomly select* between the policies. Rather, we will observe what economists call adverse selection. In the health insurance example, **adverse selection** means that individuals will select their best actions based on their private information. Hence, the more expensive policy will attract the least healthy individuals. An additional example of adverse selection, described in a seminal article by Akerlof (1970), is the "lemons" problem connected with the sale of used cars. Akerlof argued that cars depreciate so much in their first year largely because people who have the most incentive to sell their cars after only one year are those with lemons, or faulty cars. Buyers, taking into account this adverse selection of used cars, are thus unwilling to pay as much for a used car as for a new car, which is less likely to be a lemon.

Adverse selection also is important when firms issue new equity. Managers have the greatest incentive to sell stock when the stock is a lemon. This means that the incentive to issue equity is highest when management believes that the firm's stock price exceeds its intrinsic value. At these times, better informed managers know that equity

provides relatively inexpensive financing (that is, the expected return on equity is relatively low) and a new issue would thus increase the intrinsic value of existing shares. In contrast, issuing shares of stock at a price lower than what management believes they are worth provides relatively expensive financing and dilutes the intrinsic value of the firm's existing shares.

Adverse Selection Problems When Insiders Sell Shares. The incentive to retain rather than issue underpriced shares can be viewed within the context of an entrepreneur who is motivated to take his firm public in order to sell shares and diversify his portfolio.[10] To understand how an entrepreneur decides how many shares to sell, consider the situation faced by Bill Gates at the time of Microsoft's initial public offering. In deciding whether or not to sell some of his own shares, Gates has to consider:

- The diversification benefits of selling shares.
- The tax costs of selling shares (see Chapter 15).
- Whether the shares are undervalued or overvalued.

If Gates values diversification and the tax costs are not great, he will sell shares if he believes they are not substantially undervalued. Indeed, if he believes the shares are overvalued, he will sell them even if he places no value on diversification. Conversely, if Gates believes the shares are substantially undervalued, he will choose not to sell any shares even if he is extremely risk averse.

Since investors understand Gates's incentives, they monitor his tendency to sell off shares when they value Microsoft stock (both at the IPO and subsequently, in the secondary market). If Gates were to sell off almost all of his shares, which he would do to diversify optimally, Microsoft's stock would probably fall substantially because it would signal to investors that Gates no longer believes that Microsoft stock is an extraordinary investment. Gates thus faces a trade-off. By holding more shares, he provides a more favorable signal about Microsoft's prospects, which keeps the share price relatively high. However, this forces him to be less diversified than he would like to be.

To understand the price effect of a sale of Microsoft shares by Gates, it is helpful to review the issues involved in buying a used car. If you know that the car's owner is moving overseas, you might think the adverse selection problem is minimal and feel comfortable about buying the car. Similarly, if investors believe that Gates is extremely risk averse and therefore motivated to sell his shares, they will be less concerned about the adverse selection problem and be more willing to buy his shares. On the other hand, if investors believe that Gates is not very risk averse but is extremely averse to paying taxes, they would be much less willing to buy his shares.

Several decades ago, Howard Hughes (whose sophistication with corporate finance theory was documented in Chapter 14's opening vignette) sold a substantial fraction of his holdings in TWA stock. The stock price of TWA did not plummet in response to the sale because Hughes was able to credibly convince the market that he was selling TWA stock to remedy a "cash crunch" that he was personally experiencing, and not because of any adverse information he held about TWA. More recently, high-tech billionaires, like Michael Dell, founder of Dell Computer, and Bill Gates of Microsoft, have started to sell for diversification purposes without signaling poor prospects for their firms by selling a relatively small fixed percentage of their outstanding shares every quarter. Because the sales are constant and are anticipated, they do not adversely affect stock prices.

[10]These issues were first addressed in Leland and Pyle (1977).

Adverse Selection Problems When Firms Raise Money for New Investments.[11] As we discuss below, the adverse selection problem that creates problems for the seller of a used car can also make it costly for a firm to issue new equity. Firms often issue equity to raise capital to fund new investment. In the same way that it is easier to sell your car when you can convince would-be buyers that you are moving overseas, it is easier to convince investors that your stock is not overvalued if you can demonstrate that you are raising capital to fund an attractive investment project.

However, the adverse selection problem cannot always be solved by revealing the potential of a favorable investment. As a result, firms sometimes pass up good investments because of their reluctance to finance projects by issuing underpriced shares. The conditions under which a firm will pass up a positive *NPV* investment are seen in the following equations, which compare the intrinsic values of a firm's shares with and without a new investment that is financed by issuing equity.

$$\text{Share value taking the project} = \frac{PV \text{ of assets in place} + PV \text{ of new investment}}{\text{Number of original shares} + \text{Number of new shares}}$$

$$\text{Share value not taking the project} = \frac{PV \text{ of assets in place}}{\text{Number of original shares}}$$

The preceding equations show that a firm may reduce the intrinsic value of its shares if the *PV* of the new investment is low relative to the number of shares it must issue. To understand this, consider a case where management believes the firm has assets worth \$100 million with 1 million shares outstanding, suggesting that the firm's intrinsic value is \$100 per share if it does not take any new investments. If this firm's stock is selling at only \$70 per share, it will have to issue an additional 1 million shares to raise \$70 million for a project that has a value of \$90 million. The firm's intrinsic share value after taking the project would then be

$$\frac{\$100 \text{ million} + \$90 \text{ million}}{1 \text{ million} + 1 \text{ million}} = \$95 \text{ per share}$$

Hence, the firm reduces the intrinsic value of its shares by \$5 per share by taking on a positive *NPV* project. Although the project has a \$20 million positive *NPV,* the financing for the project has a negative *NPV* of −\$25 million given what the firm's managers know about the value of the firm's existing assets. The −\$25 million is the consequence of offering what managers know is a claim to \$50 million in existing assets plus \$45 million in assets from the new investment for the bargain price of \$70 million. This possibility is illustrated further in Example 19.8.

Example 19.8: Issuing Equity When Managers Know More than Investors

Olympus Corporation is currently selling at \$50 a share and has 1 million shares outstanding. The \$50 share price reflects its current business, valued at \$40 million, and an opportunity to take on an investment valued at \$30 million, which costs only \$20 million. The opportunity can be viewed as an asset with a \$10 million *NPV*.

The management of Olympus has discovered a vast amount of oil, worth \$50 million, on its property. This fact is unknown to shareholders and thus is not reflected in Olympus's current share price. If information about this oil were known to shareholders, its shares would sell for \$100 a share. Unfortunately, management has no way to reveal this information directly to the market, so management expects that its shares will be undervalued for some time.

[11] The discussion in this subsection is based on Myers and Majluf (1984).

Suppose that the firm funds its new investment by issuing 400,000 shares at $50 a share. How will this affect the intrinsic value of the firm's existing shares?

Answer: If the firm passes up the project, its shares will ultimately be worth $90 each [($40 million + $50 million)/1 million] when the information about the oil is revealed. However, if the project is taken and is financed with an equity issue, the firm's total intrinsic value will be $120 million ($40 million + $30 million + $50 million) and the total number of shares outstanding will be 1.4 million. The per share value will be $85.71 ($120 million ÷ 1.4 million) if Olympus issues shares and takes the project. Thus, the firm will choose not to invest in the positive *NPV* project if it requires issuing underpriced equity.

Using Debt Financing to Mitigate the Adverse Selection Problem. Example 19.8 illustrates why managers may choose not to issue equity when they believe that their firm's shares are underpriced. However, the example ignores the possibility that the firm can finance the project with debt. If the project can be financed with riskless debt, then the firm should take the project as long as it has a positive *NPV*. In this case, the share's intrinsic value will be equal to:

$$\text{Share value: financing project with riskless debt} = \frac{\text{Value of original assets} + \textit{NPV} \text{ of new project}}{\text{Number of shares}}$$

This value clearly increases when the company commits to a positive *NPV* project. However, the firm may still pass up the project if it is forced to issue risky debt that exposes it to the possibility of incurring financial distress costs. In this case, a firm must compare the costs of deviating from its optimal capital structure and the associated financial distress costs with the *NPV* of the particular investment project. Given this comparison, some positive *NPV* investment projects will be passed up while others will be financed with debt, causing the firm to become at least temporarily overleveraged.

Similarly, one could show that firms have an incentive to take on negative *NPV* projects and become underleveraged if it allows them to issue overpriced securities. Because of these incentives, issuing equity is considered to be an indication that a firm is overvalued. As a result, announcements of equity issues have a negative effect on a firm's stock price, which has the effect of further reducing the incentive of firms to issue equity.

This discussion suggests that managers will prefer debt to equity financing when they have favorable private information. In Example 19.8, Olympus Corporation would have been able to realize a share price of $100 if it could have financed the investment with risk-free debt. If lenders are unwilling to lend the firm additional amounts (see Chapter 16) or if the firm is unwilling to borrow more because of the financial distress costs (see Chapter 17), then undervalued firms may choose to pass up positive net present value investments.

Result 19.10 A firm may pass up a positive net present value investment project if it requires issuing underpriced equity. Since debt has a fixed claim on future cash flows, a firm's debt is less likely to be substantially undervalued. As a result, firms may prefer to finance new projects with debt rather than equity. With sufficiently high financial distress and adverse selection costs, however, firms may be better off passing up the positive *NPV* investment.

Adverse Selection and the Use of Preferred Stock. The dilution and financial distress problems that can arise when an underpriced firm finances a new project may be mitigated by issuing preferred stock. Recall from Chapter 3 that preferred stock is similar to a bond because it has a fixed payout. However, if a firm fails to meet its dividend

obligation, preferred shareholders cannot force it into bankruptcy. Hence, preferred stock will not create the problems associated with financial distress. In addition, since preferred stock offers a fixed claim, it is not likely to be as underpriced as common stock, so the dilution costs of issuing underpriced shares are much less of a problem.

For these reasons, preferred stock is a good security for firms to issue when they are having financial difficulties that they believe are temporary. If investors do not agree that the difficulties are temporary, the common stock may be underpriced, so issuing common equity may dilute the value of existing shares. In such a situation, the firm may not have taxable earnings, making debt financing less attractive. Furthermore, additional debt financing may lead to a drop in the firm's credit rating, which could create problems with the firm's nonfinancial stakeholders.

Preferred stock may be the best financing alternative in this situation because it is unlikely to be as undervalued as common stock, given its senior status and fixed dividend, and it does not increase the risk of bankruptcy as would happen when additional debt is issued.

Result 19.11 When firms are experiencing financial difficulties, they prefer equity to debt financing for a number of reasons. In particular, the tax advantages of debt may be less and the potential for suffering financial distress costs may be greater. Issuing common stock in these situations may be a problem, however, given the negative information conveyed by an equity offering. Hence, a preferred issue may offer the best source of capital.

Empirical Implications of the Adverse Selection Theory. The adverse selection theory explains a number of observations about how firms externally finance themselves. First, the reluctance of managers to issue underpriced stock helps explain why stock prices react unfavorably when firms announce their intention to issue equity. As we discuss in more detail in the section below, stock prices drop about 2 percent, on average, when firms announce the issue of new equity. The adverse selection theory also provides an explanation for Donaldson's pecking order of financing choices (see Chapters 15 and 17). Donaldson observed that firms prefer first to finance investment with retained earnings; then, when they need outside funding, they prefer to issue debt instead of equity. The adverse selection theory explains the reluctance of firms to issue equity and, in addition, suggests that firms prefer to use their retained earnings to finance investment expenditures because this allows them to retain the capacity to borrow in the future.

19.6 Empirical Evidence

Exhibit 19.3 provides a brief overview of three types of signaling theories that provide insights about financial decision making and the reaction of stock prices when firms make financing and dividend changes. This section discusses some of the empirical implications of those theories. We start by reviewing academic studies that measure the stock price responses to these financial decisions. We will then discuss the evidence on the information signaled by investment choices.

What Is an Event Study?

Academic studies that examine stock price responses to the announcements of particular information are generally referred to as **event studies**. For example, the event studies of dividend initiation announcements discussed previously were carried out as

Key Terms

adverse selection 674
event studies 678
exchange offers 681
intrinsic value 658
market-adjusted excess return 679
signals 657

Exercises

19.1. Describe how a firm's investment decisions might be made differently if its management is highly concerned about the firm's current stock price.

19.2. Why might a firm choose to increase its debt level in response to favorable information about its future prospects?

19.3. Exhibit 19.5 shows that stock prices of industrial firms react more negatively to stock issues than do utilities. Why do you think this is the case?

19.4. Classical finance theory suggests that firms take projects with positive *NPV*s regardless of the amount of cash the firm has available. However, empirical evidence suggests that the amount that firms invest is heavily dependent on their available cash flows. Why might this be?

19.5. Why might a manager close to retirement select a higher debt ratio than a manager far from retirement?

19.6. ABC Industries is considering an investment that requires the firm to issue new equity. The project will cost $100, but will add $120 to the firm's value. Although management believes the firm's value is $1,000 without the new project, outside investors value the firm at $600 without the project. If the firm currently has 100 shares outstanding, how many new shares must it issue to finance the project? Now assume that the true value of the firm will become known to the market shortly after the new equity has been issued. What will the firm's stock price be at this time if it chooses to finance this new investment? What will the stock price be if it chooses to pass up the investment?

19.7. As economies develop, disclosure laws generally get tougher and accounting information becomes more informative. Briefly describe how such changes in the quality of information affect the incentives of firms to be financed by either debt or equity.

19.8. If it was known that management was selling shares at the same time as it was increasing leverage, how would this affect the credibility of the signal? Why? What other actions or motivations by management could affect the credibility of such a signal?

19.9. The following table describes management's view of Abracadabra Corporation's future cash flows along with the consensus view of outside analysts.

	Cash Flows		
State of the Economy:	*Low*	*Average*	*High*
Management's beliefs	$400	$500	$600
Analysts' beliefs	300	400	500
Cost of distress	100	150	200

If the analysts can be convinced that management's beliefs are correct, the firm's value will increase by $200. Assume that there are no tax or other benefits from debt apart from the information the debt may convey. However, if the promised interest payments exceed the cash flows, the firm will lose $100, $150, or $200 because of financial distress, depending on the state of the economy.

Assuming that management wants to maximize the intrinsic value of the firm, how much debt will the firm take on? Now consider the possibility that management's incentives place an equal weight on the firm's intrinsic value and its current value. How much debt must the firm take on to credibly convince the analysts that their cash flow estimates are wrong? (*Hint:* Consider management's incentive to mislead analysts if the analysts' original projections are correct.)

19.10. Analysts project that Infotech, an information services company, will have the following financial data for equally probable high and low states:

	Value	
State:	*Low*	*High*
Cash	$100	$100
Fixed asset value	200	300
Growth opportunity *NPV*	100	100

Ross

ROSS, WESTERFIELD, JORDAN_ LAMB - Fundamentos de Administraç

- Conteúdo informacional dos dividendos: Cap. 17, §17.5 (p. 593–595)

ao contrário das viúvas e dos órfãos, esse grupo é muito expressivo em termos do montante de ações que possui.

Conclusão

De forma geral, quando se trata de decidir a forma de pagar dividendos e do quanto pagar, as considerações sobre a eventual tributação diferenciada para investidores individuais ou pessoas jurídicas é decisiva. Quando levamos em consideração a preferência (por qualquer motivo) por renda corrente, pode ser que os investidores estejam dispostos a pagar os impostos. Além disso, alguns investidores muito grandes, como as empresas e as instituições isentas de impostos, podem ter uma preferência bastante forte pela distribuição de altos dividendos.

Questões conceituais

17.4a Por que alguns investidores individuais preferem uma distribuição de altos dividendos?

17.4b Por que alguns investidores não individuais preferem uma distribuição de altos dividendos?

17.5 Uma solução para os fatores do mundo real?

Nas seções anteriores, apresentamos alguns fatores que favorecem uma política de dividendos baixos e outros que favorecem uma política de dividendos elevados. Nesta seção, discutimos dois conceitos importantes relacionados aos dividendos e à política de dividendos: o conteúdo informacional dos dividendos e o efeito clientela. O primeiro tópico ilustra a importância dos dividendos em geral e a importância de distinguir entre dividendos e política de dividendos. O segundo tópico sugere que, apesar das muitas considerações do mundo real discutidas, o índice de distribuição de dividendos pode não ser tão importante quanto pensávamos.

Conteúdo informacional dos dividendos

Para começar, revisaremos rapidamente alguns pontos da nossa discussão inicial. Anteriormente, examinamos três pontos de vista diferentes sobre os dividendos:

1. Com base no argumento do dividendo caseiro, a política de dividendos não é relevante.
2. Por causa dos efeitos fiscais para os investidores individuais e dos custos de novas emissões, uma política de dividendos mais baixos é melhor nos Estados Unidos. Já no Brasil, como para os investidores individuais os ganhos de capital são tributados e os dividendos não, uma política de altos dividendos é melhor.
3. Em razão do desejo por renda corrente e dos fatores relacionados, uma política de dividendos altos é melhor.
4. No Brasil, é melhor para a empresa pagar JCP e imputá-los aos dividendos, mas isso pode conflitar com o interesse de acionistas pessoas jurídicas e, nessa situação, para o acionista pessoa jurídica será melhor receber dividendos altos.

Para decidir qual desses pontos de vista é o certo, uma maneira óbvia de começar seria ver o que acontece aos preços das ações quando as empresas anunciam alterações nos dividendos. Você descobrirá que, com certa consistência, os preços das ações sobem quando o dividendo corrente é aumentado inesperadamente e, em geral, eles caem quando o dividendo é diminuído repentinamente. O que isso indica sobre qualquer um dos pontos de vista anteriores?

À primeira vista, o comportamento descrito parece coerente com o terceiro ponto de vista e incoerente com os primeiros dois, no caso norte-americano. Na verdade, muitos autores têm defendido isso. Se os preços das ações sobem em resposta a aumentos de dividendos e caem em resposta a reduções de dividendos, o mercado não estará dizendo que aprova os dividendos mais altos?

Outros autores indicam que essa observação não nos diz realmente muito sobre a política de dividendos. Todos concordam que, com tudo o mais igual, dividendos são importantes. As empresas reduzem os dividendos, mas com grande relutância. Assim, uma redução nos dividendos quase sempre é um sinal de que a empresa está com problemas.

Mais precisamente, uma redução, em geral, não é uma alteração voluntária e planejada na política de dividendos. Em vez disso, normalmente sinaliza que a administração não acha que a política de dividendos atual possa ser mantida. Como resultado, as expectativas de dividendos futuros devem ser revisadas para baixo. O valor presente dos dividendos futuros cai, assim como o preço da ação.

Nesse caso, o preço da ação diminui após uma redução nos dividendos porque, em geral, os dividendos futuros devem ser menores, e não porque a empresa mudou a porcentagem do lucro que pagará na forma de dividendos.

Para um exemplo radical, considere o caso da Borders Group, a segunda maior rede de livrarias dos Estados Unidos. Em 2008, a empresa enfrentava uma forte concorrência com a Amazon.com e a Barnes & Noble. Os resultados operacionais tinham diminuído, e o preço das ações já tinha caído mais de 50% de seu valor máximo ao longo do ano anterior. Em março de 2008, a Borders declarou que os mercados de crédito escassos estavam limitando seu poder de tomar empréstimos nos bancos. Por essa razão, a empresa anunciou que não pagaria dividendos.

Esse não foi um dia agradável para os acionistas da Borders. Em um dia típico, aproximadamente 2 milhões de ações da Borders são negociadas na Nyse. Entretanto, no dia do anúncio, mais de 26,7 milhões de ações mudaram de dono. A ação havia fechado a $7,10 no dia anterior. Quando o mercado abriu, antes do anúncio, a ação subiu para $8. Quando a empresa anunciou a omissão de dividendos, a ação caiu para $3,97, antes de fechar a $5,07, uma perda de um dia para o outro de aproximadamente 29%. Em outras palavras, a Borders perdeu quase um terço de seu valor de mercado durante o dia. Como ilustrado por esse caso, os acionistas podem reagir de maneira negativa a reduções não previstas nos dividendos. Aproveitando o exemplo, lembre-se de que, no Brasil, um anúncio como esses seria feito após o encerramento do pregão e não durante o pregão como mostrado.

Naturalmente, nem todos os anúncios de redução de dividendo resultam em desvalorizações tão acentuadas no preço da ação. Em janeiro de 2008, a MBIA, Inc., a maior seguradora de títulos do mundo, anunciou que estava reduzindo seus dividendos em 62%. Contudo, o preço da ação caiu apenas cerca de 4%. O motivo é que os investidores já esperavam tal movimento da empresa.

Do mesmo modo, um aumento inesperado nos dividendos sinaliza boas notícias. A administração aumenta os dividendos apenas quando espera um aumento nos lucros futuros, no fluxo de caixa e nas perspectivas gerais, de tal maneira que o dividendo não terá de ser reduzido mais tarde. Um aumento de dividendo é o sinal da administração para o mercado de que a empresa deve se sair bem. O preço da ação reage favoravelmente porque as expectativas de dividendos futuros são revisadas para cima, não porque a empresa aumentou o valor da distribuição do lucro.

Em ambos os casos, o preço da ação reage à variação no dividendo. A reação pode ser atribuída a variações no montante esperado de dividendos futuros, não necessariamente a uma variação na política de distribuição de dividendos. Essa reação é chamada de **efeito do conteúdo informacional** do dividendo. O fato de que as variações de dividendos veiculam informações sobre a empresa para o mercado torna difícil interpretar o efeito da política de dividendos das empresas.

efeito do conteúdo informacional
a reação do mercado a uma alteração na distribuição de dividendos de uma empresa

O efeito clientela

Em nossa discussão anterior no contexto dos Estados Unidos, vimos que alguns grupos (indivíduos ricos, por exemplo) têm um incentivo para buscar ações com pagamento baixo (ou zero). Outros grupos (as empresas, por exemplo) têm um incentivo para buscar ações com pagamento alto. Portanto, empresas com distribuições altas atraem um grupo, e empresas com distribuição baixa atraem outro.

No Brasil, podemos pensar no exemplo dos fundos de pensão. Quando os assistidos pelo fundo são jovens e pessoas de meia-idade, o fundo deveria investir em empresas com grandes oportunidades de crescimento que distribuíssem o mínimo possível de caixa e reinvestissem o máximo para aproveitar as oportunidades que se oferecem. Quando o fundo for maduro, em que saídas substanciais de caixa serão necessárias para pagar aposentadorias e pensões, os recursos do fundo deveriam privilegiar empresas maduras, pagadoras de altos dividendos. Obviamente, o argumento dos dividendos caseiros se aplica, e o fundo de pensão poderia manter seus investimentos em empresas de alto crescimento e vender ações para realizar caixa. O problema é que vendas de fundos podem afetar os preços da ação ofertada, dado o volume que podem atingir, e por isso empresas pagadoras de bons dividendos poderiam ser uma alternativa.

Esses diferentes grupos são chamados de *clientelas*, e o que descrevemos são efeitos clientela. O argumento do **efeito clientela** é que diferentes grupos de investidores desejam ter níveis diferentes de dividendos. Quando uma empresa escolhe uma determinada política de dividendos, o único efeito é atrair uma clientela específica. Se uma empresa alterar sua política de dividendos, simplesmente atrairá uma clientela diferente.

efeito clientela
o fato observável de que as ações atraem grupos específicos com base no rendimento em dividendos e nos efeitos tributários resultantes.

Ficamos apenas com um argumento simples de oferta e procura. Suponhamos que 40% de todos os investidores prefiram dividendos altos, mas apenas 20% das empresas paguem altos dividendos. Nessa situação, as empresas com altos dividendos terão pouca oferta, e os preços de suas ações subirão. Consequentemente, as empresas com baixos dividendos acharão vantajoso trocar a política até que 40% de todas as empresas tenham distribuições altas. Nesse ponto, o *mercado de ações com altos dividendos* está em equilíbrio. Firmas adicionais alterando sua política de dividendos farão esforços inúteis porque todas as clientelas estão satisfeitas. A política de dividendos de qualquer empresa individual não é relevante agora.

Para saber se você entendeu o efeito clientela, pense na seguinte afirmação: apesar do argumento teórico de que a política de dividendos é irrelevante ou de que as empresas não devem pagar dividendos, muitos investidores gostam de altos dividendos. Por isso, uma empresa pode aumentar seu preço por ação por meio de um índice de distribuição de dividendos mais alto. Isso é verdadeiro ou falso?

A resposta é "falso" se existir a clientela. Desde que um número suficiente de empresas com altos dividendos atenda aos investidores que adoram dividendos, uma empresa não poderá aumentar seu preço por ação pagando dividendos elevados. Uma clientela insatisfeita deve existir para que isso aconteça, e não há evidências de que esse seja o caso.

Questões conceituais

17.5a Como o mercado reage às alterações inesperadas nos dividendos? O que isso indica sobre os dividendos? E sobre a política de dividendos?

17.5b O que é uma clientela de dividendos? Depois de considerar tudo, você esperaria que uma empresa com risco e perspectivas de crescimento significativas, mas altamente incertas, tivesse uma distribuição de dividendos baixos ou altos?

17.6 Recompra de ações: uma alternativa aos dividendos

Até aqui em nosso capítulo, vimos os dividendos pagos em dinheiro. No entanto, os dividendos e os juros sobre capital próprio não são a única forma das empresas distribuírem dinheiro. Em vez disso, uma empresa poderia **recomprar suas ações**. As recompras têm se tornado uma ferramenta cada vez mais popular, e é enorme a quantia gasta em recompras nos Estados Unidos. Por exemplo, no primeiro trimestre de 2008, empresas norte-americanas anunciaram planos para recomprar $76 bilhões em ações, um valor abaixo dos $121,9 bilhões no quarto trimestre de 2007 e do recorde de $174 bilhões no terceiro trimestre de 2007. No todo, um recorde de $538 bilhões em planos de recompra de ações foi anunciado em 2007.

recompra de ações
a compra, por parte de uma empresa, de suas próprias ações.

de caixa livre o farão. Essa implicação normativa faz sentido econômico, e para que as empresas brasileiras a adotassem seria necessário prever, no estatuto, o não pagamento de dividendos e a regra para pagamentos de dividendos quando o negócio estiver maduro. Isso só é possível se, ao abrir o capital, o estatuto já estiver assim escrito. Como o mercado parece esperar dividendos, os estatutos geralmente preveem o mínimo que teriam que estabelecer se a assembleia no futuro deliberar sobre o quanto do lucro deve ser distribuído em dividendos.

Para encerrar nossa discussão sobre os dividendos, enfatizamos uma última vez a diferença entre os dividendos e a política de dividendos. Os dividendos são importantes porque o valor de uma ação é determinado, em última análise, pelos dividendos que ela pagará. O que não está claro é se o padrão de tempo dos dividendos (mais agora *versus* mais depois) é importante. Essa é a questão da política de dividendos, e não é fácil dar uma resposta definitiva para ela.

REVISÃO DE CONCEITOS E QUESTÕES INSTIGANTES

1. **Irrelevância da política de dividendos [OA2]** Como é possível que os dividendos sejam tão importantes e, ao mesmo tempo, a política de dividendos seja irrelevante?
2. **Recompras de ações [OA4]** Qual é o impacto de uma recompra de ações sobre o índice de endividamento de uma empresa? Isso sugere outro uso para o excedente de caixa?
3. **Cronologia dos dividendos [OA1]** Na terça-feira, 8 de dezembro, o conselho de administração da Companhia Elétrica Cidade S/A declara um dividendo de $0,75 por ação pagáveis na quarta-feira, 17 de janeiro, aos acionistas registrados na quarta-feira, 3 de janeiro. Quando é a data ex-dividendos? Se um acionista comprar ações antes dessa data, quem recebe os dividendos sobre essas ações, o comprador ou o vendedor?
4. **Dividendos alternativos [OA1]** Algumas empresas, como uma empresa britânica que oferece a seus maiores acionistas o uso gratuito de um crematório, pagam dividendos em espécie (ou seja, oferecem seus serviços aos acionistas a um custo abaixo do mercado). Os fundos de investimento devem investir em ações que pagam esses dividendos em espécie? (Os cotistas não recebem esses serviços.)
5. **Dividendos e o preço da ação [OA1]** Se os aumentos nos dividendos tendem a ser seguidos por aumentos (imediatos) nos preços das ações, como é possível dizer que a política de dividendos é irrelevante?
6. **Dividendos e o preço da ação [OA1]** Mês passado, a Companhia de Energia Elétrica de Angra dos Reis, que vinha tendo problemas com custos acima dos previstos em uma usina nuclear em construção, anunciou que estava "suspendendo temporariamente os pagamentos devido à diminuição dos fluxos de caixa associados a esse programa de investimentos". O preço da ação da empresa caiu de $28,50 para $25 quando esse anúncio foi feito. Como você interpretaria essa mudança no preço da ação (ou seja, em sua opinião, o que teria causado isso)?
7. **Planos de reinvestimento de dividendos [OA1]** Recentemente, a empresa norte-americana DRK Corporation desenvolveu um plano de reinvestimento de dividendos, ou DRIP. O plano permite que os investidores reinvistam os dividendos automaticamente na DRK em troca de novas ações. Com o tempo, os investidores da DRK poderão construir seus patrimônios reinvestindo os dividendos na compra de ações adicionais da empresa.

   Mais de mil empresas norte-americanas oferecem planos de reinvestimento. A maioria das empresas com DRIPs não cobra taxas de corretagem ou serviço. Na verdade, as ações da DRK serão compradas com um desconto de 10% sobre o preço de mercado.

   Um consultor da DRK estima que cerca de 75% dos acionistas da empresa participarão desse plano. Isso é muito mais do que a média.

   Avalie o plano de reinvestimento de dividendos da DRK. Ele aumentará a riqueza do acionista? Discuta as vantagens e as desvantagens envolvidas aqui.

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Tirole

Jean Tirole - The Theory of Corporate Finance-Princeton Univ

- Fixed-investment model (baseline for the type models): Ch. 3, §3.2 (p. 115–125)
- Variable-investment model (baseline for 6.8): Ch. 3, §3.4 (p. 127–130)
- Lemons problem, market breakdown, assets in place, pecking order: Ch. 6, §6.2 (p. 241–249)
- Certification (ratings): Ch. 6, §6.3, Application 4 (p. 249–251)
- Short-term maturities as a signal: Ch. 6, §6.3, Application 6 (p. 254–257)

of the assets in the case of failure. We show that it is optimal for investors to have priority in the case of default and for the entrepreneur to be the residual claimant (Chapter 5 will investigate another feature of debt, namely, the borrower's promise to pay fixed amounts to investors as a going concern, i.e., before liquidation; and Chapter 10 will connect debtholders' control rights with their cash-flow rights).

By focusing on a simple model of credit rationing, we do not do full justice to the corporate finance literature, which has developed a wide array of models with a similar flavor. For the sake of completeness, the supplementary section studies three broad classes of models that, through more sophisticated modeling, have aimed at deriving an interpretation of the "leftover claim" of outside investors as a standard debt claim.

3.2 The Role of Net Worth: A Simple Model of Credit Rationing

3.2.1 The Fixed-Investment Model

Variants of the following entrepreneurial model[2] will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. (We will later explore the possibility that these assets be illiquid. For example, they might be equipment or premises that are needed for the implementation of the project.) The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. (We will later observe that we can ignore the possibility that the entrepreneur consumes some of the cash and borrows more than $I - A$.)

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit");[3] or, equivalently, the entrepreneur chooses between a project with a high probability of success and another project which ceteris paribus she prefers (is easier to implement, is more fun, has greater spinoffs in the future for the entrepreneur, benefits a friend, delivers perks, is more "glamorous," etc.) but has a lower probability of success.[4]

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur.[5] In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral.[6] For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0.[7] The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. That is, we have in mind that several prospective lenders compete for issuing a loan to the borrower, and that, if the most attractive loan offer made a positive profit, the borrower could turn to an alternative lender and offer

2. This specific model is taken from Holmström and Tirole (1997). But its main idea can be found in various forms in many anterior papers.

3. See Exercise 3.20 for the continuous-effort version of the model.

4. Note that, for simplicity, we treat the entrepreneur as a unitary actor. There is an interesting question as to how moral hazard and incentives propagate down within the corporate hierarchy. Pagano and Volpin (2005) assume that benefits accrue to all company insiders, and not only to managers; in their model, managers need workers' cooperation to produce and therefore share benefits with employees.

5. For example, in the biotechnology alliance financing discussed in Section 2.4.2, the private benefit might be the entrepreneur's benefit from working on other projects (with other partners or on her own). The shift in attention then reduces the probability of success of the project under consideration.

6. Exercise 3.2 generalizes this analysis to allow for entrepreneurial risk aversion. Exercise 3.12 considers risk-averse investors.

7. The investors have rate of time preference equal to 0, which is also the market rate of interest.

to switch for a slightly lower interest rate.[8] We use the plural "lenders" even though a single lender may turn out to finance the entire loan, because we want to emphasize that lending is a passive and anonymous activity in the theories reviewed in Part II.

Let us turn to the loan contract. A contract first stipulates whether the project is financed.[9] If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure (the gross payoffs are the *ex post* monetary payoffs and take no account of past investments and private benefit). Intuitively, there is no point in specifying a positive transfer from the lenders to the borrower, as such a transfer can only weaken incentives, while it has no insurance benefit under risk neutrality. This property will be proved more rigorously and is here taken for granted. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders.[10] To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

The zero-profit constraint for the lenders can be written as

$$p_H R_l = I - A,$$

assuming that the loan agreement induces the borrower to behave (which under our assumptions will be the case). The rate of interest ι is given by

$$R_l = (1+\iota)(I-A) \quad \text{or} \quad 1+\iota = 1/p_H.$$

So, unless $p_H = 1$, the nominal rate of interest ι reflects a default premium and exceeds the expected rate of return (called r in Part VI and here normalized to 0) demanded by investors.

We summarize the timing in Figure 3.1.

We assume that the project is viable only in the absence of moral hazard. That is, the project has

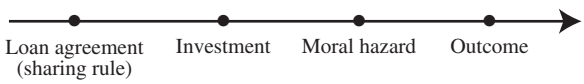


Figure 3.1

positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \tag{3.1}$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \tag{3.2}$$

It is easy to see that inequality (3.2) implies that no loan that gives an incentive to the borrower to misbehave will be granted. Indeed, rewrite (3.2) as

$$[p_L R_l - (I-A)] + [p_L R_b + B - A] < 0.$$

So, in the case of misbehavior, either the lenders must lose money in expectation, or the borrower would be better off using her cash for consumption, or both.

3.2.2 The Lenders' Credit Analysis

Because the project has negative NPV in the case of misbehavior, the loan agreement must be careful to preserve enough of a stake for the borrower in the enterprise. The borrower faces the following tradeoff once the financing has been secured: by misbehaving, she obtains private benefit B, but she reduces the probability of success from p_H to p_L. Because she has stake R_b in the firm's income (she receives R_b in the case of success and 0 in the case of failure), the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \tag{IC$_b$}$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H\left(R - \frac{B}{\Delta p}\right).$$

8. See Exercise 3.13 for the extension of the model to lender market power.

9. "Random financing" contracts, in which the borrower brings equity in exchange for a probability between 0 and 1 of being financed may in some cases be optimal when the investment size is fixed (as it is here) or more generally in the presence of indivisibilities or increasing returns to scale (see Exercise 3.1). For simplicity, we focus on deterministic contracts.

10. The lenders' net payoff is thus $R_l - (I-A)$ in the case of success, and $-(I-A)$ in the case of failure. The borrower's net payoff is thus $R_b - A$ in the case of success, and $-A$ in the case of failure, to which, in both cases, must be added a private benefit B if shirking occurs.

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) \geqslant I - A, \qquad (\mathrm{IR_l})$$

where "$\mathrm{IR_l}$" stands for the lenders' individual rationality constraint (which we will also often call the "breakeven constraint" or the "participation constraint"). Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_{\mathrm{H}}\frac{B}{\Delta p} - (p_{\mathrm{H}}R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \quad \Longleftrightarrow \quad p_{\mathrm{H}}R - I < p_{\mathrm{H}}\frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. Condition (3.4) says that the NPV is smaller than the minimum expected rent that must be left to the borrower to provide her with an incentive to behave.

Thus, *the borrower must have enough assets in order to be granted a loan*. Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. With insufficient assets, the entrepreneur must borrow a large amount and therefore pledge a large fraction of the return in the case of success. The entrepreneur then keeps only a small fraction of the monetary gain and is demotivated. The two parties cannot find a loan agreement that both induces effort (which requires a high compensation for the borrower in the case of success) and allows the lenders to recoup their investment. There is *credit rationing*. A rationed borrower may be willing to give a high fraction of the return to the lenders,[11] which here is equivalent to be willing to pay a high interest rate. But the lenders do not want to grant such a loan.

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_{l} to competitive investors so as not to leave them with a surplus:

$$p_{\mathrm{H}}R_{\mathrm{l}} = I - A.$$

Her stake,

$$R_{\mathrm{b}} = R - R_{\mathrm{l}} = R - \frac{I - A}{p_{\mathrm{H}}} \geqslant R - \frac{I - \bar{A}}{p_{\mathrm{H}}} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

As the conventional wisdom goes, "one only lends to the rich." The threshold $\bar{A}$ has a natural interpretation. As noted earlier, the term $p_{\mathrm{H}}B/\Delta p$ is nothing but the minimum expected monetary payoff to be left to the borrower to preserve incentives; it will be called the *agency rent*. The borrower must make an initial contribution at least equal to $\bar{A}$ so as to reduce the agency rent net of the initial downpayment A to at most the monetary profit $p_{\mathrm{H}}R - I$ of the project.

Using the breakeven condition for the lenders ($p_{\mathrm{H}}R_{\mathrm{l}} = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_{\mathrm{b}} = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_{\mathrm{H}}R_{\mathrm{b}} - A = p_{\mathrm{H}}(R - R_{\mathrm{l}}) - A & \\ \qquad = p_{\mathrm{H}}R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded*.[12]

So, the borrower's utility jumps up at $A = \bar{A}$. While the discontinuity is an artefact of the rigidity of the level of investment, the fact that 1 unit of assets may be worth more than 1 to the borrower in a situation of asymmetric information is quite general. Indeed, in the continuous investment version of this model to be developed in Section 3.4, we will see that for the borrower *assets or net worth have a shadow value exceeding* 1.

Determinants of credit rationing. To sum up, two factors may make a firm credit-constrained in this model:[13]

(i) low amount of cash on hand (low A);

11. This will be the case if A is small.

12. This property holds only in equilibrium. Were the entrepreneur to deviate and misbehave, the entrepreneur's (off-the-equilibrium-path) utility would exceed the smaller (off-the-equilibrium-path) NPV (at least for A close to $\bar{A}$), since the lenders would lose money.

13. The market interest rate, here normalized at 0, is another determinant of the strength of the balance sheet. More generally, the pledgeable income must exceed the investors' outlay times $(1 + r)$,

(ii) high agency cost, where the agency cost can be measured, fixing the project's NPV, $p_H R$, by the combination of the private benefit B and the likelihood ratio $\Delta p / p_H$.

The entrepreneur's ability to borrow is limited by the nonpledgeability of some $(p_H B/\Delta p)$ of the value to investors. Here moral hazard is determined by two factors: the private benefit B that the entrepreneur can enjoy by misbehaving, and the extent to which the verifiable performance reveals such misbehavior. The informativeness of the performance variable regarding effort is defined by the likelihood ratio $(\Delta p/p_H) = (p_H - p_L)/p_H$.[14] This ratio measures the proportional reduction in the probability of success when the entrepreneur misbehaves and is therefore also a measure of the marginal productivity of effort by the borrower. The higher the likelihood ratio, the more informative about effort choice the outcome is ("the better the performance measurement"), and the easier the access to outside financing (in the sense that the minimum net worth $\bar{A}$ decreases). In the model of this section, the pledgeable income never exceeds $p_H R - B$, since the entrepreneur can always take her private benefit B, but may be much smaller when performance measurement is poor, i.e., the likelihood ratio is low.

In practice, the agency cost is influenced not only by the project's and the entrepreneur's characteristics, but also by the surrounding legal, regulatory, and corporate environment. Countries with strong investor protection limit the managers' ability to squander investor money and thereby exhibit lower agency costs; relatedly, the firms' ability to cross-list in jurisdictions with good shareholder protection is expected to reduce their agency cost and therefore to facilitate financing.[15]

Remark (full investment of entrepreneurial assets). We have assumed that the borrower invests her entire wealth. However, it is easy to see that this is an optimal choice for the borrower. Would the borrower want to consume $c \leqslant A$ and invest only $A - c$? If the project is still funded, the borrower still obtains the entire social surplus $p_H R - I$. On the other hand, it becomes more difficult to obtain a loan. Now, the entrepreneur's initial assets must exceed $\bar{A} + c$ in order for the project to be funded. Therefore the entrepreneur cannot gain by not investing her entire wealth in the project.[16]

Remark (high-powered incentive scheme). Earlier we claimed that risk neutrality implies that the absence of reward for the entrepreneur in the case of failure involves no loss of generality. Suppose, more generally, that the entrepreneur receives R_b^S in the case of success and R_b^F in the case of failure, where

$$p_H R_b^S + (1 - p_H) R_b^F \geqslant p_L R_b^S + (1 - p_L) R_b^F + B$$
$$\iff \quad (\Delta p)(R_b^S - R_b^F) \geqslant B$$

in order to discourage the entrepreneur from misbehaving. The investors' income is then

$$p_H(R - R_b^S) + (1 - p_H)(-R_b^F) \leqslant p_H\left(R - \frac{B}{\Delta p}\right) - R_b^F \leqslant \mathcal{P}.$$

Rewarding the entrepreneur in the case of failure implies a uniform upward shift in her minimum incentive-compatible pay structure and an overall reduction in what can be pledged to investors (note the analogy with the previously considered case of an initial consumption c). By contrast, the entrepreneur's utility, provided that she can secure funding, is not affected: because the investors break even, the entire surplus goes to the entrepreneur, who receives

$$U_b = p_H R - I,$$

regardless of the choice of R_b^F. We thus conclude that rewarding the entrepreneur in the case of failure cannot raise her utility, but can compromise financing.

where r is the rate of interest:

$$p_H\left(R - \frac{B}{\Delta p}\right) \geqslant (1 + r)(I - A).$$

Thus, keeping the investment cost I fixed, an increase in the rate of interest r is equivalent to a decrease in the cash on hand.

14. The likelihood ratio is often defined as p_H/p_L. The two notions are obviously equivalent.

15. See, for example, the empirical confirmations by Doidge et al. (2004), Miller (1999), Pagano et al. (2001), and Reese and Weisbach (2002).

16. This reasoning relies, of course, on the borrower's putting equal weight on current and future consumption. If the borrower had immediate consumption needs, she would put some of A aside for consumption. We invite the reader to extend the analysis to the more general specification in which the borrower consumes c_0 at the start and c_1 after the outcome is realized, and has utility the expectation of $u_0(c_0) + u_1(c_1)$, where the functions $u_0(\cdot)$ and $u_1(\cdot)$ are increasing and concave. (The basic insights are unaltered. See also Exercise 3.2.)

Remark (value and investor value). Because the essence of corporate finance is that investors cannot appropriate the full benefit attached to the investments they enable, we must distinguish two slices in the overall cake: that for the insiders and the rest for the outsiders (the decomposition must be finer if there are multiple categories of each). In this book, "value" or "total value" refers to the total cake, while "investor value" refers to the investors' slice; in the barebones model of this section, these two values are pR and pR_l for probability of success p once the investment has been sunk (of course, one needs to subtract I and $I - A$, respectively, if one wants to obtain the corresponding net or *ex ante* magnitudes). The empirical literature often uses the phrase "value" for what we call here "investor value," but this should not create confusion.

Remark (risk taking). Moral hazard here refers to the possibility that the borrower takes an action that reduces investor value (and total value as well). There is no risk taking. We will come back to risk taking in subsequent chapters, but the reader may want to consult Exercises 3.15, 3.16, and 4.15 for three simple ways of introducing risk taking in the context of this simple model.

3.2.3 Do Investors Hold Debt or Equity?

We interpreted the loan agreement as a profit-sharing contract. It turns out that with two levels of profit, 0 and R, the lenders' claim can be thought of as being either debt or equity: put differently, there is here no difference between risky debt and equity. The debt interpretation goes as follows: the borrower must reimburse R_l or else go bankrupt. In the case of reimbursement the borrower keeps the residual $R - R_l$. Alternatively, the two parties can define shares in an all-equity venture. The entrepreneur and the investors hold fractions R_b/R and R_l/R, respectively, of equity. These are called "inside equity" and "outside equity."

This feature of the two-outcome model is both a weakness and a strength. A serious weakness is that it cannot, as it stands, account for the richness of existing securities; but we will show how to extend it in order to generate a more realistic diversity of claims. A strength of this modeling is that it will enable us to analyze a number of key ideas without being held back by the need to specify whether one is analyzing debt, equity, or an alternative claim. Some readers may find it surprising that a lack of predictive power relative to the structure of outside claims may constitute a strength. To clarify this point, it is worth pointing out that many phenomena in corporate finance have wider scope than that defined by the context in which they were discovered. Let us provide some illustrations in support of this view:

(a) As we will study in Chapter 5, Easterbrook (1984) and Jensen (1986) have argued that it is optimal to require cash-rich firms to pay out income on a regular basis, thereby forcing them to return to the capital market. The payment takes the form of a dividend in Easterbrook and of a short-term debt obligation in Jensen. The starting point for both analyses, namely, the desire to pump free cash flow out of the firm, is the same.

(b) The foundations for the soft-budget-constraint problem, also studied in Chapter 5, do not rely on outside claims being debt or equity. While it is usually analyzed in the context of specific assumptions on the financial structure, its logic is quite general.

(c) The literatures on monitoring of a firm by a large shareholder and by a bank holding debt claims have much in common. They are both concerned with the monitor's incentive to supervise and with the impact of monitoring on the firm's behavior.

(d) The idea of using dispersed claimholders to extract rents from third parties (see Chapters 7 and 11) has been developed in separate literatures on debt and on equity.

Thus, abstracting in a first step from the complex issues associated with the diversity of outside claims may generate a better focus on, and a more rigorous analysis of the fundamentals of such phenomena. A richer analysis can then be obtained from the introduction of further modeling features that motivate a diversity of outside claims.

3.2.4 Dilution and Overborrowing

Recall from Section 2.3.3 (see also Fama and Miller 1972) that debt contracts include negative covenants

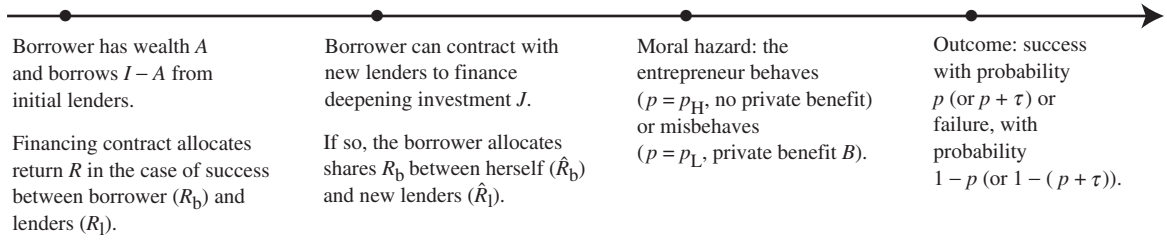


Figure 3.2

prohibiting the dilution of creditors' claims through the issue of new securities, especially ones with equal or higher seniority. There are two basic reasons for such covenants. First, creditors obviously do not want the borrower to issue claims that have a higher or the same seniority as theirs, as this reduces the amount they can collect if the firm defaults. Second, and more subtly, the issue of new securities may alter managerial incentives and the size of the pie.

Let us illustrate the second reason in our simple context. Consider the borrowing contract above in which the lenders take claim R_l in the case of success and the borrower an incentive-compatibility claim $R_b \geqslant B/\Delta p$. Now suppose that there is an opportunity for a "deepening investment." This investment costs an extra J and increases the probability of success uniformly by τ. That is, the probability of success becomes $p_H + \tau$ if the entrepreneur behaves and $p_L + \tau$ if the entrepreneur misbehaves.[17] Assume that this deepening investment is inefficient in that its net cost C_1 is positive, or put differently the expected increase in profit is smaller than J:

$$C_1 \equiv J - \tau R > 0.$$

The timing goes as in Figure 3.2.

We assume away any negative covenant prohibiting further borrowing and so the borrower can contract with new lenders.[18] However, in the case of new financing, initial lenders are not formally diluted in that they keep their stake R_l in success when the borrower contracts with new lenders. So the first motivation for inserting a covenant that prohibits the issuing of new securities is absent.

Note first that it is not in the interest of the borrower to contract with new investors if this results in the same effort, i.e., in no taking of private benefit. Intuitively, the new investment reduces total value by C_1, and so someone must lose in the process. Because the value of the initial investors' claim is increased (to $(p_H+\tau)R_l$) if the borrower still behaves, either the entrepreneur or the new investors must lose, which is impossible because the losing party would refuse to write the second financing contract. So assume that the new financing contract disincentivizes the borrower. This reduced incentive results in a second cost:

$$C_2 \equiv (\Delta p)R - B > 0.$$

As described in the timing, let $\hat{R}_b$ and $\hat{R}_l$ denote the new stake of the borrower and the stake of the new lenders, with

$$\hat{R}_b + \hat{R}_l = R_b.$$

Assuming that the new lenders are competitive, then

$$(p_L + \tau)\hat{R}_l = J.$$

The entrepreneur gains from overborrowing if and only if

$$(p_L + \tau)\hat{R}_b + B > p_H R_b,$$

or, using the breakeven condition for the new investors,

$$[(p_L + \tau)R_b - J] + B > p_H R_b.$$

After some manipulations, this condition becomes

$$[p_H - (p_L + \tau)]R_l > C_1 + C_2.$$

This necessary and sufficient condition for the deepening investment to be financed has a simple

17. This additivity property is convenient because it separates the incentive compatibility constraint from the impact of the new investment.

18. More generally, the division of the pie ($R_l + R_b = R$) is not made contingent on the event of a deepening investment.

interpretation. The right-hand side is the total cost of refinancing: direct cost plus incentive cost. The left-hand side of the inequality is the *externality* on the initial investors. Thus the total cost must be smaller than the loss of value for the initial investors.

When the borrower's balance sheet (as measured by A, say) improves, R_b increases, R_l decreases, and so this inequality is less likely to be satisfied. Put differently, in the absence of negative covenant, *overborrowing is more likely to happen with weak borrowers*.

Let us conclude this analysis of overborrowing with a few remarks. First, overborrowing in this situation can alternatively be avoided by forcing the entrepreneur not to dilute her own claim; this requirement is usually included in compensation contracts, although there have been attempts to evade it through derivative contracts (see Section 1.2.2). Second, the financing contracts need not be signed sequentially: simultaneous contracts also give rise to an overborrowing problem (see Bizer and DeMarzo 1992; Segal 1999). Third, the overborrowing problem arises with a vengeance in the context of sovereign borrowing, in which it is hard to specify a limit on indebtedness of the sovereign, if only because there are many different ways for a government to add new liabilities (see Bolton and Jeanne (2004) for an analysis of sovereign borrowing with the possibility of dilution). Finally, *in a multiperiod financing context, uncoordinated lending further leads to excessively short maturity structures of debt*, as investors scramble to obtain priority over other investors (see Exercise 5.9).

3.2.5 Boosting the Ability to Borrow: Reputational Capital and Capability

Recall from Chapter 2 that lenders do not only look at tangible assets such as cash, land, and equipment. Ceteris paribus, they are more likely to issue a loan if the borrower has a good reputation, as was stressed in particular by Diamond (1991). The role of this intangible capital is easily analyzed in the credit rationing model.

Suppose, for example, that the borrower has less attractive opportunities for misbehavior, in that the private benefit B from misbehaving is reduced to $b < B$.[19] This may have several interpretations. Along the lines of the "effort interpretation" of moral hazard, one might imagine that the project falls well within the core competency of the entrepreneur and therefore demands less attention or supervision of the subordinates: the task is just easier for the entrepreneur. Alternatively, one could imagine that the entrepreneur has less attractive outside options (focusing on other, separate projects of her own) or opportunities for fraud and embezzlement (e.g., it is harder to buy inputs at an inflated price from a friend or family).

With reduced scope for moral hazard, the asset threshold is accordingly lower: from equation (3.3),

$$\bar{A}(b) < \bar{A}(B),$$

where

$$\bar{A}(\beta) \equiv p_H \frac{\beta}{\Delta p} - (p_H R - I),$$

and thus

$$\bar{A}(B) - \bar{A}(b) = \frac{p_H}{\Delta p}(B - b) > 0.$$

In this sense, a "more reliable borrower" (that is, a borrower who has a lower private benefit from misbehaving) is more likely to obtain a loan.

How does this fit with the idea that a good reputation helps raise external finance? Suppose now that the private benefit (B or b) is not directly observed by the lenders, who only have the borrower's track record at their disposal. That is, the lenders know whether the borrower's past projects have been successful or whether past loans have been reimbursed. They use this information to update their beliefs about the reliability of the borrower. A better track record is an (imperfect) indicator of good reliability, that is, in our example, of a low private benefit from misbehaving.

Consider an entrepreneur who got a loan for a first project, and may in the future have new projects that will also call for outside financing. Let us further assume that these future projects are not yet well-defined, and focus on short-term finance. (Chapter 5 will analyze long-term loans.) In this situation, the entrepreneur should adopt a long-term perspective.

19. We could alternatively analyze the impact of a higher probability of success or of changes in other variables, with similar insights. The focus on the private benefit allows a cleaner analysis because changes in the private benefit keep the NPV of the project constant.

That is, she should not content herself with comparing the private benefit and the monetary payoff attached to the first project; she should also take into account the fact that a current success will bring two further benefits:

A *retained-earnings benefit*: even under symmetric information between the parties about the entrepreneur's reliability, a current success helps the entrepreneur build up net worth. This net worth has a shadow value; a unit of income is valued above 1 by the entrepreneur if there is a probability of credit rationing in the future. This benefit is studied in Exercise 3.11.

A *reputational benefit*: if, furthermore, the lenders have incomplete information about the entrepreneur's reliability, their updating of beliefs about this reliability confers an extra benefit on the entrepreneur in the case of success. Reputation complements net worth in reducing the probability of future credit rationing.[20]

An implication of the existence of this reputational benefit is that an unreliable borrower who would have no incentive to behave were her unreliability known to the lenders may have an incentive to behave today in order to get a loan tomorrow. The analysis of the situation becomes more complex once we realize that lenders are unlikely to be fools and understand that unreliable borrowers may have an incentive to masquerade as reliable ones. A proper study of reputational capital requires some (at least intuitive) understanding of dynamic games with incomplete information (see Exercise 6.3). We hope that the idea that reputational capital can substitute for net worth to thwart credit rationing is clear enough. There is indeed empirical evidence that reputation helps borrowers to obtain credit as well as better terms (see, for instance, Banerjee and Duflo's (2000) study of the Indian software industry).

Remark (information sharing). The impact of reputational capital is stronger, the more widely the information about borrower performance is disseminated. Padilla and Pagano (2000) observe that information sharing among lenders reinforces the borrowers' incentives to perform and argue that this may account for the fact that lenders (banks, finance companies, and retailers) spontaneously provide information about past defaults, delays in payment, current debt exposure, and riskiness of their borrowers to credit bureaus and credit-rating agencies, and therefore to their competitors. They develop a model in which lenders may share information even when this may encourage consumer poaching and thus enhanced *ex post* competition.

3.2.6 Making Efficient Use of Information to Reduce the Agency Cost

A basic theoretical result in the economics of agency, due to Holmström (1979), states that making economic agents accountable for events over which they have no control does not help with moral-hazard problems and generally worsens incentives. Roughly speaking, one should try to use the most informative or precise measurement of the agent's economic activity, or what is called in statistics a "summary" or "sufficient statistic."[21] This result underlies much of the thinking about managerial compensation, for example, the quest for good metrics to reward employees (based on customer satisfaction, reduction of unit costs, sales, etc.) or division managers (like EVA (economic value added) or balanced scorecard methods). More to the point for our context, it offers theoretical foundations for the use of benchmarking. Benchmarking, also called relative performance evaluation, consists in comparing the performance of, say, a firm with that of similar firms, to better assess managerial accomplishments. For example, a car producer's good financial performance is less indicative of good management if other car producers also do well than if the automobile industry is in a

20. Things are actually a bit more complicated than this dual benefit suggests: the reputational benefit depends on the borrower's equilibrium behavior (which itself depends on the retained-earnings benefit) and not only on the reputational one. Technically, if the retained-earnings benefit is strong enough to induce a high-private-benefit entrepreneur to behave, then success brings no reputational benefit.

21. A good introduction to sufficient statistics is Chapter 9 of DeGroot (1970). Suppose that one observes two variables x and y, and that one is trying to infer a third, unobservable variable z. The joint distribution of x and y, given z, is $f(x, y \mid z)$. The variable x is a sufficient statistic for (x, y) if the posterior distribution of z conditional on the observation of x and y depends only on x. To recognize sufficient statistics, a necessary and sufficient condition is the factorization criterion, that is, the existence of functions g and h such that $f(x, y \mid z) = g(x, y)h(x, z)$. A simple computation then shows that the distribution of z conditional on x and y does not depend on y.

recession. Or, a high price fetched by the stock of a software or biotechnology start-up company in an initial public offering (IPO) is not foolproof evidence of good entrepreneurship and careful venture capital monitoring if this price is reached during a stock price bubble.

We will come back a few times in this book to the issue of the quality of performance measurement and how it affects the ability to receive financing.[22] Let us just observe that in our context, the ability to raise financing is enhanced by conditioning entrepreneurial compensation on the performance measure with the highest available likelihood ratio.[23]

Let us provide a first illustration of this principle.

Benchmarking. A possible reinterpretation of our model is that there are three states of nature.

(i) Favorable state (probability p_L). The environment is sufficiently favorable that the project will succeed regardless of the entrepreneur's effort.

(ii) Unfavorable state (probability $1 - p_H$). The environment is harsh and the project will fail even if the entrepreneur does her best.

(iii) Intermediate state (probability $\Delta p = p_H - p_L$). Success is not guaranteed, but is reached provided the entrepreneur exerts effort.

Of course, no one *ex ante* knows which state prevails. The financing and effort decisions are chosen in the ignorance of the state of nature.[24]

Suppose now that one will learn *ex post* whether the state was favorable or not (i.e., intermediate or unfavorable), say, by looking at a less promising firm in the same industry that succeeds only if circumstances are favorable. Consider the following compensation scheme:

- the entrepreneur receives 0 if the state is favorable;
- the entrepreneur otherwise receives R_b in the case of success and 0 in the case of failure.

The incentive constraint is still

$$(\Delta p)R_b \geqslant B \qquad (IC_b)$$

since the entrepreneur's stake is still R_b in the state of nature in which she affects profit. The pledgeable income, however, has increased since one no longer pays the entrepreneur for being lucky: now the maximal pledgeable income is

$$p_H R - (\Delta p)\Big[\min_{\{IC_b\}} R_b\Big] = p_H R - B,$$

where $[\min_{\{IC_b\}} R_b]$ denotes the smallest reward R_b that ensures incentive compatibility.

Next, let us assume that the firm's performance can be compared with that of an identical firm facing the same state of nature. Assuming that the entrepreneur in the other firm behaves, then "success" in the other firm provides information that the state is either favorable or intermediate, while "failure" in the other firm reveals an unfavorable state. Then, conditional on the entrepreneur failing, one learns either that she was unlucky or that she failed because she misbehaved. In this case, the pledgeable income cannot be increased by benchmarking if one abides by the entrepreneur's limited liability:[25] when

22. See, for example, Section 4.4 and Chapter 9.

23. For example, it would never come to one's mind to condition the entrepreneur's compensation on the weather in Bali, on the outcome of the soccer World Cup, or on other "irrelevant" variables. Why? Let us be a bit more technical here. In the notation of footnote 21, let (x, y) denote the verifiable state of nature (which includes, but is not limited to, the firm's profit $x \in \{0, R\}$), on which the entrepreneur's reward R_b can be conditioned. Thus, let $R_b(x, y)$ denote the state-contingent compensation specified by the financing contract. Suppose that the firm's profit x is a sufficient statistic for (x, y) when assessing the entrepreneur's effort, which we will here call $z \in \{L, H\}$ (see footnote 21 for the definition of a sufficient statistic). The density of the verifiable state (x, y) for a given effort z can be factorized: $f(x, y \mid z) = g(x, y)h(x, z)$. Thus for a choice of effort $z \in \{L, H\}$, the entrepreneur's expected reward is

$$\int_x \int_y R_b(x, y)g(x, y)h(x, z)\,dx\,dy = \int_x \hat{R}_b(x)h(x, z)\,dx,$$

where $\hat{R}_b(x) \equiv \int_y R_b(x, y)g(x, y)\,dy$.

So, a contract that rewards the entrepreneur solely as a function for profit ($\hat{R}_b(x)$) can do at least as well as a more general contract. And, in general, it can do better (in our context, it does strictly better in particular if $\int_y R_b(0, y)g(0, y) > 0$ and if a strictly positive borrower payoff in the case of success jeopardizes financing). Added risk is bad when the limited liability constraint is binding (and would be bad if the agent were risk averse even if she is not protected by limited liability).

24. Information accrues *ex post* through profit realization. Still the state is not learned *ex post* in the basic model.

25. Benchmarking could become relevant again in this example if we relaxed the limited liability constraint by introducing reputational concerns such as a stigma that affects future borrowing or other future relationships of the borrower or else costly nonmonetary penalties (jail or costly collateral pledging as in Chapter 4). Then, the observation that both entrepreneurs fail implies that the state of nature was unfavorable and so stigmas and/or nonmonetary penalties are not in order, unlike the situation in which only one entrepreneur fails (which implies that she misbehaved).

the entrepreneur fails, she already receives 0. On the other hand, it would be optimal to punish the entrepreneur harshly when she fails and the benchmark firm succeeds.[26]

3.2.7 Sensitivity of Investment to Cash Flow: A First Look

Recall from Section 2.5 the empirical finding that investment is sensitive to cash flow. An interesting issue is whether this "investment–cash flow sensitivity" increases with the extent to which the firm is financially constrained. Fazzari et al. (1988) use *a priori* measures of financial constraints and find that the sensitivity of investment to cash flow is particularly large for firms that have trouble raising external funds (for example, firms facing high agency costs). Kaplan and Zingales (1997) argue that there is no theoretical basis for this relationship and present empirical evidence that differs from that of Fazzari et al.

Although the model in this chapter is static while the empirical evidence relates to ongoing concerns (multistage financing is studied in Chapter 5), it can shed some light on the debate. We can imagine that cash on hand A includes the cash flow accruing from the firm's previous activity and see how investment reacts to a small change in the cash flow.[27]

There is a sense in which Fazzari et al. (1988) are right on the theoretical front: the firms whose investment is boosted by a small increase in cash flow are the marginal firms, i.e., those whose cash on hand A lies just below $\bar{A} = I - \rho_0$, where ρ_0 denotes the pledgeable income:

$$\rho_0 \equiv p_H\left(R - \frac{B}{\Delta p}\right) = p_H R - \frac{p_H B}{\Delta p}.$$

Firms with more cash or a lower agency cost do not modify their investment behavior as their investment was already unconstrained.

Suppose, however, that firms are heterogeneous in the two dimensions: cash A and pledgeable income ρ_0 (we normalize the investment I to be the same for all). Assume for simplicity that these two variables are independently distributed (there is no reason for this to be the case: for example, firms with higher pledgeable income may have been able to invest more in the past and be richer today). Let $G(A)$ denote the (continuous) cumulative distribution of cash among firms in the economy, with density $g(A)$. Because only firms with cash on hand A satisfying $\rho_0 \geqslant I - A$ receive financing, aggregate investment among firms with pledgeable income ρ_0 is

$$\mathcal{I}(\rho_0) \equiv [1 - G(I - \rho_0)]I.$$

Now, consider a small, uniform increase in cash δA for all firms. Then, investment among firms characterized by ρ_0 increases by

$$\delta\mathcal{I}(\rho_0) = g(I - \rho_0)I\delta A.$$

And so

$$\frac{\partial}{\partial \rho_0}(\delta\mathcal{I}(\rho_0)) = -g'(I - \rho_0)I\delta A.$$

If the density is decreasing ($g' < 0$), the sensitivity of investment to cash flow is higher for firms with a *low* agency cost (a high ρ_0) as in Kaplan and Zingales; intuitively, the cutoff $\bar{A}$ for firms with a low agency cost is low, and so with a decreasing density there are a lot of marginal firms. With an increasing density ($g' > 0$), the sensitivity of investment to cash flow is higher for firms with a *high* agency cost (a low ρ_0), as in Fazzari et al. Thus, unless one has more precise information about the actual heterogeneity of firms, it is difficult to predict how the sensitivity of investment to cash flow varies with an *a priori* measure of financial constraints (a proxy for (minus) ρ_0).

26. This is one aspect in which a "limited liability model" differs from a "risk-aversion model." In the rest of Section 3.2, we might as well have assumed that the entrepreneur is very risk averse at her subsistence level, normalized at zero consumption, that is, her utility falls very quickly at that level. Suppose at the extreme that the entrepreneur gets $-\infty$ when receiving a negative income. Then, provided that $p_H < 1$ (the entrepreneur may behave and be unlucky), it would not be optimal to set rewards below the subsistence level.

27. This thought experiment in a sense consists in looking at a single period of an ongoing firm that engages in short-term borrowing from investors. There are two reasons why this is only a first step toward an understanding of the sensitivity of investment to cash flow. First, if the firm anticipates that it may be credit-constrained tomorrow, the shadow value of 1 unit of profit at the end of the period exceeds 1, as it may help overcome financing problems in the future. More importantly, this shadow value may vary with current investment. Second, the description of the financial arrangements as a sequence of short-term borrowing contracts misses important long-term financing features (credit lines, debt–equity ratio, maturity structure of debt, etc.) that have an important impact on financial constraints (see Chapter 5).

3.3 Debt Overhang

Following Myers (1977), a number of contributions have studied situations in which a borrower is debt-ridden and unable to raise funds for an otherwise profitable project. The borrower is then said to suffer from debt overhang. The framework just developed suggests two possible interpretations of debt overhang. The first interpretation pursued below is a mere reinterpretation of the credit rationing analysis above: previous investors' collateral claim on the firm's assets reduces the net worth to below the threshold asset level for financing the new investment. Furthermore, the new project overall produces too little pledgeable income and so investment does not take place even if previous investors are willing to renegotiate their claim. The second and more interesting interpretation, and that stressed by the literature, emphasizes the need for renegotiating past liabilities in order to enable new investments.

3.3.1 Decrease in Net Worth

First, the borrower may have a positive-NPV project that would be financed in the absence of any previous debt obligation, but is denied financing due to such an obligation. Namely, suppose that (i) the entrepreneur has A in cash or collateral, but owes D from previous borrowing to a group of investors whom we will call the "initial investors," (ii) the initial investors have insisted on a covenant specifying that the borrower cannot raise more funds without their consent, and (iii) the borrower's assets A are pledged to the initial investors as collateral in case of default. If[28]

$$A > \bar{A} > A - D \geqslant 0,$$

the project would have been financed in the absence of previous borrowing but is not undertaken, since investors as a whole, that is, the initial investors and new investors (who can, of course, be the initial investors themselves), cannot recoup the cost of their investment $(I - A)$ plus the previous debt obligation (D) while they can receive D by seizing the collateral. More precisely, suppose the borrowers, the initial investors, and the new investors enter an agreement so as to finance the project. Because initial investors can secure themselves D by seizing the collateral, they must receive an expected payment at least equal to D under this agreement. Because the pledgeable income net of the investment cost is equal to

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) - I,$$

new investors obtain at most

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) - I - D + A = A - D - \bar{A} < 0,$$

which contradicts the fact that rational investors must at least break even.

3.3.2 Lack of Renegotiation[29]

Second, and more interestingly, suppose that

(i) the project is sufficiently profitable to attract funds even if the borrower has zero net worth,

$$\bar{A} < 0;$$

(ii) the borrower has previously been granted a long-term loan and is due to reimburse D "at the end," that is, when the outcome of the project (if financed) occurs;

(iii) this long-term debt obligation is contractually *senior* to any claim that the borrower might issue (a senior claim is a claim that must be paid before the borrower or any other claimholder receives any money);

(iv) the borrower has no cash ($A = 0$); and

(v) the debt overhang problem is sufficiently serious as not to be overcome by the expected profitability of the new project, or, put differently, the "slack" in pledgeable income, $-\bar{A}$, is smaller than what has to be paid back to previous investors, $p_{\mathrm{H}}D$, if the project is funded:

$$\bar{A} + p_{\mathrm{H}}D > 0.$$

28. Recall that

$$\bar{A} = p_{\mathrm{H}}\frac{B}{\Delta p} - (p_{\mathrm{H}}R - I)$$

is the minimum net worth to obtain financing.

29. The notion that renegotiation breakdowns generate debt overhang is central to Myers's (1977) original analysis, and also underlies that in Hart and Moore (1995) and Bhattacharya and Faure-Grimaud (2001). We will describe the debt overhang situation as one in which a new investment cannot be financed solely because renegotiation with previous debtholders proves infeasible. Debt overhang is generally described in the literature as a situation in which a firm may not be able to continue because it cannot renegotiate with its creditors. It is clear that the two situations are formally equivalent. The act of spending money to let a distressed firm continue is equivalent to an investment.

Because the borrower has no cash, initial investors receive nothing if the project is not financed. So, they are willing to participate in the financing of the project as long as they break even on this investment. For example, they can forgive existing debt, finance the investment I, and demand the entire cash-flow rights attached to external shares, that is, $R - B/\Delta p$ in the case of success. The initial investors then obtain

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) - I = -\bar{A} > 0.$$

The borrower is willing to go along with this arrangement, which allows her to continue and obtain rent $p_{\mathrm{H}}B/\Delta p$ in expectation rather than 0 if the project is not financed.

Suppose next that the initial investors have no cash and thus cannot directly finance the investment I. The borrower then needs to turn to new investors. Are the latter willing to finance the project? Because the initial debt is senior, and because the borrower needs to keep a minimum stake in the firm in order to commit to behave, at most

$$R - \frac{B}{\Delta p} - D$$

can be pledged to new investors in the case of success (and 0 in the case of failure). New investors are willing to enter an agreement to finance the project if and only if

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p} - D\right) \geqslant I$$

or

$$\bar{A} + p_{\mathrm{H}}D \leqslant 0,$$

which contradicts assumption (v).

To sum up, *the borrower cannot raise funds from new investors if she does not renegotiate some debt forgiveness from initial investors*. If renegotiation with initial investors is infeasible, gains from trade between the borrower and the community of investors may not be realized. Renegotiation breakdown creates debt overhang.

The possibility of debt overhang is often invoked in contexts in which "initial investors" stand for "corporate bondholders." It is often thought that because they are dispersed, and despite the existence of some coordinating mechanisms (nomination of a bond trustee, possibility for the firm to offer new securities in exchange for the bonds), bondholders have trouble renegotiating their claim when the borrower faces distress and requires some debt forgiveness.

In contrast, let us assume that initial investors are able to act collectively and renegotiate their initial claims. Because $\bar{A} < 0$, we know that there exists some renegotiated arrangement that is agreeable to all parties (borrower, initial investors, new investors), who would all get nothing if they failed to reach an agreement. Suppose, for example, that the initial investors accept a reduction in the face value of the debt from D to $d < D$, where

$$\bar{A} + p_{\mathrm{H}}d = 0.$$

Then new investors receive

$$\left(R - \frac{B}{\Delta p} - d\right)$$

in the case of success and are therefore willing to invest, since

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p} - d\right) = I$$

is equivalent to their breakeven constraint (3.3):

$$p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right) = I - \bar{A}.$$

Initial investors benefit from forgiving some of their claim as they now get

$$p_{\mathrm{H}}d = -\bar{A} > 0.$$

Lastly, the borrower can undertake the project and obtains rent $p_{\mathrm{H}}B/\Delta p > 0$.

Debt renegotiation thus allows the project to be undertaken and all parties to share the resulting gains from trade. How these gains from trade are actually shared depends, of course, on the relative bargaining power of the borrower and the initial investors (the new investors being assumed to be competitive and thus to just break even). The arrangement described above corresponds to the renegotiation that is most favorable to the initial investors. But, by varying continuously the relative bargaining power of the borrower and initial investors, one can generate any level of debt forgiveness from $D - d$ (the most favorable to initial investors) to D (the least favorable to them).

3.4 Borrowing Capacity: The Equity Multiplier

3.4.1 The Continuous-Investment Model

The continuous-investment model of this section is the polar opposite of the fixed-investment one. The fixed-investment model depicts a situation in which returns are sharply decreasing beyond some investment level. In contrast, we now assume that there are constant returns to scale in the investment technology. An investment $I \in [0, \infty)$ yields income RI, proportional to I, in the case of success, and 0 in the case of failure. The borrower's private benefit from misbehaving is also taken to be proportional to investment. As before, the borrower has a choice between behaving, in which case she derives no private benefit and the probability of success is p_H, and misbehaving, that is, enjoying private benefit BI, and reducing the probability of success to $p_L = p_H - \Delta p < p_H$. (One can also analyze the intermediate case of a continuous investment with general decreasing returns to scale (see Exercise 3.5).)

The borrower initially has cash A, and must therefore borrow $I - A$ to finance a project of size I. A loan agreement specifies that the lenders (who as before are assumed to make no profit) and the borrower receive 0 each in the case of failure, and R_l and R_b, respectively, in the case of success, where $R_l + R_b = RI$.

As in Section 3.2, we assume that investment has positive NPV (net present value), here per unit of investment, if the borrower behaves,

$$p_H R > 1, \tag{3.7}$$

but negative NPV otherwise,

$$1 > p_L R + B, \tag{3.8}$$

so that unless one can control the agency problem the investment cannot be funded. We also make an assumption that guarantees that the equilibrium investment is finite:

$$p_H R < 1 + \frac{p_H B}{\Delta p}. \tag{3.9}$$

Like inequality (3.5) in Section 3.2, inequality (3.9) has a simple interpretation: the expected net revenue per unit of investment, $p_H R - 1$, is lower than the per-unit agency cost, $p_H B / \Delta p$.

Finally, we keep assuming that the capital market is competitive. The analysis is very similar when the borrower faces a lender with market power, except that the resulting investment scale is smaller (see Exercise 3.13).

3.4.2 The Lenders' Credit Analysis

Following the steps of Section 3.2, the incentive compatibility and the breakeven conditions are

$$(\Delta p) R_b \geqslant BI \tag{IC$_b$}$$

and

$$p_H (RI - R_b) \geqslant I - A. \tag{IR$_l$}$$

In equilibrium, competitive lenders make no profit on the contract that is most advantageous for the borrower; the borrower's net utility is therefore equal to the social surplus brought about by the investment:

$$U_b = (p_H R - 1) I. \tag{3.10}$$

From (3.10) it is optimal for the borrower to invest as much as possible. The upper bound on investment and in turn her borrowing capacity ("outside financing capacity" or "debt capacity") are determined by constraints (IC$_b$) and (IR$_l$). Substituting (IC$_b$) into (IR$_l$), we obtain

$$I \leqslant kA, \tag{3.11}$$

where

$$k = \frac{1}{1 - p_H (R - B/\Delta p)} > 1. \tag{3.12}$$

The denominator of k is positive from (3.9). Furthermore, conditions (3.7) and (3.8) imply that $(\Delta p) R > B$, and therefore that the denominator of k is smaller than 1. This is important: the fact that $k > 1$ shows that the borrower can lever her wealth, k being the multiplier.

The multiplier is smaller, the higher the private benefit (B) and the lower the likelihood ratio ($\Delta p / p_H$, fixing p_H and thus the profitability of the investment), which are our two measures of the agency cost.

Conditions (3.7) and (3.10) furthermore imply that it is optimal for the borrower to invest k times her cash A, that is, to borrow $d = (k - 1)$ times her level of cash, where

$$d = \frac{p_H (R - B/\Delta p)}{1 - p_H (R - B/\Delta p)}. \tag{3.13}$$

The maximum loan, dA, is called "*borrowing capacity.*"[30]

Another important concept (which will be used, for example, in computing the value of retained earnings in a dynamic context) is the *shadow value v of equity (here cash).* The entrepreneur derives gross utility $v > 1$ from one more unit of equity. Letting $U_b^g \equiv A + U_b$ denote the borrower's gross utility, and using (3.10) and (3.11), we have

$$U_b^g \equiv vA, \tag{3.14}$$

where the shadow value of equity is

$$v = \frac{p_H B/\Delta p}{1 - p_H(R - B/\Delta p)} > 1. \tag{3.15}$$

As one would expect (in the relevant range defined by (3.7)–(3.9)), the borrowing capacity increases with per-unit income R and decreases with the extent of the moral-hazard problem (measured by the borrower's private benefit or the inverse of the likelihood ratio). The shadow value of equity increases with per-unit income R and also with the extent of the moral-hazard problem.[31]

Finally, let us introduce some notation that will be used repeatedly throughout the book. Let

$$\rho_1 \equiv p_H R$$

denote the expected payoff per unit of investment and

$$\rho_0 \equiv p_H\left(R - \frac{B}{\Delta p}\right)$$

denote the expected pledgeable income per unit of investment. Assumptions (3.7) and (3.9) can be rewritten as

$$\rho_1 > 1 > \rho_0.$$

The equity multiplier is then

$$k = \frac{1}{1 - \rho_0}, \tag{3.11′}$$

the debt capacity per unit of net worth

$$d = \frac{\rho_0}{1 - \rho_0}, \tag{3.12′}$$

and the borrower's gross utility

$$U_b^g = vA = \frac{\rho_1 - \rho_0}{1 - \rho_0}A. \tag{3.14′}$$

The borrower's net utility can then be written as

$$U_b^n = U_b = U_b^g - A = \frac{\rho_1 - 1}{1 - \rho_0}A = (\rho_1 - 1)I,$$

as one could have expected.

Remark (factors that keep the investment bounded). Condition (3.9) (the condition that the pledgeable income per unit of investment is smaller than 1) was needed in order to keep the investment finite in this constant return to scale environment. Such a condition is no longer needed if the price of output and therefore the revenue in the case of success is not fixed but rather depends on, say, industry investment. An increase in per-firm investment then lowers the market price, reducing both value and pledgeable income (see Exercise 3.17 for more detail).

Remark (sensitivity of investment to cash flow). Let us briefly return to the sensitivity of investment to cash flow. In the variable-investment model,

$$\frac{\partial}{\partial \rho_0}\left(\frac{\partial I}{\partial A}\right) = \frac{1}{(1 - \rho_0)^2} > 0,$$

and so firms with a low agency cost, which are therefore less financially constrained, exhibit a higher sensitivity. Intuitively, such firms have a high multiplier and their investment is therefore more sensitive to available cash.

3.4.3 Collateral Values: Outside Debt and the Maximal Incentives Principle

We now return to the indeterminacy of the financial structure (debt or equity) discussed earlier. It turns out that this indeterminacy was an artefact of the absence of profit in the case of failure.[32]

Thus, assume that, for investment size I, the profit is $R^S I$ in the case of success and $R^F I$ in the case of failure, where R^F is now positive. $R^F I$ can be thought of as the salvage value of assets and

$$RI \equiv (R^S - R^F)I$$

30. Note also that the "gearing ratio" $g = d/k = p_H R - p_H B/\Delta p$ is less than 1, and that the debt-over-inside-equity ratio is equal to d.

31. The shadow value is here constant with wealth A. With a decreasing-returns-to-scale technology, v depends on wealth and $v'(A) < 0$: the marginal wealth enables less and less profitable marginal investments as wealth increases (see Exercise 3.5).

32. The clearest illustration of this point is for the variable-investment model, which is why we treat this here. The same point can be made in a slightly different form (as some indeterminacy may remain) in the fixed-investment version (see Exercise 3.18).

as the increase in profit brought about by success. One would expect R^F to be larger when secondary asset markets are liquid.[33]

The model is otherwise the same as in the rest of Section 3.4: the private benefit (BI in the case of misbehavior, 0 otherwise) is also proportional to investment.

The generalization of the condition that the NPV per unit of investment is positive while the pledgeable income per unit of investment is negative ($p_H R > 1 > p_H(R - B/\Delta p)$) is

$$p_H R + R^F > 1 > p_H\left(R - \frac{B}{\Delta p}\right) + R^F.$$

A contract specifies an investment level I and a sharing rule, or equivalently a reward for the entrepreneur for each performance level: $\{R_b^S, R_b^F\}$, with $R_b^S, R_b^F \geqslant 0$ due to limited liability.

The optimal contract maximizes the entrepreneur's expected compensation,

$$U_b = \max_{\{R_b^S, R_b^F, I\}} \{p_H R_b^S + (1 - p_H) R_b^F - A\},$$

subject to two constraints (that will turn out to be binding at the optimum): the entrepreneur's incentive constraint,

$$(\Delta p)(R_b^S - R_b^F) \geqslant BI,$$

and the investors' breakeven constraint,

$$p_H(R^S I - R_b^S) + (1 - p_H)(R^F I - R_b^F) \geqslant I - A.$$

To show that the investors' breakeven constraint is binding, note that, if it were not, then the entrepreneur could increase R_b^S and R_b^F by an equal and small amount without affecting the incentive compatibility constraint. This uniform increase in compensation would raise the entrepreneur's payoff. As is now familiar, we conclude that the investors receive no surplus, and so (by substituting the breakeven constraint into the objective function) the entrepreneur's utility is equal to the NPV:

$$U_b = (p_H R + R^F - 1) I.$$

Because the NPV per unit of investment is positive, the entrepreneur therefore chooses the highest possible investment.

Next, note that the incentive constraint is binding (otherwise, the optimal investment would be infinite, which would violate the two constraints combined).

Lastly, suppose that $R_b^F > 0$ at the optimum. And consider a small increase $\delta R_b^S > 0$ in managerial compensation in the case of success together with a small decrease $\delta R_b^F < 0$ in the case of failure that keeps the investors' profitability constant:

$$p_H \delta R_b^S + (1 - p_H)\delta R_b^F = 0.$$

This small change (which is feasible only if $R_b^F > 0$) also keeps the objective function constant. But the incentive constraint is now slack, a contradiction. We thus conclude that at the optimum

$$R_b^F = 0.$$

Hence, an all-equity firm cannot be optimal: in the absence of debt, the entrepreneur would receive $R^F I$ times her share of stocks in the firm, and therefore would be rewarded even in the case of failure. By contrast, investors' holding debt $D \geqslant R^F I$ is an optimal financial structure. Using the fact that the two constraints are binding, the borrowing capacity is given by

$$R^F I + p_H\left(R - \frac{B}{\Delta p}\right) I = I - A$$

or

$$I = \frac{A}{1 - [p_H(R - B/\Delta p) + R^F]}. \qquad (3.16)$$

Predictions. The variable-investment model of this section is, of course, much too simplistic to provide even a stylized account of capital structure and investment. It, however, delivers three interesting preliminary insights.

• *Firms with lower agency costs borrow more.* As in Section 3.2.2, the firm's outside financing capacity is higher, the lower the agency cost as measured either by the private benefit B or by (the inverse of) the likelihood ratio $\Delta p/p_H$ (keeping p_H and therefore profitability constant).

• *The investors' holding safe debt plus some equity maximizes the entrepreneur's stake in the project and thereby her incentives.* (We will investigate the generality of this insight in Section 3.5.)

Decomposing the investors' claim into safe debt (which repays $R^F I$) and risky equity (which repays in

33. Several chapters in this book (primarily Chapter 14) will investigate the determinants of asset prices in secondary markets.

expectation $p_H[R - B/\Delta p]I$), the leverage ratios,

$$\frac{\text{debt}}{\text{total equity}} = \frac{R^F I}{p_H R I} = \frac{R^F}{p_H R}$$

and

$$\frac{\text{debt}}{\text{outside equity}} = \frac{R^F I}{p_H(R - B/\Delta p)I} = \frac{R^F}{p_H(R - B/\Delta p)},$$

are both constant in this simple-minded model.

• *Credit rationing is more binding for firms with less tangible assets or assets that have a lower value in liquidation* (there is indeed substantial evidence in this direction: see Chapter 2). To see this, let us decrease the value of tangible assets while keeping the NPV per unit of investment constant: that is, keeping other parameters constant, let us consider a decrease from R^F to $\hat{R}^F$ ($\hat{R}^F < R^F$) of the per-unit salvage value and an increase from p_H to $p_H + \tau$ ($\tau > 0$) of the probability of success such that

$$\hat{R}^F + (p_H + \tau)R = R^F + p_H R.$$

In order to keep the agency problem invariant (so as not to interfere with the first prediction), let us assume that the probability of success in the case of misbehavior becomes $p_L + \tau$. The borrower's incentive compatibility constraint is then unchanged as

$$[(p_H + \tau) - (p_L + \tau)](R_b^S - R_b^F) = (\Delta p)(R_b^S - R_b^F).$$

The analysis is unchanged and the new investment becomes

$$\hat{I} = \frac{A}{1 - [(p_H + \tau)(R - B/\Delta p) + \hat{R}^F]} < I.$$

Thus, ceteris paribus, tangible assets facilitate financing.[34]

3.4.4 Going Forward

This chapter offered a first glance at the basic conflict between value and pledgeable income. When pressed to produce returns to attract investors, borrowers first offer them a large debt repayment or a higher share of profits (Section 3.2). This policy is, however, limited by entrepreneurial moral hazard and must be supplemented by costly "concessions." Technically, with competitive investors, the total value (NPV) goes to the entrepreneur, who aims at maximizing this value subject to the constraint that the pledgeable income be sufficient to enable the investors to recoup their investments. The resulting policy (charter, covenants, governance structure, etc.) therefore sacrifices value to generate enough pledgeable income if the breakeven constraint condition is binding.

The variable-investment model of Section 3.4 pointed at such an elementary concession: a limited investment size. Indeed, with constant returns to scale, it would be optimal for the firm to grow without bounds, but pleasing investors requires a limited size (all the more so, as we have seen, as the agency problem is important and as assets are intangible). The rest of the book will provide further illustrations of the idea that entrepreneurs must sometimes "bend over backwards" in order to attract investors: costly collateral pledging, restricted exit options, short maturity structures, enlisting of active and speculative monitors, allocations of control rights to equityholders and debtholders, limits on takeover defenses, and so forth.

34. Rather than increase the probability of success, we could have increased the payoff R in the case of success. Then investment would have been invariant (see equation (3.16)). As increases in the value of the risky component are in general associated with both types of changes, the conclusion that tangible assets facilitate financing is robust.

Supplementary Sections

3.5 Related Models of Credit Rationing: Inside Equity and Outside Debt

This supplementary section reviews three classic, alternative models of credit rationing. These models are a bit more complex than the basic credit rationing model developed in this chapter and this supplementary section is accordingly more technical than the text. They are not relegated to the supplementary section because they are deemed "less important." Rather, the reader should recall from the introduction that we want to conduct controlled experiments throughout the book. Using the same simple and tractable model throughout allows us to concentrate on the key insights of the theory without getting bogged down by extraneous modeling changes. This is the motivation for setting these models aside. It should furthermore be borne in mind that these

adopted a mechanism-design approach. Biais et al. (2002) generalize the optimal-mechanism-design approach to situations in which there is an agency problem between the underwriter and the issuer (as in Baron 1982).[9]

Investors have no informational advantage over issuers. While most informational asymmetries relate to insiders' private knowledge about assets in place and prospects, it is easy to envision situations in which the asymmetry of information operates in the reverse direction, namely, in which investors are better informed on some dimensions. For example, venture capitalists are usually better able than unseasoned entrepreneurs to assess a business model or prospects of a product. In this chapter we will simplify the analysis by assuming that insiders are better informed than investors.[10]

No signal sent to third parties. This chapter focuses on the information conveyed by the issuance to investors. For conciseness, we do not cover an interesting literature that analyzes the informational impact of financial decisions on third parties, such as product-market competitors or suppliers (see Gertner et al. 1988; Poitevin 1989; Bhattacharya and Chiesa 1995; Yosha 1995). For instance, a firm may be eager to signal to investors that the demand for its product is high, as this may allow it to obtain more financing, but still be reluctant to convey such information to potential entrants in that market, whose entry it wants to deter. In contrast, there is no tension for the firm when signaling that it has low costs simultaneously to the capital and product markets when it wants to deter potential entrants.[11]

6.2 Implications of the Lemons Problem and of Market Breakdown

A number of important insights can be gleaned from the following barebones model, in which the borrower has private information about the probability of success.

Privately-known-prospects model. A borrower/entrepreneur has no funds ($A = 0$) to finance a project costing I. The project yields R in the case of success and 0 in the case of failure. The borrower and the lenders are risk neutral, and the borrower is protected by limited liability. The interest rate in the economy is normalized at 0.

The borrower can be one of two types. A good borrower has a probability of success equal to p. A bad borrower has a probability of success q. Assume that $p > q$ and that $pR > I$ (at least the good type is creditworthy). There are two subcases, which we will treat separately:

either $pR > I > qR$
(only the good type is creditworthy),

or $pR > qR > I$
(both types are creditworthy).

The borrower has private information about her type. The capital market, which is competitive and demands an expected rate of return equal to 0, puts probabilities α and $1 - \alpha$ on the borrower being a good or a bad type, respectively. Under asymmetric information, the capital market does not know whether it faces a "p-borrower" (a good borrower) or a "q-borrower" (a bad borrower).[12] Let

$$m \equiv \alpha p + (1 - \alpha)q$$

denote the investors' prior probability of success.

Note that we have left out for the moment moral hazard in the definition of the privately-known-prospects model. The coexistence of moral hazard

9. Another well-known contribution on competition among asymmetrically informed investors is Broecker (1990), who assumes that investors receive private signals about the firm's profitability (but are still less well informed than the borrower) and compete in reimbursement rules for the borrower's business. See also Milgrom and Weber's classic paper (1982) on auctions with common values, and the large subsequent literature.

10. In Inderst and Müller (2005b), a borrower applies to a lender for a loan. The initial contract is drawn under symmetric information. The lender then acquires private, soft information about the quality of the borrower. Because the lender does not internalize the borrower's rent from being funded, the lender denies rationally, but inefficiently, credit for a range of signals. In another recent paper, Inderst and Müller (2005a) add collateral and show that this improves the efficiency of the lender's credit decision by flattening the borrower's repayment schedule.

11. There is a separate literature on the disclosure of proprietary information, arguing that private financing may make it possible to reveal information to an investor without revealing it to competitors (see Campbell 1979; Campbell and Kracaw 1980). In Bhattacharya and Ritter (1983), the firm chooses how much information to reveal; it attempts to reveal its true value to investors and does not reveal all the information that its competitors would like to learn.

12. Here we present the model in terms of a single borrower whose quality is unknown. Equivalently, the model represents a situation in which there are lots of entrepreneurs, a fraction α of which are high-quality ones, and in which investors are unable to tell borrowers apart in terms of quality.

with adverse selection is not necessary for most applications (for which one can therefore ignore private benefits, $B = 0$ in the notation of the book, and thereby remove the moral-hazard component) since adverse selection by itself creates an agency cost and a concomitant credit rationing, and triggers a number of interesting institutional responses. Ignoring moral hazard therefore simplifies the presentation. (In Application 6, however, we will add *ex post* moral hazard in the context of *ex ante* private information about the likelihood of a liquidity shock; in that application, moral hazard will generate a rent from continuation, and *ex post* credit rationing, and thereby create a cost of financing through short-maturity liabilities.) Note also that we assume that the entrepreneur has no cash on hand ($A = 0$), and so she cannot signal her trust in the project by investing her personal wealth into it. Cash on hand will play a key role in Application 8 below.

6.2.1 Market Breakdown and Cross-Subsidization

6.2.1.1 Symmetric Information

To set a benchmark, first consider financing when the investors know the project's prospects.

The good entrepreneur obtains financing. One optimal arrangement[13] for her is to secure the highest level of compensation, R_b^G in the case of success, consistent with investors' breaking even on average:

$$p(R - R_b^G) = I.$$

If $qR < I$, the bad borrower does not want to invest because, under symmetric information, she would receive the NPV, $qR - I < 0$ if she could secure funding. Besides, she cannot obtain financing anyway because the pledgeable income, qR, is smaller than the investors' outlay, I.

If $qR > I$, then the bad borrower receives funding and secures compensation R_b^B in the case of success, where

$$q(R - R_b^B) = I.$$

Clearly,

$$R_b^B < R_b^G.$$

6.2.1.2 Asymmetric Information

The symmetric-information outcome, however, is not robust to asymmetric information, as the bad borrower can, by mimicking the good borrower, derive utility qR_b^G that is greater than that (either 0 or qR_b^B) she obtains by revealing her type.[14]

Let us assume that the only feasible financial contracts are contracts that give the borrower a compensation $R_b \geqslant 0$ in the case of success and 0 in the case of failure. (The validity of this assumption will be discussed in the remark below on the optimality of contracts.) Such contracts necessarily pool the two types of borrower as each prefers receiving financing to not being funded, and conditional on being funded, prefers contracts with a higher compensation. The investors' profit for such a contract is therefore on average:

$$[\alpha p + (1 - \alpha)q](R - R_b) - I = m(R - R_b) - I.$$

No lending: $mR < I$. This case can arise only if the bad borrower is not creditworthy. It then arises whenever the probability that the borrower is a bad borrower is large enough, or

$$\alpha < \alpha^*,$$

where

$$\alpha^*(pR - I) + (1 - \alpha^*)(qR - I) = 0.$$

Because the borrower cannot receive a negative compensation ($R_b \geqslant 0$), investors lose money if they choose to finance the project. Accordingly they do not and the market breaks down.

The good borrower is therefore hurt by the suspicion that she might be a bad one. There is *underinvestment.*

13. Here there is some indeterminacy as to the way the entrepreneur is compensated: the contract can specify any reward $R_b \leqslant R_b^G$ in the case of success, together with, for example, a lump-sum payment (sign-up fee or advances) $T \geqslant 0$ such that investors break even:

$$p(R - R_b) = I + T.$$

Equivalently, the entrepreneur could receive no lump-sum payment up front and receive cash even in the case of failure.

Our choice of contract, in which the borrower receives nothing in the case of failure, will facilitate the comparison with the outcome under asymmetric information.

14. The same lack of incentive compatibility holds *a fortiori* for any of the contracts that are optimal for the good borrower under symmetric information (see the previous footnote), as the reader will check. As we will later observe, the bad borrower is least tempted to choose the good borrower's contract if the latter rewards the borrower only for a good outcome.

Lending: $mR \geqslant I$. This case corresponds either to the situation in which both types are creditworthy or to that in which the bad borrower is not creditworthy but $\alpha \geqslant \alpha^*$.[15]

The borrower's compensation R_b is then set so that investors break even *on average*:

$$m(R - R_b) = I.$$

This implies that, *ex post*, investors make money on the good type ($p(R - R_b) > I$) and lose money on the bad type ($q(R - R_b) < I$): there is *cross-subsidization*.

Note also that

$$R_b < R_b^G$$

(and $R_b > R_b^B$ if the bad borrower is creditworthy). The good borrower is still hurt by the presence of bad ones, although to a lesser extent than when the market breaks down. The good borrower must content herself with a lower compensation (i.e., a higher cost of capital) in the case of success than under symmetric information. Put differently, and interpreting the investors' share as a risky loan with nominal interest rate r such that $R - R_b = (1 + r)I$, then $r > r^G$, where r^G is the rate of interest that the good borrower could obtain under symmetric information: $R - R_b^G = (1 + r^G)I$.

When the bad borrower is not creditworthy, then the outcome is *overinvestment*, as was pointed out in particular by De Meza and Webb (1987), one of the early papers in this literature. Adverse selection (i.e., asymmetric information) reduces the quality of loans.

Remark (a measure of adverse selection). The condition

$$mR \geqslant I$$

can be rewritten as

$$\left[1 - (1 - \alpha)\left(\frac{p - q}{p}\right)\right] pR \geqslant I.$$

We can thus define an index of adverse selection:

$$\chi \equiv (1 - \alpha)\left(\frac{p - q}{p}\right).$$

In the absence of signaling possibility, the good borrowers' pledgeable income, pR, is discounted by the presence of bad borrowers. The discount is measured by the product of the probability of bad types, $1-\alpha$, times the likelihood ratio, $(p-q)/p$.[16] This discount is the counterpart of the agency cost that obtains under moral hazard (and is equal to the product of the private benefit B divided by the likelihood ratio $(p_H - p_L)/p_H$).[17]

Alternatively, we can measure the cost incurred by the good borrower due to asymmetric information. Instead of receiving the NPV,

$$pR - I,$$

attached to her type, she receives

$$pR_b = p\left(R - \frac{I}{m}\right)$$

or, after some manipulation,[18]

$$pR_b = (pR - I) - \frac{\chi}{1 - \chi} I.$$

Remark (optimality of contracts). Whether the market breaks down or not, a good borrower is hurt by the presence of bad borrowers and therefore would like to separate from bad borrowers if she could. Could she do better than demanding some compensation R_b in the case of success and 0 in the case of failure? Relatedly, could an investor make money by offering a more sophisticated contract to the entrepreneur? The answer to these questions (which are studied in Section 6.5) turns out to be "no" when both types are creditworthy. Intuitively, lending is then efficient and so contractual innovations, keeping investor profitability constant, just amount to redistributing wealth between the good and bad borrowers. A contract that rewards the borrower only in the case of success best reflects the good borrower's comparative advantage, as she is more likely

15. The former situation can be subsumed in the latter one by setting $\alpha^* = 0$.

16. The likelihood ratio can be defined by $(p - q)/p$, $(p - q)/q$, or p/q, indifferently. That $(1 - \alpha)$ enters the measure of adverse selection comes from the fact that good borrowers cannot be distinguished from bad ones in this section. As we will see in Section 6.3, the likelihood ratio, but not the prior α, plays a role in the characterization of a separating equilibrium (the prior plays a role, however, in determining whether the separating equilibrium is unique or dominated by a pooling outcome).

17. The agency cost in the moral-hazard case was expressed in absolute terms while it is here convenient to write it as a fraction of total income so as to let the likelihood ratio appear.

18. Note that this expression holds only when financing can be secured, i.e., when $(1 - \chi)pR \geqslant I$. Under this restriction one indeed checks that $pR_b \geqslant 0$.

to succeed than the bad one. It thereby minimizes the subsidizing of the bad borrower by the good one.

By contrast, when the bad borrower is not creditworthy, the pooling allocation implies overinvestment. It would be more efficient to give a lump-sum payment to bad borrowers to "go away" and accept not to invest; this policy, however, raises concerns about its feasibility (see Section 6.5).

6.2.2 Extensions and Applications

Application 1: Market Timing

Firms tend to issue shares when stock[19] prices are high.[20] As discussed in Section 2.5, there are several possible reasons for this. A commonly advanced one is that adverse selection becomes less relevant during booms.

To see this, let us assume that the probability of success is the sum of the firm's type (p or q, good or bad) and a publicly observable shift parameter $\tau \gtrless 0$ that indexes the firm's, the industry's, or the economy's publicly observable prospects: the probabilities of success are then $p + \tau$ and $q + \tau$ for the good and bad borrowers, respectively. The condition for financing becomes

$$[\alpha(p + \tau) + (1 - \alpha)(q + \tau)]R > I$$

or

$$(m + \tau)R > I.$$

Thus the better the market conditions (the larger τ is), the more likely it is that firms can obtain financing. During booms, the intrinsic value of the project becomes large relative to the lemons problem.[21] The reader will indeed check that the index χ of adverse selection is smaller when market conditions improve.

Application 2: Assets in Place, the Negative Stock Price Reaction, and the Going-Public Decision

Let us next suppose that the entrepreneur already owns a project that, without further investment, will succeed with probability p or q, yielding profit R. As before, the entrepreneur knows the probability of success while the investors put probability α on p and $(1 - \alpha)$ on q. Thus, in the absence of further information (and so the investors' expectation of the probability of success is m), the assets in place are undervalued (respectively, overvalued) if the true probability of success is p (respectively, q).

For computational simplicity, we will assume that the entrepreneur initially owns all shares. But nothing is altered if she owns only a fraction of the shares. By "stock price reaction upon the announcement of an equity issue," we mean the difference between the total value of shares (whoever owns them) before and after the announcement. This notion corresponds to the approach taken by event studies in empirical work.

An equity offering may be motivated by a profitable "deepening investment" (more generally, the key feature is that one cannot contract on the cash flow generated by this investment separately from that generated by assets in place: the incomes generated by the two are intertwined or fungible[22]). At cost I, the probability of success can be raised by an amount τ such that

$$\tau R > I.$$

That is, investing is efficient for both types of borrowers. Note that we assume for the moment that the increase in profitability is uniform across types: the probability of success becomes $p + \tau$ for a good borrower and $q + \tau$ for a bad one.

The entrepreneur, however, has no cash on hand. Accordingly, the full amount I must be raised from investors. The entrepreneur must therefore issue new shares, thereby reducing the fraction of shares she owns.

A key insight is that relinquishing shares to investors is relatively less costly to the borrower with overvalued assets in place (the bad borrower) than to

19. Note that we have not yet distinguished between risky debt and equity. See Application 3 below, though.

20. More generally, equity market timing is the practice of issuing shares at a high price and repurchasing them at a low price. Also, "market timing" sometimes refers to the attempt by borrowers to sell equity when it is overvalued. We here mean that borrowers issue equity during good times.

21. We derived this result in the case of a separable production function (additive in probabilities). More generally, an increase in the average probability of success facilitates financing.

Note also that the more general point is that credit rationing is alleviated during booms, whether it is due to adverse selection or moral hazard.

22. Otherwise, it would be optimal for the good borrower to engage in project finance so as to avoid having to cross-subsidize the bad one.

the borrower with undervalued assets in place (the good borrower). Thus, if the good borrower conducts an equity offering, so does the bad one.

Let us therefore investigate the possibility of an (efficient) pooling equilibrium. The entrepreneur must offer a stake R_1 in success to the investors such that

$$[\alpha(p+\tau)+(1-\alpha)(q+\tau)]R_1 = I \iff (m+\tau)R_1 = I,$$

where, as earlier, $m \equiv \alpha p + (1-\alpha)q$ is the prior mean probability of success. There exists a unique R_1, $0 < R_1 < R$, satisfying this condition.

The good borrower, though, can guarantee herself pR by not diluting her stake.[23] Thus, she is willing to issue new shares only if

$$(p+\tau)(R-R_1) \geqslant pR \iff \tau R \geqslant \frac{p+\tau}{m+\tau} I. \quad (6.1)$$

After some manipulation, condition (6.1) can be rewritten to show that the value of investment, $\tau R - I$, must exceed some strictly positive hurdle,

$$\tau R - I \geqslant \frac{\chi_\tau}{1-\chi_\tau} I,$$

where χ_τ is the post-investment index of adverse selection,

$$\chi_\tau = \frac{(1-\alpha)[(p+\tau)-(q+\tau)]}{p+\tau} = \frac{(1-\alpha)(p-q)}{p+\tau}$$

(so $\chi_0 = \chi$).

Condition (6.1) is always satisfied if there is little adverse selection (χ_τ is close to 0) or if the deepening investment is very profitable ($\tau R/I$ is large).

We are thus led to consider two situations:

Pooling equilibrium. If condition (6.1) holds, then both types conduct an equity offering.[24] If the accrual of this deepening investment is anticipated,[25] the total value of shares *before and after* the seasoned equity offering is

$$(m+\tau)R - I.$$

There is no stock price reaction to the offering, which is perfectly anticipated and uninformative.

Separating equilibrium. More interestingly, suppose that condition (6.1) is violated. The good borrower then does not raise funds. The bad borrower still does, but under market conditions that are not as favorable as in a pooling equilibrium. Because the investors know that the equity offering reveals overvalued assets, they demand a higher stake $R_1^B > R_1$ such that

$$(q+\tau)R_1^B = I.$$

The good borrower does not want to raise funds because

$$(p+\tau)(R-R_1^B) < pR \iff \tau R < \frac{p+\tau}{q+\tau} I, \quad (6.2)$$

which holds if condition (6.1) is violated.

The announcement of a seasoned equity offering then leads to a *negative stock price reaction*. The preannouncement total value of shares is[26]

$$V_0 = \alpha[pR] + (1-\alpha)[(q+\tau)R - I].$$

After the announcement, it becomes

$$V_1 = (q+\tau)R - I.$$

Hence,

$$V_0 > V_1 \iff pR > (q+\tau)R - I.$$

But we know that

$$pR > (p+\tau)\left(R - \frac{I}{q+\tau}\right),$$

and so *a fortiori*

$$V_0 > V_1.$$

23. That this "reservation utility" depends on the borrower's type is the essential difference with the barebones model. Here, in the jargon of incentive theory, "reservation utilities are type-contingent." See Jullien (2000) for the state-of-the-art treatment of adverse selection with type-contingent reservation utilities.

24. This pooling equilibrium is not unique whenever

$$(p+\tau)I/(m+\tau) \leqslant \tau R \leqslant (p+\tau)I/(q+\tau);$$

indeed, if investors believe that an equity offering comes from a bad borrower and $\tau R \leqslant (p+\tau)I/(q+\tau)$, then the good type indeed prefers not to raise funds. However, the pooling equilibrium is the *Pareto-dominant* equilibrium (it is the best equilibrium for both the good and the bad borrower), and so we will focus on it.

Note also that if condition (6.1) holds, then the bad borrower definitely prefers to raise funds since the analogous condition for her is

$$\tau R \geqslant \frac{q+\tau}{m+\tau} I,$$

which is always satisfied.

25. Otherwise, the news of the existence of an investment opportunity by itself raises the value of shares.

26. As in the previous footnote, note that we assume that the investment opportunity is perfectly anticipated by the capital market. Otherwise, the issue of new securities could convey good news about the firm's opportunity set and the concomitant boost in share price might dominate the effect unveiled here.

Combining both cases, we see that the pooling equilibrium (condition (6.1)) is more likely to obtain if the project being financed is more valuable (τ increases or I decreases). We therefore conclude that price reaction on average should be less negative in booms.

Furthermore, and again combining the two cases, *the negative price reaction is smaller when the volume of equity offering, as measured by the amount collected in the offering,*[27] *is large.* Actually, in our example, the price reaction is 0 when both types issue shares. More generally, with a continuum of types, the price reaction is always negative, as long as some types refrain from issuing equity (see Exercise 6.5).

Remark (correlation between value of assets in place and profitability of investment). The analysis can be straightforwardly extended to allow for increases in the probability of success to be positively or negatively correlated with the value of assets in place. Let τ_G and τ_B denote the increases in the probability of success for the good and bad types, respectively. Investors know the values τ_G and τ_B, but do not know which obtains (otherwise they would also know whether the borrower is good or bad if $\tau_G \neq \tau_B$). Assume $p + \tau_G > q + \tau_B$ and so who is a "good borrower" does not vary with investment. The average increase τ is equal to $\alpha\tau_G + (1-\alpha)\tau_B$. The condition for both types conducting a seasoned equity offering is now

$$(p + \tau_G)\left(R - \frac{I}{m + \tau}\right) \geqslant pR.$$

An increase in correlation corresponds to an increase in τ_G keeping τ constant. Thus, the good borrower is more likely to issue shares, the higher the correlation, as might have been expected.

Remark (going-public decision). Although too simplistic, this model sheds some light on the going-public decision. Think about the firm's resorting to the capital market as a process through which an entrepreneur (or more generally an entrepreneur and a close set of well-informed financiers: venture capitalist, friends, or family holding an equity-like stake) decides to tap further financing and dilute her own stake in order to expand. Then the entrepreneur will tend to remain private when optimistic about the firm's prospects. Of course, the model abstracts from many interesting issues (studied later in the book) associated with the going-public process, such as the certification by an investment banker, the acceptance of strong disclosure requirements, and possibly the loss of control over the firm. But its basic point—that entrepreneurs who feel that assets in place are undervalued by the market tend to forgo profitable investment opportunities and to remain private—is a robust one (see Chemmanur and Fulghieri 1999).

Application 3: Pecking-Order Hypothesis

An important theme in corporate finance is that adverse selection calls for the issuance of debt claims. As we discussed in the introduction, Myers (1984) and Myers and Majluf (1984) have formulated a pecking-order hypothesis that places debt as the preferred source of external financing. Recall that these authors argue that sources of financing can be ranked according to their information intensity, from low to high information intensity: (1) internal finance (entrepreneur's cash, retained earnings), (2) debt, (3) junior debt, convertibles, and (4) equity.

The pecking-order hypothesis is based on the investors' concern about the value of the claim they acquire. It is clear, for example, that *default-free debt* creates no concern for investors as to the value of their claim. We first provide conditions under which debt is indeed the preferred source of financing under asymmetric information about the firm's prospects,[28] and then discuss the robustness of the pecking-order hypothesis.

As discussed in Chapter 3, there is no distinction between debt and equity claims when the profit is either R or 0. Let us therefore add a salvage value of the assets R^F: the profit in the case of failure is $R^F > 0$ and that in the case of success is $R^S = R^F + R$, where R still denotes the profit increment. Except for the introduction of a salvage value, the model is

27. This amount is I in the pooling equilibrium and $(1-\alpha)I$ on average in the separating one.

28. We know that under moral hazard and risk neutrality, the entrepreneur should offer a debt contract to investors so as to mitigate the moral-hazard problem (see Sections 3.4 and 3.5). We show that the same point holds under adverse selection, even when there is no moral hazard.

otherwise that of Section 6.2.1: there are no assets in place. The investment cost I must be entirely defrayed by the investors. The probability of success is p for a good borrower (probability α) and q for a bad one (probability $1-\alpha$). The prior mean probability of success is $m \equiv \alpha p + (1-\alpha)q$.

Let us assume that

$$mR^S + (1-m)R^F > I$$

and so there is enough pledgeable income to secure funding even when the bad borrower pools with the good one.

Let $\{R_b^S, R_b^F\}$ denote the (nonnegative) rewards of the borrower in the cases of success and failure. Assuming that the borrower receives funding, the investors' breakeven condition is

$$m(R^S - R_b^S) + (1-m)(R^F - R_b^F) \geqslant I.$$

The good borrower maximizes her expected payoff

$$pR_b^S + (1-p)R_b^F$$

subject to the breakeven constraint. At the optimum, the investors' breakeven condition is satisfied with equality. It can be rewritten as

$$[p - (1-\alpha)(p-q)](R^S - R_b^S) + [1 - p + (1-\alpha)(p-q)](R^F - R_b^F) = I.$$

The good borrower's utility is then equal to

$$pR_b^S + (1-p)R_b^F = [pR^S + (1-p)R^F - I] - (1-\alpha)(p-q)[(R^S - R_b^S) - (R^F - R_b^F)].$$

On the right-hand side of this equality, the first term in brackets represents the NPV of the good borrower, namely, what she would receive under symmetric information. The second term as usual refers to the adverse-selection discount.

The good borrower wants to minimize this discount while satisfying the investors' breakeven constraint.[29] Because the discount increases with R_b^F and decreases with R_b^S, the good borrower sets

$$R_b^F = 0.$$

Then, R_b^S is determined by the investors' breakeven constraint:

$$m(R^S - R_b^S) + (1-m)R^F = I.$$

To sum up this analysis, the borrower commits the entire salvage value as safe debt issued to investors. The borrower further issues risky equity with stake $R^S - R_b^S$ in the case of success (and 0 in the case of failure) so as to make up for the shortfall in pledgeable income:

$$m(R^S - R_b^S) = I - R^F.$$

Thus, the firm first issues safe debt with a debt obligation D given by

$$D = R^F,$$

and, second, supplements the capital thus raised through an equity issue entitling shareholders to a fraction R_1/R of profits in excess of R^F, where

$$mR_1 = I - D.$$

Note that the borrower must issue more equity, the more acute the adverse-selection problem (the lower m is) or the higher the investment cost.

Intuitively, the borrower starts by issuing the claim that is least exposed to adverse selection, here the safe-debt claim. Doing so allows the good borrower to *minimize the cross-subsidization* with

29. Alternatively, we can use Lagrangian techniques. Let μ denote the shadow price of the investors' breakeven constraint, and $\mathcal{L}$ the Lagrangian of the program:

$$\mathcal{L} \equiv pR_b^S + (1-p)R_b^F + \mu[m(R^S - R_b^S) + (1-m)(R^F - R_b^F) - I].$$

Then

$$\frac{\partial \mathcal{L}}{\partial R_b^S} = p - \mu m \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial R_b^F} = (1-p) - \mu(1-m).$$

Because

$$\frac{p}{m} > 1 > \frac{1-p}{1-m},$$

necessarily,

$$\frac{\partial \mathcal{L}}{\partial R_b^F} \geqslant 0 \quad \text{implies that} \quad \frac{\partial \mathcal{L}}{\partial R_b^S} > 0,$$

and, conversely,

$$\frac{\partial \mathcal{L}}{\partial R_b^S} \leqslant 0 \quad \text{implies that} \quad \frac{\partial \mathcal{L}}{\partial R_b^F} < 0.$$

Thus we are led to consider two cases (the second is studied only for the sake of completeness): (i) $\partial \mathcal{L}/\partial R_b^F < 0$ (the most interesting case). Then $R_b^F = 0$. (ii) $\partial \mathcal{L}/\partial R_b^F \geqslant 0$. In this case, $\partial \mathcal{L}/\partial R_b^S > 0$. And so, if there is no bound on R_b^S, R_b^S must be increased as much as possible (and R_b^F must decrease accordingly to keep the breakeven constraint satisfied) until $R_b^F = 0$, in which case we are back to case (i). But it is probably more reasonable to add the constraint that $R_b^S \leqslant R$. Otherwise, the borrower could in the case of failure borrow R from a third party and reimburse this third party from the reward, R_b^S, received from the apparent "success." Thus, case (ii) corresponds to the uninteresting case in which $I < R^F$, that is, the investment is "self-financing." In this case, the entrepreneur issues only safe debt. The pecking order still applies, although in a rather trivial way.

the bad borrower. The more sensitive the investors' claim to the borrower's private information, the higher the return that the investors demand from a good borrower to make up for the money they lose on the bad one. As we will observe in Section 6.3, *this principle of issuing low-information-intensity claims carries over to situations in which the good borrower has the means, and not only the incentive, to separate from the bad one.*

How robust is the debt bias to the specification of the income space? Section 6.6 considers the case of a *continuum* of possible incomes. It builds on Innes (1990, see Section 3.5) and DeMarzo and Duffie (1999).[30] It derives conditions (basically, the conditions obtained by Innes in the moral-hazard, no-adverse-selection setup)[31] under which a good borrower separates from a bad one by offering a standard debt contract.

Are low-information-intensity claims always debt claims? The debt bias principle must be qualified in four important respects:

Insurance. First, forces other than signaling may alter the nature of the securities issued. This point is well illustrated by the Leland–Pyle–Rothschild–Stiglitz model of diversification by a risk-averse entrepreneur, reviewed in Application 8. We will derive conditions under which the bad borrower obtains full insurance, and even the good borrower is partially insured. Their contracts cannot therefore be viewed as insider equity contracts.

Exit strategy. Second, and more interestingly, the issue may not only serve the "*ex post*" goal of obtaining the best possible terms for the issuer at the issuing date. The issue may also reflect an "*ex ante*" objective of providing the issuer with good incentives to create value before the issuing date. As we alluded to in Section 4.4 and will emphasize in Chapter 9, it may then be optimal for the issuer to commit to float information-intensive securities because such securities induce value measurement by the market and allow insiders to be compensated for their past performance; that is, the floating of information-intensive securities enables partial or full exit strategies.

Nature of informational asymmetry. Third, what constitutes a low-information-intensity claim depends on the form of informational asymmetry. We have seen that, when information relates to the probability of success, signaling tends to result in the issuance of a standard debt contract.

Suppose that the asymmetry of information is also related to the *riskiness* of the distribution, and that the good borrower has a less risky distribution than the bad one. Then it is clear that a debt contract may no longer reflect the good borrower's comparative advantage; for, the debt contract provides the bad borrower with a substantial rent when the income is very high.

To illustrate this point in a trivial manner, suppose that there are three possible levels of income: low, middle, and high. A good type always obtains the middle income. A bad type obtains either the low or the high income. The firm's expected income is higher for the good type. The good type then signals herself by issuing a claim that distributes everything to investors when the firm's income is either low or high, but less than the firm's income when the firm obtains the middle income. Such a claim, which may not violate the monotonicity of the investors' claim with the firm's income, does not resemble a debt claim because it distributes the firm's income to investors when income is high.

A more sophisticated illustration of the principle that low-information-intensity securities need not be debt claims is Stein's (1992) rationalization of convertible bonds as reducing the investors' exposure to low-profitability, high-risk borrowers when the former observe signals about the borrower's type after purchasing the securities.

Rent extraction. We have assumed that the entrepreneur or manager faces a competitive financial market. Investors cannot then attempt to extract the good borrower's rent. The pecking-order hypothesis

30. DeMarzo and Duffie consider a "hidden-knowledge" model rather than an "adverse-selection" one (that is, the issuer learns her information *after* the contract is signed) and look at a variable investment scale. They also make an assumption that is weaker than the monotone likelihood ratio property assumed in the appendix.

Other papers that argue that debt contracts are a natural response to adverse selection include Allen and Gale (1992) and Nachman and Noe (1994), which both use Banks and Sobel's (1987) "divinity refinement" to select pooling at a debt contract. For more on security design under adverse selection, see, in particular, Boot and Thakor (1993) and Demange and Laroque (1995).

31. For readers who have covered Section 3.5, the optimality of a debt claim for investors depends on the assumption that the investors' claim is monotonic.

actually states that the good borrower maximizes this rent by issuing low-information-intensity securities, thereby minimizing the cross-subsidization of the bad borrower.

Suppose in contrast that investors have some market power. For example, they may have control over the managerial position; or a venture capitalist or a large investor might have a smaller informational handicap vis-à-vis the borrower than other investors. Then the investors will want to extract some of the good borrower's rent. Rent extraction is best performed when the borrower's stake is least sensitive to her private information—the case of a fixed compensation[32]—that is, when the investors' stake (which is complementary to that of the borrower) is derived from *high*-information-intensity securities!

Of course, providing the borrower with a fixed stake, namely, a wage that is not contingent on performance is not desirable when the borrower must exert effort. There is then an incentive-rent extraction tradeoff (see Laffont and Tirole 1986). Furthermore, there is now scope for separation: confident borrowers will tend to select high-powered incentive schemes along the lines of the pecking-order hypothesis, while less confident ones will go for safer compensation (higher fixed wage, lower volume of stock options). To use an analogy, regulated utilities that are confident in their ability to reduce cost tend to choose price caps or sliding-scale plans rather than low-powered cost-of-service regulation.[33]

6.3 Dissipative Signals

Section 6.2 focused on environments in which good borrowers could not separate from bad ones (except by forgoing profitable investment opportunities, when there are assets in place). In practice, borrowers often try to convey the quality of the securities they issue through "dissipative signals"; these dissipative signals are the counterpart in an adverse-selection context of the "value-decreasing concessions" in the moral-hazard context. This section describes some frequently used dissipative signals, without any attempt at exhaustivity.

Application 4 considers the reduction in the asymmetry of information between borrower and lenders through the costly certification by an informed investor or other party or through a disclosure policy. Applications 5–9 then analyze how the good borrower may try to signal her residual private information (that is, the information that is still private after certification and disclosure) through financial structure choices. The key theme in those applications is that, in order to separate, the good borrower must offer contractual terms that do not appeal to a bad one and allow lenders to break when they know that they are facing a good borrower. This will lead us to the general principle that, as in the pecking-order theory, the response to the lemons problem is the issuance of low-information-intensity securities, i.e., securities for which investors are not "too exposed" to errors in their assessment of the borrower's type.[34]

Application 4: Certification

As we have seen, adverse selection in general leads to cross-subsidization or market breakdown, which are costly to good borrowers or issuers. Therefore, good issuers have an incentive to try to mitigate the investors' informational disadvantage. The asymmetry of information can be reduced through disclosure to investors of information about the firm's prospects. Another form of disclosure bears on past repayments (see Exercise 6.7, based on Padilla and Pagano (1997), on information sharing among lenders). But, while disclosure is not to be neglected, it is most effective for "hard information," that is, information that can be verified by the investors once disclosed by the issuer.[35] Disclosure is a less effective

32. Using the notation of Application 3, the borrower's utility is $\theta R_b^S + (1-\theta)R_b^F$, where $\theta \in \{p, q\}$. The derivative of this utility with respect to θ is $R_b^S - R_b^F$. And so the utility (rent) grows most slowly (actually not at all) with the borrower's type when $R_b^S = R_b^F$.

33. Yermack (1997) analyzes stock option awards to CEOs of large U.S. corporations between 1992 and 1994. He finds that the average cumulative abnormal stock return in the 50 days following the award is slightly above 2% (the award is not disclosed until several months after the fiscal year ends, so the market cannot react to the news of a more incentivized CEO). Yermack's interpretation is that managers who receive private information about impending improvements in corporate performance may influence compensation committees towards more performance-based compensation. The story is thus a bargaining analog of the compensation-menu theory just alluded to.

34. This definition of a low-information-intensity security is vague. The supplementary section gives a general and rigorous definition. Besides, what constitutes a low-information-intensity security will become clear in specific applications.

35. See Grossman (1980), Grossman and Hart (1980), Milgrom (1981), and Milgrom and Roberts (1986) for the theory of disclosure.

means of reducing informational asymmetries if the information is "soft," that is, cannot be verified by the investors.

Lending by an informed party (whether a bank, a peer, or a trade creditor) is a signal that the informed party is confident about the possibility of repayment. Such "informed lending" is therefore likely to bring along less well-informed investors.[36] Monitoring will be studied in depth in a moral-hazard context in Part III of this book. Let us here mention that similar ideas have been developed in an adverse-selection context; for example, Ghatak and Kali (2001) analyze "positive associative matching" in a world of joint liability (see also Section 4.5 of this book); when entrepreneurs are made liable for the loans issued to other entrepreneurs through cross guarantees, good borrowers have a strong incentive to associate themselves with a safe partner.

More generally, issuers can reduce informational asymmetries by borrowing from well-informed investors or by asking them to certify the quality of the issue. There is a large variety of certifying agents: underwriters,[37] rating agencies, auditors, venture capitalists. Of course, it must be the case that the certifying agent has an incentive to become well-informed about the firm's prospects and to take actions that properly convey their information to the prospective investors. The "actions" can be a rating, a report, or a subscription to the issue (or, in the case of a venture capitalist, the action of keeping a non-negligible stake in the firm).[38] And, in all cases, reputation helps keep the certifier honest (indeed, reputation is the only such incentive for a rating agency, which does not take a stake in the firm). We refer to Baron (1982), Raviv (1989), and Chapter 9 for a discussion of monitors' incentives. Here we content ourselves with a simple analysis in which the certification is modeled in reduced form as the purchase, at cost $c > 0$, of a signal that perfectly reveals the borrower's type.

Recall that in the privately-known-prospects model (without assets in place) and in the absence of certification, funding, if any, implies an entrepreneurial reward R_b in the case of success given by

$$m(R - R_b) = I,$$

where $m = \alpha p + (1 - \alpha)q$ is the prior mean probability of success. Let us assume that $mR > I$ and so funding is indeed feasible; the good borrower is then concerned by cross-subsidization.[39]

Suppose that at cost c, the borrower can have access to a reputable certifier who then provides accurate evidence regarding the quality of the project; that is, other investors will then know whether the borrower's probability of success is p or q.[40] (Note that the borrower has no cash to pay the certifier up front. One can imagine that the borrower gives the certifier shares in the firm; these shares can further ensure that the certifier will incur the monitoring cost (see Chapter 9).)

A bad borrower obviously has no incentive to pay a cost c to reveal to the capital market that the probability of success is only q. By resorting to a certifier,

A key insight of this literature is that hard information is, under weak conditions, disclosed if it is known to be held by the issuer. The intuition is that a good issuer benefits from disclosing and thus discloses. An average issuer must then disclose not to be pooled with bad ones. And so even a bad issuer can disclose. A limitation on disclosure occurs when the issuer may or may not have the hard information. An issuer with bad information may then claim not to have any information (see, for example, Tirole (1986) and Okuno-Fujiwara et al. (1990) for models with this feature).

These models assume that information once disclosed is assimilated by investors. Fishman and Hagerty (2003) study an interesting model of disclosure in which a fraction of investors do assimilate the disclosed information while the remaining fraction only observe that there has been disclosure. They show that there may be an equilibrium with no voluntary disclosure, that investors, but not issuers, should support mandatory disclosure, and that mandatory disclosure rules are more likely with regards to information that is difficult to understand.

Finally, Dewatripont and Tirole (2005) study the efficacy of communication that is neither hard nor soft in that its understanding by the receiver depends on the sender's and the receiver's efforts to communicate, which in equilibrium depend on the congruence of their objectives.

36. See, for example, Rochet and Tirole (1996a,b) for investigations of this idea in the context of interbank loans.

37. In the United States underwriters are employed in over 80% of the offerings (Smith 1977). In contrast, according to Marsh (1979), 99% of the new equity in the United Kingdom in the mid 1970s was raised through rights offers (in which current shareholders receive a right from the firm giving them an option to purchase additional shares at a prespecified price).

38. There is, for example, a large empirical literature on certification in initial public offerings. See, for example, Megginson and Weiss (1991) and the references therein.

39. What follows holds *a fortiori* in the "no lending" case. In this case, the good borrower receives 0 in the absence of certification. Hence, provided that certification is feasible (i.e., $pR \geqslant I + c$), the good borrower will be certified.

40. Exercise 6.6 allows for noisy signals about borrower quality.

a good borrower can obtain compensation $\hat{R}_b^G$ in the case of success given by

$$p(R - \hat{R}_b^G) = I + c.$$

The good borrower prefers to resort to a certifier if and only if[41]

$$\hat{R}_b^G > R_b \quad \Longleftrightarrow \quad R - \frac{I+c}{p} > R - \frac{I}{m},$$

or, after some manipulation,

$$\frac{c}{I+c} < (1-\alpha)\left(\frac{p-q}{p}\right).$$

This latter condition compares the certification cost, c, expressed as a fraction of the amount of funds to be raised, $I+c$, with our measure of adverse selection χ, which, recall, is equal to the probability of a bad type, $1-\alpha$, times the likelihood ratio, $(p-q)/p$.

Chemmanur and Fulghieri (1994) argue that the cost of diversification "c" may be the lack of diversification of the certifier when the latter, unlike here, is risk averse. They consider a firm's decision to raise external finance either by placing shares privately with a risk-averse large investor such as a venture capitalist or selling shares to a wider constituency, for example, through an IPO, assuming that information acquisition is needed to raise funds. The issuer then trades off the risk premium demanded by the large investor, and the duplication of information under decentralized monitoring in a wider capital market.[42]

Application 5: Costly Collateral Pledging

This section studies the possibility of signaling by pledging collateral (Besanko and Thakor 1987; Bester 1985, 1987; Chan and Kanatas 1985).[43] It builds on the idea, already exploited in Chapter 4, that collateral is valued less highly by the lenders than by the borrower. It shows how a borrower may want to pledge collateral, even though she would not need to do so if information were symmetric. To give the gist of the argument in the simplest possible setting, we extend the privately-known-prospects model as follows: while the borrower still has no cash on hand (in the notation of previous chapters, $A = 0$), she has (a sizeable amount of) assets that can be pledged to investors. That such assets are more valuable to the borrower than to the investors (see Section 4.3 for a fuller discussion), is formalized in the usual way: a transfer of assets valued $C \geqslant 0$ by the borrower has value βC, $0 \leqslant \beta < 1$, for the investors.

Assumption 6.1. *Under symmetric information even the bad borrower does not need to pledge collateral to receive funding:*

$$0 < \tilde{V} \equiv qR - I < V \equiv pR - I.$$

Symmetric information. If the lenders knew the borrower's prospects, the borrower's utility would be equal to the project's NPV, V for the good borrower and $\tilde{V}$ for the bad one, since the project has positive NPV and the entire income is pledgeable (see Section 3.2). To obtain utility V under symmetric information, the good borrower would demand a reward R_b^G in the case of success such that lenders break even when the probability of success is p:[44]

$$p(R - R_b^G) = I.$$

Indeed, her utility would then be equal to

$$pR_b^G = pR - I = V.$$

41. To be more precise, multiple equilibria coexist over a range of parameters, namely,

$$(1-\alpha)\left(\frac{p-q}{p}\right) \leqslant \frac{c}{I+c} \leqslant \frac{p-q}{p}.$$

Then, "no certification" and "certification of the good borrower only" are both equilibria (there also exists a third equilibrium, in which the good borrower randomizes between being certified and not being certified). The equilibrium is unique only if we focus on *Pareto-dominant* equilibria. In the range with multiple equilibria, both types are better off if the good borrower does not get certified (the "no certification" equilibrium) as the lack of certification then carries no stigma.

42. Lerner and Tirole (2005) analyze forum shopping, that is, the choice of congruence between the certifier on the one hand and the certified agent (here, the issuer) and the buyers (here, the investors) on the other. In the financial context of investment banking, relationship banking, venture capital, or ratings, the congruence is determined by the financial stake, if any, of the certifier in the issuer and by the certifier's willingness to attract future issuers' business. In the basic model, the issuer has no private information about the quality of issued securities; the issuer chooses a level of congruence as well as concessions made to investors (for example, price, collateral pledging, or control rights) and the certifier studies the quality. Issuers with *a priori* more attractive offerings choose more complacent certifiers and make fewer concessions. When the issuer has private information (that is correlated with the certifier's future assessment), then in a separating equilibrium, confident issuers (the "good borrowers") select tougher (less complacent) certifiers than under symmetric information, and also tougher ones than less confident issuers.

43. See Coco (2000) for a survey of the use of collateral.

44. Again, this contract is not uniquely optimal. Any compensation scheme that lets the investors break even and does not give a negative income to the borrower in any state of nature is optimal.

the case of failure, if she wants to separate from the bad one.[48]

Second, we have not yet investigated uniqueness. There might exist other separating, pooling, or hybrid equilibria. Section 6.7 shows that the allocation $\{R_b^*, C^*\}$ for the good type and $\{R_b^B, 0\}$ for the bad type is the unique (perfect Bayesian) outcome when the capital market's prior belief that the borrower is good is lower than some threshold, that is, if and only if[49]

$$\alpha \leqslant \alpha^* \quad \text{for some } \alpha^*,\ 0 < \alpha^* < 1.$$

Remark (signaling through weak entrenchment). As was shown in Section 4.3.6, "posting one's job as collateral" is formally akin to posting more familiar forms of collateral. Assume that the manager has private information about her quality rather than about the quality of the current project. A good manager would like to convey her information to investors. Because a good manager is more likely than a bad one to deliver a high performance and see her appointment renewed, she can use a low degree of entrenchment, in the sense of a low protection against managerial turnover, to signal her quality. In practice, the composition of the board of directors and the design of takeover defenses affect the ease with which shareholders can remove existing management. Furthermore, managerial turnover is (both theoretically and empirically) associated with bad news about firm performance. In this context, a good manager, who is less likely to fail, bears a lower cost from jeopardizing her job in the case of failure than a bad one. Thus, weak protection against managerial turnover is an effective signaling device.

The previous analysis of collateral pledging showed that a good borrower both demands a higher reward R_b in the case of success and posts a higher level of collateral in the case of failure. Relabeling the variables, the analysis thus also predicts a *negative covariation between managerial equity and job protection*: a confident manager will opt both for low job protection, as we just argued, and high-powered incentives (i.e., high sensitivity of compensation to performance).[50] This prediction seems to be supported empirically; in particular, Subramanian et al. (2002) find that managers with the steeper incentives are also more likely to be fired after a poor performance.

Application 6: Short-Term Maturities

Chapter 5 showed that firms that generate too little cash flow to meet their liquidity needs do not want to adopt a wait-and-see attitude but rather should secure resources early on in order not to face credit rationing at intermediate stages. This section shows that in a situation of asymmetric information about the firm's prospects, a firm may want to signal its creditworthiness by securing less resources (liquidity) for the future than would be efficient under symmetric information. In essence, a good borrower can convey that she is confident about the firm's prospects and that she is not afraid of going back to the capital market at an intermediate stage.

Let us consider the following variant of the model of debt maturities set up in Section 5.2. At date 0 the entrepreneur has a project of fixed size I, has wealth A, and must borrow $I - A$. At date 1, the investment yields a deterministic and verifiable short-term (date-1) profit $r > 0$. With probability λ, continuation requires reinvesting ρ (the liquidation value

48. Technically, the separating allocation $\{R_b^*, C^*\}$ for the good type and $\{R_b^B, 0\}$ for the bad type is the "low-information-intensity optimum," that is, the allocation that maximizes the good borrower's utility subject to the investors' breakeven condition (or, more generally, subject to the capital market not losing money on any type) and to the bad borrower not receiving a rent.

49. Intuitively, the only way for the good type to obtain a higher utility than that of the separating allocation is to relax the mimicking constraint by letting the bad borrower obtain more than $\tilde{V}$ when mimicking. This implies, however, that the investors lose money on the bad borrower (and thus the bad borrower must pool with the good one). The cross-subsidization is, however, costly to the good borrower, as the profit made by investors on the good borrower must offset their loss on the bad borrower times the ratio $(1 - \alpha)/\alpha$ of bad to good borrowers.

Note furthermore that the good borrower can guarantee herself the separating payoff pR_b^*. (The following reasoning paraphrases that in the supplementary section for the reader who will have skipped that section.) It suffices that she offers a pair of options $\{R_b^*, C^*\}$ and $\{R_b^B, 0\}$, from which she will choose after the investors agree to finance the project. The investors are guaranteed to break even regardless of the borrower's type, since the good borrower will choose $\{R_b^*, C^*\}$ from (6.5), and the bad borrower will choose $\{R_b^B, 0\}$ from the program above. On the other hand, if α is high, it becomes optimal for the good borrower to pool with the bad one: see Section 6.7 for a description as well as for a computation of the best equilibrium for the good type.

50. This assumes that the manager does not have so much cash on hand that her number of shares allows her to control the board of directors. If this were the case (e.g., as in family firms), then high stakes would also be associated with strong entrenchment.

is equal to 0); with probability $1-\lambda$, continuation requires no reinvestment.[51]

In the case of continuation, the firm at date 2 yields R in the case of success and 0 in the case of failure. The probability of success is p_H in the case of good behavior and $p_L = p_H - \Delta p$ in the case of misbehavior (which yields private benefit B). Moral hazard is introduced in order to create an entrepreneurial rent from continuation, or, equivalently, a cost for the entrepreneur associated with early termination. Put differently, moral hazard introduces a friction in the date-1 refinancing market. Let

$$\rho_1 \equiv p_H R \quad \text{and} \quad \rho_0 \equiv p_H\left(R - \frac{B}{\Delta p}\right)$$

denote the continuation NPV and pledgeable income, respectively.

Assume that

$$\min\{\rho_1, r\} > \rho > \rho_0.$$

The left inequality states that, viewed at date 1, continuation is always a positive-NPV proposition ($\rho_1 > \rho$) and that the short-term income suffices to meet the liquidity shock ($r > \rho$). The right inequality implies that in the absence of retentions (date-1 income that is not redistributed to investors) or credit line, the borrower cannot meet the liquidity shock by returning to the capital market ($\rho > \rho_0$). Finally, assume that the project cannot be financed in the absence of a positive probability of continuation,

$$I > A + r,$$

and that the project has a positive NPV,

$$r + \rho_1 - \lambda\rho > I.$$

Figure 6.1 summarizes the timing.

Symmetric information. We will consider an asymmetry of information about the probability λ of a liquidity shock. But suppose, first, that the entrepreneur and the investors are symmetrically informed, and that

$$I - A \leqslant r + \rho_0 - \lambda\rho,$$

which implies that investors are willing to finance the project even with certain continuation.

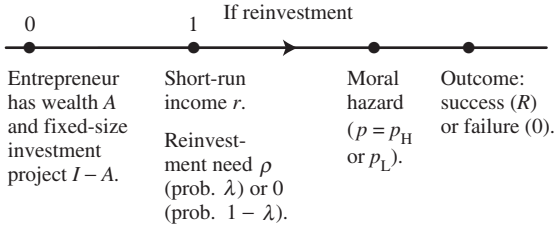


Figure 6.1

Let us show that, under symmetric information, the contract between the investors and the entrepreneur allows the latter to always bring the project to completion, and that this optimum can be implemented by a level of short-term debt

$$d \leqslant r - (\rho - \rho_0),$$

a reinvestment of remaining earnings, if any, in bonds (yielding a zero rate of interest at date 2), a reward R_b for the entrepreneur in the case of success, and the remaining income going to the investors at date 2.

Letting x denote the probability of continuation in the case of a shock, the NPV is

$$U_b = r + [1 - \lambda + \lambda x]\rho_1 - \lambda x\rho - I.$$

Hence, $x = 1$ is optimal. The condition

$$d \leqslant r - (\rho - \rho_0)$$

allows continuation even in the case of a liquidity shock: the borrower can use retentions $(r-d)$ together with what can be raised in the capital market (ρ_0) to meet shock ρ.

Asymmetric information. Assume now that the investors are imperfectly informed about the probability of shock. This probability is

$$\begin{aligned} \lambda &\quad \text{with probability } \alpha, \\ \tilde{\lambda} > \lambda &\quad \text{with probability } 1-\alpha. \end{aligned}$$

The entrepreneur knows which obtains. Assume that even the bad borrower (whose probability of a shock is $\tilde{\lambda}$) can continue with probability 1 under symmetric information:

$$I - A \leqslant r + \rho_0 - \tilde{\lambda}\rho.$$

Thus the good borrower is concerned solely by cross-subsidies to the bad one.

51. As usual, the entrepreneur is risk neutral and protected by limited liability; and the investors are risk neutral and demand a 0 rate of return.

We focus (without loss of generality) on contracts specifying a short-term debt $d \in [0, r]$ at date 1, a reward $R_b^+ \geqslant B/\Delta p$ in the case of no shock and success, and a reward $R_b^- = B/\Delta p$ in the case of a shock, continuation, and success (R_b^+ and R_b^- are the only incomes received by the entrepreneur, who receives nothing in the case of failure or early termination). Intuitively, large rewards R_b^- (i.e., in excess of the incentive payment $B/\Delta p$) in the case of a shock, continuation, and success are relatively more attractive to the bad borrower and so will not be used by the good borrower, who has a relative preference for being rewarded more in the absence of shock. The rationale for focusing on such contracts as well as the equilibrium analysis are provided in Section 6.8. The main predictions of the model are as follows:

(i) In a separating equilibrium the bad borrower gets her symmetric-information allocation and therefore continues with probability $x = 1$. By contrast, the good borrower uses a suboptimally low probability of continuation in order to separate from the bad borrower: $x < 1$. Liquidation is as costly to her as to a bad borrower when a shock occurs but is relatively less costly overall, as the shock occurs less often.

(ii) The good borrower grants herself a higher reward R_b^+ in the absence of a shock and in the case of success than under symmetric information: because she reduces her liquidity hoarding relative to the symmetric-information case, investors are willing to increase her compensation. But, as usual, the good borrower is worse off than under symmetric information; she sacrifices continuation, which is a more efficient "currency," that is, a more efficient form of "payment" to the entrepreneur than monetary compensation (as long as $\rho < \rho_1$ and $R_b^+ \geqslant B/\Delta p$).

(iii) There exists a threshold α^* such that the separating equilibrium described above is the unique equilibrium whenever $\alpha \leqslant \alpha^*$. Other equilibria exist when $\alpha > \alpha^*$; they involve some pooling and are preferred by both types to the separating equilibrium.

Returning to the first implication, the discrete-shock model has a slightly awkward feature: the random probability of continuation in the case of a shock. This can be implemented either through a "random credit line" or, equivalently, through a "random debt": $d \geqslant r - (\rho - \rho_0)$ with probability $1 - x$ (precluding reinvestment in the case of a shock since $\rho + d > r + \rho_0$) and $d < r - (\rho - \rho_0)$ with probability x. In this sense, the debt is larger than under symmetric information (for which $x = 1$). The particular conclusion of a stochastic debt is rather unrealistic, but it is an artefact of the discrete-shock version: with a continuum of shocks (a continuous distribution $F(\rho)$ as in Chapter 5), the *short-term debt* d for the good borrower is deterministic and *larger than under symmetric information* (we leave it to the reader to demonstrate this property).

Relationship to the literature. The idea that short-term debt can be used as a signal of high-quality borrowing, which was first explored in a different context by Diamond (1991, 1993),[52] relates to a more general theme in the economics of adverse selection. Namely, (costly) short-term contracting may be a way of signaling that one is confident about the future and that one does not fear having to recontract at later stages. In Aghion and Bolton (1987), a supplier has superior information about the probability of entry of a competitor and would like to signal that this probability of entry is low in order to obtain better terms of trade when contracting today with buyers. Aghion and Bolton show that the supplier can signal to buyers that entry is unlikely by offering a contract specifying no penalty for breach if the buyer later switches to a different supplier; this is, in essence, a short-term contract. The point is that imposing no penalty for breach is less costly to the supplier when the probability of entry by a rival is small and so the "sorting condition" is satisfied. Hermalin (2002) considers a labor relationship in which a long-term contract between an employer and an employee specifying a penalty for breach induces the employer to provide general purpose on-the-job training to the employee. Hermalin shows that a worker with private information about her talent may want to signal a high talent by offering no

52. Ross (1977) also modeled debt as a signal of quality. Ross's model was not concerned with the maturity structure, but rather with the cost imposed on managers by bankruptcy. Under costly bankruptcy, issuing debt is relatively less costly for a borrower who knows that the probability of low profit is small.

penalty for breach in order to prove that she is not afraid of going back to the labor market, even though such a short-term contract deprives her of on-the-job training. In Diamond (1991), a borrower enters into a short-term borrowing contract in order to signal her creditworthiness. Diamond's model, unlike the one considered here, assumes that cash flows are not verifiable (but they are observable). Diamond shows that borrowers with high (respectively, intermediate, low) ratings use short- (respectively, long-, short-) term debt, where the rating refers to the *ex ante* probability of a good type.[53]

Finally, we have assumed, as elsewhere in this book, that entrepreneurs are rational. Landier and Thesmar (2004) study a competitive credit market in which optimistic and realistic entrepreneurs coexist. To some extent, optimistic entrepreneurs are akin to the confident borrowers (p-borrowers) of our adverse-selection model. Indeed, optimistic borrowers in Landier and Thesmar opt for shorter debt maturities than realistic entrepreneurs, as they (mistakenly) believe that they are unlikely to face difficult circumstances; relatedly, they are more willing to transfer control in such circumstances (for contingent transfers of control, see Chapter 10). Some features are different in a behavioral world, though. First, investors obviously pay the optimistic entrepreneurs "with dreams," yielding abnormally low returns to entrepreneurship (investors, however, do not benefit from the entrepreneurs' irrationality, since competition in the financial market drives investor profits to 0). Second, contracts may end up being contingent on variables that the borrower has no control over, violating a standard principle of agency theory.[54] Landier and Thesmar test their model on French entrepreneurship data and find a positive correlation between optimistic expectation errors (that they measure by comparing reported entrepreneurial expectations on future business growth and actual performance) and the use of short-term debt.

Application 7: Payout Policy

Large and well-established firms distribute a substantial fraction of their earnings in payouts (dividends and stock repurchases). For example, in 1999, U.S. corporations paid $350 billion in dividends and repurchases, plus an extra $400 billion on liquidation dividends associated with mergers and acquisitions. Indeed, most firms pay dividends while also raising debt or equity.

Payout behavior exhibits well-known patterns.[55] A key pattern for this chapter is that payout announcements affect stock prices and convey information beyond that contained in earnings announcements. The firm's stock price substantially increases (respectively, decreases) upon the announcement of an increase (respectively, decrease) in payout. This reaction is particularly strong for low-capitalization firms. All this suggests that dividends convey information held by the firm's insiders, but not by the stock market. This application focuses on this pattern and more generally on the *level* of payout; it thereby neglects interesting questions related to the choice of payout *structure* between dividends and share repurchases.[56]

Financial economists have repeatedly argued that dividends are used by a firm's insiders as signals. In particular, Bernheim and Wantz (1995) provide evidence that dividends are often motivated by signaling concerns rather than a disposal of free cash

53. Similar ideas have also been expressed in a screening setup, i.e., a setup in which the uninformed parties make the offers. In particular, in Michelacci and Suarez (2004), firms post employment contracts and learn the workers' abilities only after the workers have taken the job. Fixing the wage in advance has the benefit of eliminating holdup problems associated with bargaining after relationship-specific investments have been sunk by the parties. Alternatively, the firms can leave scope for recontracting or bargaining; this helps them address the adverse-selection problem, as high-ability workers, whose wage is higher under *ex post* bargaining than that of low-ability workers, may find such open-ended contracts more attractive than a fixed-wage contract. As a result, contracts tend to be too open-ended, which reduces aggregate income.

54. Namely, the sufficient statistic theorem (see Chapter 3).

55. See, for example, Allen and Michaely (1995, 2004) for exhaustive overviews and Karpoff and Thorley (1992) for a brief survey of the main facts. A large literature has been preoccupied with the firms' motivation to pay dividends, whether for signaling or for other reasons. Papers in this strand of research include Allen et al. (2000), Araujo et al. (2004), Benartzi et al. (1997), Bernheim (1991), and Healy and Palepu (1988).

56. A well-known puzzle is why corporations have traditionally favored dividends even in countries where the latter are taxed at the ordinary tax rate while share repurchases are taxed on a capital gain basis, which, combined with the ability to postpone the realization of capital gains, results in a lower effective tax rate (share repurchases caught up with dividends in the late 1990s).

Another interesting fact is that dividends are smoother (vary less over time) than share repurchases. Theories of why firms may opt for dissipative dividends include Ofer and Thakor (1987) and Hausch and Seward (1993).

$p\delta R_b^S + (1-p)\delta R_b^F = 0$ does not affect $U_b(c)$ and $U_l(c)$ and reduces $\tilde{U}_b(c)$. Third, suppose that $x < 1$. Then increasing x slightly, keeping xR_b^S constant, does not affect $U_b(c)$ and $\tilde{U}_b(c)$ and increases $U_l(c)$ (since $pR > I$). So, $x = 1$ given that $\tilde{U}_b(c) = 0$ (as was shown in the text, the full information solution does not hold under asymmetric information, and so the constraint $\tilde{U}_b(c) \leqslant 0$ must be binding). Hence, we can take $x = 1$, and because $\tilde{U}_b(c) = 0$

$$qR_b^S = \mathcal{A}.$$

We conclude that the low-information-intensity optimum is the allocation derived in Section 6.3.

Equilibrium uniqueness. We saw in the supplementary section that the issuance game admits a unique (perfect Bayesian) equilibrium if and only if the low-information-intensity optimum is interim efficient. We must therefore examine Program II (see the supplementary section). First, we minimize the investors' loss $\mathcal{L}(\tilde{\mathcal{R}})$ on the bad borrower when the bad borrower has net utility $\tilde{\mathcal{R}}$:

$$\min_{\{\tilde{x},\tilde{R}_b^S,\tilde{R}_b^F,\tilde{\mathcal{A}}\}} \mathcal{L}(\tilde{\mathcal{R}}) = -[\tilde{x}[q(R-\tilde{R}_b^S) + (1-q)(-\tilde{R}_b^F) - I] + \tilde{\mathcal{A}}]$$

s.t.

$$\tilde{x}[q\tilde{R}_b^S + (1-q)\tilde{R}_b^F] - \tilde{\mathcal{A}} \geqslant \tilde{\mathcal{R}},$$

where the notation mimics that just employed. And so

$$\begin{aligned}\mathcal{L}(\tilde{\mathcal{R}}) &= -\tilde{x}(qR - I) + \tilde{\mathcal{R}}\\ &= \tilde{\mathcal{R}}\end{aligned}$$

at the optimum (since $qR < I$).

The next program is the same as that for the low-information-intensity optimum except that (i) the breakeven condition is tightened by $(1-\alpha)\tilde{\mathcal{R}}$, and (ii) the mimicking condition is relaxed by $\tilde{\mathcal{R}}$:

$$\max_{\{x,\tilde{\mathcal{R}}\}} U_b(c)$$

s.t.

$$\alpha U_l(c) - (1-\alpha)\tilde{\mathcal{R}} \geqslant 0,$$

$$\tilde{U}_b(c) \leqslant \tilde{\mathcal{R}}.$$

By the same reasoning as for the low-information-intensity program, we can content ourselves with contractual terms c specifying $R_b^F = 0$. Then one can solve this program with respect to $(xR_b^S, x, \mathcal{A})$ rather than $(x, R_b^S, \mathcal{A})$ (it is a bit simpler) and show that

$$\mathcal{A} = A$$

and

$$\text{either } qR_b^S = A \text{ or } \tilde{R} > 0.$$

In sum, the good borrower can either leave no rent to the bad borrower and set

$$R_b^S = \frac{A}{q}$$

as in the separating allocation; or she can set $\tilde{\mathcal{R}} > 0$ and then R_b^S is determined by the investors' breakeven constraint:

$$\alpha[p(R - R_b^S) - (I - A)] + (1-\alpha)[-\mathcal{L}(\tilde{\mathcal{R}})] = 0,$$

where

$$\mathcal{L}(\tilde{\mathcal{R}}) = \tilde{\mathcal{R}} = qR_b^S - A.$$

And so

$$pR - I + A = \left[p + \frac{1-\alpha}{\alpha}q\right]R_b^S - \frac{1-\alpha}{\alpha}A.$$

She then gets a higher utility (whether she is a good or bad borrower) than in the separating equilibrium.

6.10 Exercises

Exercise 6.1 (privately known private benefit and market breakdown). Section 6.2 illustrated the possibility of market breakdown without the possibility of signaling. This exercise supplies another illustration. Let us consider the fixed-investment model of Section 3.2 and assume that *only the borrower knows the private benefit associated with misbehavior*. When the borrower has private information about this parameter, lenders are concerned that this private benefit might be high and induce the borrower to misbehave. In the parlance of information economics, the "bad types" are the types of borrower with high private benefit. We study the case of two possible levels of private benefit (see Exercise 6.2 for an extension to a continuum of possible types). The borrower wants to finance a fixed-size project costing I, and, for simplicity, has no equity ($A = 0$). The project yields R (success) or 0 (failure). The probability of success is p_H or p_L, depending on whether the

borrower works or shirks, with $\Delta p \equiv p_H - p_L > 0$. There is no private benefit when working. The private benefit B enjoyed by the borrower when shirking is either $B_L > 0$ or $B_H > B_L$. The borrower will be labeled a "good borrower" when $B = B_L$ and a "bad borrower" when $B = B_H$. At the date of contracting, the borrower knows the level of her private benefit, while the capital market puts (common knowledge) probabilities α that the borrower is a good borrower and $1-\alpha$ that she is a bad borrower. All other parameters are common knowledge between the borrower and the lenders.

To make things interesting, let us assume that under asymmetric information, the lenders are uncertain about whether the project should be funded:

$$p_H\left(R - \frac{B_H}{\Delta p}\right) < I < p_H\left(R - \frac{B_L}{\Delta p}\right). \tag{1}$$

Assume that investors cannot break even if the borrower shirks:

$$p_L R < I. \tag{2}$$

(i) Note that the investor cannot finance only good borrowers. Assume that the entrepreneur receives no reward in the case of failure (this is indeed optimal); consider the effect of rewards R_b in the case of success that are (a) smaller than $B_L/\Delta p$, (b) larger than $B_H/\Delta p$, (c) between these two values.

(ii) Show that there exists α^*, $0 < \alpha^* < 1$, such that

- no financing occurs if $\alpha < \alpha^*$,
- financing is an equilibrium if $\alpha \geqslant \alpha^*$.

(iii) Describe the "cross-subsidies" between types that occur when borrowing is feasible.

Exercise 6.2 (more on pooling in credit markets). Consider the model of Exercise 6.1, in which the borrower has private information about her benefit of misbehaving, except that the borrower's type is drawn from a continuous distribution instead of a binary one. We will also assume that there is a monopoly lender, who makes a credit offer to the borrower. The borrower has no equity ($A = 0$).

Only the borrower knows the private benefit B of misbehaving. The lender only knows that this private benefit is drawn from an *ex ante* cumulative distribution $H(B)$ on an interval $[0, \bar{B}]$ (so, $H(0) = 0$, $H(\bar{B}) = 1$). (Alternatively, one can imagine that lenders face a population of borrowers with characteristic B distributed according to distribution H, and are unable to tell different types of borrower apart in their credit analysis.) The lender knows all other parameters. For a loan agreement specifying share R_b for the borrower in the case of success, and 0 in the case of failure, show that the lender's expected profit is

$$U_l = H((\Delta p)R_b)p_H(R - R_b) + [1 - H((\Delta p)R_b)]p_L(R - R_b) - I.$$

Show that

- the proportion of "high-quality borrowers" (that is, of borrowers who behave) is endogenous and increases with R_b;[78]
- adverse selection reduces the quality of lending (if lending occurs, which as we will see cannot be taken for granted);
- there is an externality among different types of borrower, in that the low-quality types (B large) force the lender to charge an interest rate that generates strictly positive profit on high-quality types (those with small B);
- the credit market may "break down," that is, it may be the case that no credit is extended at all even though the borrower may be creditworthy (that is, have a low private benefit). To illustrate this, suppose that $p_L = 0$ and H is uniform ($H(B) = B/\bar{B}$). Show that if

$$\frac{p_H^2}{\bar{B}}\frac{R^2}{4} < I$$

(which is the case for $\bar{B}$ large enough), no loan agreement can enable the lender to recoup on average his investment.

Exercise 6.3 (reputational capital). Consider the fixed-investment model. All parameters are common knowledge between the borrower and the investors, except the private benefit which is known only to the borrower. The private benefit is equal to B with probability $1-\alpha$ and to b with probability α, where $B > b > 0$.

78. In this model, the loan agreement attracts all types of borrowers if it attracts any type willing to behave. It is easy to find variants of the model in which this is not the case and an increase in R_b attracts higher-quality borrowers, where "higher-quality" refers to an *ex ante* selection effect and not only to an *ex post* behavior like here.

(i) Consider first the one-period adverse-selection problem. Suppose that the borrower has assets $A > 0$ such that

$$p_{\mathrm{H}}\left(R - \frac{b}{\Delta p}\right) > I - A > \max\left[p_{\mathrm{H}}\left(R - \frac{B}{\Delta p}\right), p_{\mathrm{L}}R\right].$$

Show that the project receives funding if and only if

$$(p_{\mathrm{H}} - (1-\alpha)\Delta p)\left(R - \frac{b}{\Delta p}\right) \geqslant I - A.$$

(ii) Suppose now that there are two periods ($t = 1, 2$). The second period is described as in question (i), except that the belief $\tilde{\alpha}$ at date 2 is the posterior belief updated from the prior belief α, and that the borrower has cash A only if she has been successful at date 1 (and has 0 and is not funded if she has been unsuccessful). So, suppose that the first-period project is funded and that the borrower receives at the end of date 1 a reward A when successful and 0 when unsuccessful. The first-period funding is project finance and does not specify any funding for the second project. Suppose for notational simplicity that the private benefit is the same (B or b) in period 1 and in period 2. Let Δp_1 denote the increase in the probability of success when diligent in period 1. Assume that

$$\begin{aligned} b &< (\Delta p_1)A < B \\ &< (\Delta p_1)\left[p_{\mathrm{L}}\left(R - \frac{I-A}{p_{\mathrm{H}} - (1-\alpha_{\mathrm{S}})\Delta p}\right) + B\right] \end{aligned}$$

and

$$\begin{aligned} &(p_{\mathrm{H}} - (1-\alpha)\Delta p)\left(R - \frac{b}{\Delta p}\right) \\ &\qquad < I - A \\ &\qquad < (p_{\mathrm{H}} - (1-\alpha_{\mathrm{S}})\Delta p)\left(R - \frac{b}{\Delta p}\right), \end{aligned}$$

where $1 - \alpha_{\mathrm{S}} \equiv (1-\alpha)p_{\mathrm{L}}/((1-\alpha)p_{\mathrm{L}} + \alpha p_{\mathrm{H}})$.

A "pooling equilibrium" is an equilibrium in which the borrower's first-period effort is independent of her private benefit. A "separating equilibrium" is (here) an equilibrium in which the b-type works and the B-type shirks in period 1. A "semiseparating" equilibrium is (here) an equilibrium in which in period 1 the b-type works and the B-type randomizes between working and shirking.

- Show that there exists no pooling and no separating equilibrium.
- Compute the semiseparating equilibrium. Does this model formalize the notion of reputational capital?

Exercise 6.4 (equilibrium uniqueness in the suboptimal risk-sharing model). In the suboptimal risk-sharing model of Application 8, prove the claim made in the text that the low-information-intensity optimum depicted by $\{\mathrm{S}, \mathrm{B}\}$ in Figure 6.3 is interim efficient if and only if the belief that the borrower is a good borrower lies below some threshold α^*, $0 < \alpha^* < 1$. (Verify the weak-monotonic-profit condition in the supplementary section, and show that α^* is in the interior of the interval $[0, 1]$.)

Exercise 6.5 (asymmetric information about the value of assets in place and the negative stock price reaction to equity offerings with a continuum of types). Consider the privately-known-prospects model of Application 2 in Section 6.2.2, but with a continuum of types. The entrepreneur already owns a project, which with probability p yields profit R and probability $1 - p$ profit 0. The probability p is private information of the borrower. From the point of view of the investors, p is drawn from cumulative distribution $F(p)$ with continuous density $f(p) > 0$ on some interval $[\underline{p}, \bar{p}]$. Assume that the distribution has monotone hazard rates:

$$\frac{f(p)}{F(p)} \quad \text{is decreasing in } p$$

and

$$\frac{f(p)}{1 - F(p)} \quad \text{is increasing in } p.$$

(This assumption, which is satisfied by most usual distributions, is known to imply that the truncated means $m^-(p)$ and $m^+(p)$ have slope less than 1:

$$0 < (m^-(p))' \equiv \frac{\mathrm{d}}{\mathrm{d}p}[E(\tilde{p} \mid \tilde{p} \leqslant p)] \leqslant 1$$

and

$$0 < (m^+(p))' \equiv \frac{\mathrm{d}}{\mathrm{d}p}[E(\tilde{p} \mid \tilde{p} \geqslant p)] \leqslant 1$$

(see, for example, An 1998).)

The model is otherwise as in Section 6.2.2. A seasoned offering may be motivated by a profitable deepening investment: at cost I, the probability of success can be raised by an amount τ such that

$$\tau R > I$$

(of course, we need to assume that $\bar{p} + \tau \leqslant 1$). The entrepreneur has no cash on hand, is risk neutral, and is protected by limited liability. The investors are risk neutral and demand a rate of return equal to 0.

(i) Show that in any equilibrium, only types $p \leqslant p^*$, for some cutoff p^*, raise funds and finance the deepening investment.

(ii) Show that $p^* > \underline{p}$ and that if $p^* < \bar{p}$, then

$$\frac{\tau R}{I} = \frac{p^* + \tau}{m^-(p^*) + \tau}.$$

Show that if the benefits from investment are "not too large," in that

$$\frac{\tau R}{I} < \frac{\bar{p} + \tau}{E[p] + \tau},$$

then indeed $p^* < \bar{p}$.

Show that if there are multiple equilibria, the one with the highest cutoff p^* Pareto-dominates (is better for all types than) the other equilibria.

(iii) Is there a negative stock price reaction upon announcement of an equity issue?

(iv) Focusing on an interior Pareto-dominant equilibrium, show that, when τ increases, the volume of equity issues increases.

Exercise 6.6 (adverse selection and ratings). A borrower has assets A and must find financing for a fixed investment $I > A$. As usual, the project yields R (success) or 0 (failure). The borrower is protected by limited liability. The probability of success is p_H or p_L, depending on whether the borrower works or shirks, with $\Delta p \equiv p_H - p_L > 0$. There is no private benefit when working. The private benefit enjoyed by the borrower when shirking is either b (with probability α) or B (with probability $1 - \alpha$). At the date of contracting, the borrower knows her private benefit, but the market (which is risk neutral and charges a 0 average rate of interest) does not know it. Assume that $p_L R + B < I$ (the project is always inefficient if the borrower shirks) and that

$$p_H\left(R - \frac{B}{\Delta p}\right) < I - A < p_H\left(R - \frac{b}{\Delta p}\right) \qquad (1)$$

and

$$[\alpha p_H + (1-\alpha)p_L]\left[R - \frac{b}{\Delta p}\right] < I - A. \qquad (2)$$

(i) Interpret conditions (1) and (2) and show that there is no lending in equilibrium.

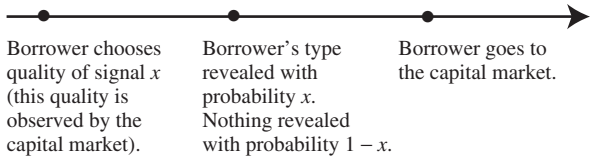


Figure 6.5

(ii) Suppose now that the borrower can at cost $r(x) = rx$ (which is paid from the cash endowment A) purchase a signal with quality $x \in [0,1]$. (This quality can be interpreted as the reputation or the number of rating agencies that the borrower contracts with.) With probability x, the signal reveals the borrower's type (b or B) perfectly; with probability $1 - x$, the signal reveals nothing. The financial market observes both the quality x of the signal chosen by the borrower and the outcome of the signal (full or no information). The borrower then offers a contract that gives the borrower R_b and the lenders $R - R_b$ in the case of success (so, a contract is the choice of an $R_b \in [0, R]$). The timing is summarized in Figure 6.5.

Look for a pure strategy, *separating* equilibrium, that is, an equilibrium in which the two types pick different signal qualities.[79]

- Argue that the bad borrower (borrower B) does not purchase a signal in a separating equilibrium.
- Argue that the good borrower (borrower b) borrows under the same conditions regardless of the signal's realization, in a separating equilibrium.
- Show that the good borrower chooses signal quality $x \in (0,1)$ given by

$$A = x(A - rx) + (1-x)\left[p_L\left(R - \frac{I - A + rx}{p_H}\right) + B\right].$$

- Show that this separating equilibrium exists only if r is "not too large."

Exercise 6.7 (endogenous communication among lenders). Padilla and Pagano (1997) and others have observed that information sharing about creditworthiness is widespread among lenders (banks,

79. One will assume that if the signal reveals the borrower's type, the investors put probability 1 on this type, even when they put weight 0 on the corresponding type after observing the quality of the signal.

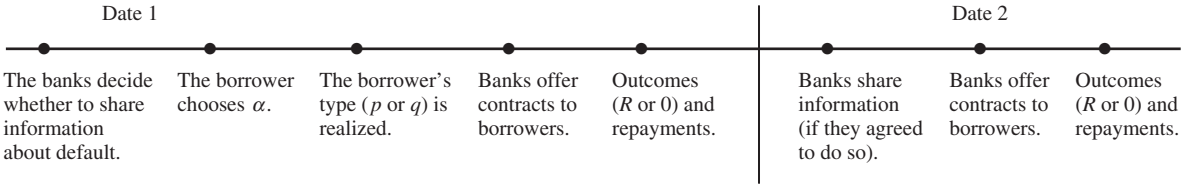


Figure 6.6

suppliers, etc.). For example, Dun & Bradstreet Information Services, one of the leading rating agencies, collects information from thousands of banks. Similarly, over 600,000 suppliers communicate information about delays and defaults by their customers; and credit bureaux centralize information about the consumer credit markets.

Padilla and Pagano (see also Pagano and Jappelli (1993) and the references therein) argue that information sharing has both costs and benefits for the banks. By sharing information, they reduce their differentiation and compete more with each other. But this competition protects their borrowers' investment and therefore enhances opportunities for lending. In a sense, the "tax rate" (the markup that banks can charge borrowers) decreases but the "tax base" (the creditworthiness of borrowers) expands. This exercise builds on the Padilla–Pagano model.

There are two periods ($t = 1, 2$). The discount factor between the two periods is δ. A risk-neutral borrower protected by limited liability has no cash on hand ($A = 0$). Each period, the borrower has a project with investment cost I. The project delivers at the end of the period R or 0. There is no moral hazard. The probability of success is p if the entrepreneur is talented (which has probability α), and q if she is not (which has probability $1 - \alpha$). We will assume that the market rate of interest in the economy is 0, that the lenders are risk neutral, and that only the good type is creditworthy:

$$pR > I > qR.$$

The date-1 and date-2 projects (if financed) are correlated and yield the same profit (they both succeed or both fail).

There are n towns. Each town has one bank and one borrower. The "local bank" has local expertise and thereby learns the local borrower's type; the other banks, the "foreign banks," learn nothing (and therefore have beliefs α that the entrepreneur is talented) at date 1. At date 2, the foreign banks learn

- only whether the borrower was financed at date 1, if there is no information sharing among banks;
- whether the borrower was financed at date 1 and whether she repaid (i.e., whether she was successful), if there is information sharing about riskiness.

In other words, information sharing is feasible on hard data (repayments), but not on soft data (assessment of ability).

Padilla and Pagano add two twists to the model. First, banks decide *ex ante* whether they will communicate information about default and they make this decision public. Second, the borrower's type may be endogenous (in which it refers more to an investment in the projects or industry than in "pure talent"): at increasing and convex cost $C(\alpha)$ $\{C' > 0, C'' > 0, C(0) = 0, C'(0) = 0, C'(1) = \infty\}$, the borrower develops a p project with probability α and a q project with probability $1 - \alpha$. C can be viewed as an investment cost and represents a nonmonetary cost borne by the borrower.

Contracts between banks and borrowers are *short-term* contracts. These contracts just specify a payment R_b for the borrower in the case of success during the period (and 0 in the case of failure). Furthermore, in each period, banks simultaneously make take-it-or-leave-it offers to borrowers. And at date 2, the incumbent bank (the bank that has lent at date 1) makes its offer after the other banks.

The timing is summarized in Figure 6.6.

(i) Suppose first that the probability α of being a p-type is exogenous (there is no borrower investment), that $[\alpha p + (1-\alpha)q]R - I + \delta(\alpha p + (1-\alpha)q)(R - I) < 0$, and that $qR - I + \delta q(R - I) < 0$. Show that the banks prefer not to share information.

(ii) Next, suppose that the borrower chooses α. Assuming that the two assumptions made in (i) still hold in the relevant range of αs (for example, $\alpha \in [0, \bar{\alpha}]$, where $\bar{\alpha}$ satisfies the conditions), show that the banks choose to share information.

Exercise 6.8 (pecking order with variable investment). Consider the privately-known-prospects model with risk neutrality and variable investment. For investment I, the realized income is either $R^S I$ (in the case of success) or $R^F I$ (in the case of failure), where $R^S > R^F \geqslant 0$. A good borrower has probability p_H of success when working and p_L when shirking; similarly, a bad borrower has probability q_H of success when working and q_L when shirking, where $p_H - p_L = \Delta p = q_H - q_L$, for simplicity. The entrepreneur's private benefit is 0 when working and BI when shirking. The entrepreneur is risk neutral and protected by limited liability; the investors are risk neutral and demand a rate of return equal to 0.

(i) Let $\tilde{U}_b^{SI}$ denote the bad borrower's gross utility under symmetric information.[80] Consider the problem of maximizing the good borrower's utility subject to the investors' breaking even on that borrower, to the mimicking constraint that the good borrower's terms not be preferred by the bad borrower to her symmetric-information terms, and to the no-shirking constraint. Let $\{R_b^S, R_b^F\}$ denote the (nonnegative) rewards of the good borrower in the cases of success and failure. Write the separating program.

(ii) Show that $R_b^F = 0$.

(iii) (Only if you have read the supplementary section.) Show that the separating outcome is the only perfect Bayesian equilibrium of the issuance game if and only if $\alpha \leqslant \alpha^*$ for some threshold α^*.

Exercise 6.9 (herd behavior). It is often argued that the managers of industrial companies, banks, or mutual funds are prone to herd.[81] They engage in similar investments with sometimes little evidence that their strategy is the most profitable. An economic agent may indeed select a popular strategy against her own information that another strategy may be more profitable. A number of contributions have demonstrated that herding behavior may actually be individually rational even though it is often collectively inefficient. The literature on herding behavior starts with the seminal contributions of Banerjee (1992), Bikhchandani et al. (1992), Scharfstein and Stein (1990), and Welch (1992); see Bikhchandani and Sharma (2001) for a survey of applications of this literature to financial markets.

There are several variants of the following basic argument. Consider first a sequence of agents $i = 1, 2, \ldots$ choosing sequentially between strategies A and B. Agents receive their own signals; they observe previous decisions but not the others' signals. Suppose that agents 1 and 2 have, on the basis of their own information, selected A. Agent 3, observing the first two choices, may well then select strategy A even if her own signal favors the choice of B. Agent 4, not knowing agent 3's motivation to choose A, may then also choose strategy A even if his own signal points toward the choice of B. And so forth. It may therefore be the case that all agents choose A, even though the cumulative evidence, if it were shared, would indicate that B is the best choice.

The literature also analyzes herd behavior in situations in which agents have principals (that is, they are not full residual claimants for the consequences of their choices). In particular, such agents may adopt herd behaviors because of *reputational concerns* (see Chapter 7). Suppose, for instance, that a manager's job is rather secure; herding with the managers of other firms is then likely to be attractive to the manager: if the strategy fails, the manager has the excuse that other managers also got it wrong ("it was hard to predict"). Choosing an unpopular strategy, even if one's information points in that direction, is risky, as there will be no excuse if it fails. The literature on herd behavior has also investigated the use of benchmarking by principals in *explicit incentives* (compensation contracts) rather than in implicit ones (career concerns).

Let us build an example of herding behavior in the context of the privately-known-prospects model of Section 6.2. There are two entrepreneurs, $i = 1, 2$, operating in different markets, but whose optimal

80. This utility was derived in Section 3.4.2. It is equal to

$$\left[1 + \frac{q_H R - 1}{1 - q_H(R - B/\Delta p)}\right] A$$

if $q_H R \geqslant 1$, and to A otherwise.

81. One of the first empirical papers on herding behavior is Lakonishok et al. (1992). The large empirical literature on the topic includes Chevalier and Ellison (1999).

strategy is correlated. There are two periods, $t = 1, 2$. Entrepreneur i can raise funds only at date $t = i$ (so they secure funding sequentially). A project yields R when it succeeds and 0 when it fails. The entrepreneurs are risk neutral and protected by limited liability; the investors are risk neutral and demand a rate of return equal to 0. The entrepreneurs have no net worth or cash initially.

The two entrepreneurs each have to choose between strategy A and B. Strategies differ in their probability of success. A borrowing contract with investors specifies both the managerial compensation R_b in the case of success (and 0 in the case of failure) and the strategy that the entrepreneur will select.[82] Crucially, entrepreneur 2 and her potential investors observe the date-1 financing contract for entrepreneur 1. Entrepreneurs, but not investors, learn the state of nature.

Consider the following stochastic structure.

Unfavorable environment (probability $1 - \alpha$). The probabilities of success are, with equal probabilities, $(q, 0)$ for one project and $(0, q)$ for the other, where the first element is entrepreneur 1's probability of success and the second entrepreneur 2's. So entrepreneurs necessarily choose different projects if they apply for funding.

Favorable environment (probability α). With probability θ, the best project is the same for both and has probability of success p; the worst project for both has probability of success r, where

$$p > \max\{q, r\}.$$

With probability $1 - \theta$, the two entrepreneurs' best strategies differ: the probabilities of success are (p, r) and (r, p), respectively, for entrepreneur 1's and entrepreneur 2's best strategy (which are A or B with equal probabilities). Thus θ is the probability of correlation of the best strategies in a favorable environment; this probability is equal to 0 in the unfavorable environment.

Let $m \equiv \alpha p + (1 - \alpha)q$ and assume that

$$qR > I.$$

Show that funding and herding (with probability $\alpha(1 - \theta)$, entrepreneur 2 chooses entrepreneur 1's best strategy even though it does not maximize her probability of success) is an equilibrium behavior as long as

$$r\left[R - \frac{I}{\theta p + (1 - \theta)r}\right] \geqslant p\left[R - \frac{I}{q}\right].$$

Note that entrepreneur 2 is on average worse off than in an hypothetical situation in which investors did not observe the strategy of entrepreneur 1 (or that in which the optimal strategies were uncorrelated).

Exercise 6.10 (maturity structure). At date 0 the entrepreneur has cash on hand A and needs to finance an investment of fixed size I. At date 1, a deterministic income r accrues; a liquidity shock must be met in order for the firm to continue. Liquidation yields nothing. The probability of success in the case of continuation depends on a date-1 effort: for a good borrower, this probability is p_H or p_L depending on whether she behaves (no private benefit) or misbehaves (private benefit B); similarly, for a bad borrower, it is q_H or q_L. We assume that

$$p_H - p_L = q_H - q_L = \Delta p,$$

and so the incentive compatibility constraint in the case of continuation is the same for both types of borrower:

$$(p_H - p_L)R_b = (q_H - q_L)R_b = (\Delta p)R_b \geqslant B,$$

where R_b is the borrower's reward in the case of success.

The borrower knows at the date of contracting whether she is a "p-type" or a "q-type." Let

$$\rho_0^G \equiv p_H\left(R - \frac{B}{\Delta p}\right) \quad \text{and} \quad \rho_0^B = q_H\left(R - \frac{B}{\Delta p}\right)$$

denote the date-1 pledgeable incomes for the good and bad types.

The liquidity shock is deterministic and equal to ρ. Information is asymmetric at date 0, but the capital market learns the borrower's type perfectly at date 1, before the liquidity shock has to be met. Assume that

$$\rho_0^G > \rho > \rho_0^B.$$

Suppose further that under symmetric information only the good borrower is creditworthy (provided that she is incentivized to behave).

Assume that $r < I - A < r + [\rho_0^G - \rho]$. Show that the good borrower can costlessly signal her type.

82. One will assume therefore that the first entrepreneur cannot condition her financing contract on the later choice of strategy by the second entrepreneur.